

11.5 The Geochemistry of Acid Mine Drainage

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[†]Deceased

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11.5.1 Introduction

11.5.1.1 Scale of the Problem

Mine wastes are the largest volume of materials handled in the world (ICOLD, 1996). The generation of acidic drainage and the release of water containing high concentrations of dissolved metals from these wastes are an environmental problem of international scale. Acidic drainage is caused by the oxidation of sulfide minerals exposed to atmospheric oxygen. Although acid drainage is commonly associated with the extraction and processing of sulfide-bearing metalliferous ore deposits and sulfide-rich coal, acidic drainage can occur wherever sulfide minerals are excavated and exposed to atmospheric oxygen. Engineering projects, including road construction, airport development, and foundation excavation, are examples of civil projects that have resulted in the generation of acidic drainage. On US Forest Service lands, there are between 20 000 and 50 000 mines releasing acidic drainage (USDA, 1993). Kleinmann et al. (1991) estimated that more than 6400 km of rivers and streams in the Eastern United States have been adversely affected by mine-drainage water. Between 8000 and 16 000 km of streams have been affected by metal mining in the Western United States. The annual worldwide production of mine wastes exceeded 4.5 billion tonnes in 1982 (ICOLD, 1996). In Canada alone, there are approximately 980 million tonnes of mine wastes, with the potential to cause acidic drainage (Government of Canada,

1991). The estimated costs for remediating mine wastes internationally total in the tens of billions of dollars (Feasby, 1993).

11.5.1.2 Overview of the Mining Process and Sources of Low-Quality Drainage

The recovery of metals from sulfide-rich ore bodies proceeds through a series of steps: from mining to crushing to mineral recovery (i.e., concentration), followed typically by smelting of the sulfide concentrates and thence to metal refining; although the nature of the ore body dictates the processes used to extract metals from ores, each of these steps generates a waste stream. The volumes of the waste streams can be large. For example, the production of 1 tonne of copper typically requires the excavation and processing of 100 tonnes of ores, not including the removal of overburden or rock to access the ore. Each of the steps of metal production can lead to the generation of low-quality water.

11.5.1.3 Sources of Low-Quality Drainage

11.5.1.3.1 Mine workings

Minerals are typically excavated by underground mining, strip mining, or open-pit mining. The selection of the mining technique is dictated by the physical structure, location, and grade or value of the ore body and by the characteristics of the

adjacent geological materials. Although open-pit mining and underground mining are the two most common mining techniques, placer mining and solution mining also have been used for mineral extraction. Placer mining involves excavation of river or stream sediments and separation of valuable minerals by gravity, by selective flotation, or by chemical extraction. Most solution mining is by heap leaching in which the extractant solution is trickled over broken ore on the surface or in underground workings; less common is injection into underground geological formations. The consequence of the excavation of open-pits and other mining-related disturbances is that sulfide minerals previously isolated from the atmosphere are exposed to oxygen. The oxidation of sulfide minerals ensues.

11.5.1.3.2 Open-pits

Open-pit mining is a surface mining technique employed to extract ores from large deposits that are relatively close to the surface. By design, open-pit mining can expose large surface areas of wall rock to atmospheric oxygen. If the wall rocks contain iron sulfide minerals, open-pits can be an important source of acidic drainage during and following mining as long as they remain exposed. Open-pits are typically several hundreds of meters deep and wide. For example, the Kennecott Copper Mine in Utah, United States, is 4 km wide and 1.2 km deep. Thus, large surfaces of rocks are exposed as potential sources of acidic drainage.

11.5.1.3.3 Waste rock

Open-pit and underground mining result in the excavation of large volumes of rock to gain access to ore bodies. After the ore body is accessed, ore for processing is separated from the host rock on the basis of economic cutoff values. Ore of higher metal grade is processed, and rock below the cutoff grade is put to waste. Frequently, ore is segregated into high-grade and low-grade ore stockpiles. Ore is the material that will yield a profit; thus, the metal contents of the discrimination between high-grade and low-grade ores will vary with the costs of mining activities and the value of the metals extracted. The waste rock from mining operations may be used in construction activities at the mine site. Excess waste rock is deposited in waste-rock piles whose composition differs greatly from mine to mine because of variations in ore-deposit and host-rock mineralogy and because of differences in the processing techniques and ore-grade cutoff values. Daily production of waste rock in Canada is estimated to be 1 000 000 tonnes (Government of Canada, 1991). Because of the large volume of rock excavated in open-pit operations, waste-rock piles may be tens of hectares in area and tens of meters in height (Ritchie, 1994).

11.5.1.3.4 Mill tailings

The ore extracted in most nonferrous, metalliferous mining operations is rich in base or precious metals, but the ore minerals are generally too dilute for direct processing using metallurgical techniques. Thus, most ores are processed through comminution steps that involve crushing and grinding to a fine-grain size and concentration steps such as gravity, magnetic, and flotation for beneficiation or upgrading of the ore minerals in the matrix. The grain size of the milled rock is dictated by the process used for mineral recovery. Typical grain

sizes range from 25 μm to 1.0 mm. At many plants, differential flotation is used to separate the valuable sulfide minerals containing base or precious metals from others (e.g., pyrite $[\text{FeS}_2]$ or pyrrhotite $[\text{Fe}_{0.875-1}\text{S}]$) that have no to little commercial value. A flotation concentrator may contain several circuits for the selective recovery of a variety of metal sulfides and the production of a series of metal-sulfide concentrates. The concentrate from the flotation step is retained for further metallurgical processing. Mill tailings are the residual material, including sulfide gangue minerals, that is discharged to tailings impoundments, typically as a slurry of water and finely ground rock. Mineral particles are removed from the slurry by gravity sedimentation, and the water is often recycled to the mill or discharged.

The ratio of tailings to concentrate can be very large, particularly at gold and precious-metal mines at which the concentrate may represent only a small fraction (e.g., $\sim 1\%$) of the ore processed. The mining industry produces immense amounts of mine tailings. The mass of tailings produced daily in Canada is estimated to be 950 000 tonnes (Government of Canada, 1991). Tailings impoundments may be very large. For example, the Inco Ltd. Central Tailings Disposal Area covers an area of 25 km^2 with tailings up to 50 m in depth, and the ultimate capacity is more than 725 million tonnes (Puro et al., 1995).

Mill tailings are typically retained in impoundments. The retaining dams of many impoundments are constructed of coarse-grained tailings or of tailings combined with waste rock. These types of impoundments are designed to drain, thereby enhancing their structural integrity but resulting in the development of a thick zone of only partial saturation. The entry of gas-phase oxygen into the unsaturated tailings results in sulfide-mineral oxidation and in the release of low-quality drainage.

11.5.1.3.5 Wastes from extractive metallurgy operations

Metallurgical processing following milling (i.e., comminution and concentrating) involves extractive metallurgy operations such as hydrometallurgy, pyrometallurgy, and electrometallurgy. Hydrometallurgy includes leaching operations performed in aqueous media using various chemicals and oxidizing agents (e.g., pressure leaching, pressure oxidation (POX), heap leaching, and bioleaching/oxidation). Resultant oxidation and leaching products are fine-grained residue formed during the process, which includes compounds such as elemental sulfur, various sulfate compounds, jarosite, goethite, and hematite.

Pyrometallurgy operations such as smelting and roasting involve processing of ores and other materials at high temperatures (i.e., often $>1000^\circ\text{C}$) and in the absence of aqueous media. Environmental issues related to pyrometallurgy involve gaseous and particulate emissions (e.g., SO_2 , CO_2 , CO, Hg, As-, Sb-, Se-, and Te-oxides). Significant amounts of SO_2 gas is produced from the oxidation of sulfide minerals during pyrometallurgical treatments. Although most of the SO_2 is captured and converted to sulfuric acid or elemental sulfur and/or reacted with lime to form gypsum, the quantities of roaster and smelting gases are significant. In addition, smelting operations produce wastes such as slags and sludge that are disposed in waste impoundments and sludge ponds. Electrometallurgy operations utilizing electrical energy for electrolysis include electrorefining and electrowinning unit operations. Wastes contain small quantities of fine-grained residue and slimes incorporating impurity elements originating from the anodes.

11.5.2 Mineralogy of Ore Deposits

11.5.2.1 Coal Deposits

Coal is a fossil fuel that represents the remains of flora that accumulated as peat in swamps and bogs during geological and prehistoric times. The accumulation occurred in submerged conditions, thereby preventing complete decay of the organic material to $\text{CO}_2 + \text{H}_2\text{O}$ during the early stage of maturation. Reflecting its origin, coal occurs in beds, maximum thicknesses approximate 90 m, and typical mining thicknesses are 1–4 m.

The uses of coal include electricity generation, steelmaking, and cement production. The carbon content of coal varies from about 70% to 95 wt% with most of the remainder consisting of O, H, N, and S. The O content generally ranges from about 2% to 20%, and the major change that occurs during coalification is a decrease in the O content and an increase in the carbon content. With this change, the physical properties and thermal yield per unit weight also change, and various classifications have been devised to reflect those properties. A common commercial subdivision is into 'brown coal' and 'hard coal,' which in turn is an indication of the degree of induration. Coals can also be grouped under four types with decreasing energy content: anthracite, bituminous, subbituminous, and lignite.

Proven coal reserves are estimated to be 826 billion tonnes (World Energy Council, 2009). The bituminous type forms about 52% of the world reserves followed by subbituminous type at 30% and lignite at 17%. The reserves are about equally distributed among Europe and Asia region (33%), Asia Pacific region (31.4%), and North America (29.8%) with the United States, Russia, China, and India having the biggest reserves. According to International Energy Agency (2010) figures, current world production is about 5990 Mt (million metric tonnes) of hard coal and 913 Mt of brown coal. Germany, at about 160 Mt, is the largest producer of brown coal, with Russia a distant second. China's production of 2971 Mt of hard coal is the world's highest, followed by the United States (919 Mt). India ranks third and produces 526 Mt. In recent years, the United States has undergone a pronounced shift to mining of Western coal, predominantly subbituminous, which has a lower average content of sulfur than coals from Appalachia. Wyoming alone now accounts for about a third of all US production. According to the statistics given by BP Statistical Review of World Energy (2010), the production and

consumption of coal in the Asia Pacific region have more than doubled within the last decade (i.e., 1000–2200 Mt oil equivalent in 1999). This is in sharp contrast to the figures from North America, and Europe and Eurasia regions where the production and consumption figures have decreased slightly.

Environmental concerns have been focused on the gaseous (oxides of sulfur and nitrogen and greenhouse gases such as carbon dioxide and methane), particulate, and trace-element emissions (e.g., mercury, selenium, and arsenic), on the environmental quality of the ash and slag residues, and on acidic drainage that may ensue as a consequence of the exposure of mining-related wastes and mine openings to atmospheric weathering. Many of the environmentally least desirable aspects concerning the utilization of coal are related to the presence of mineral matter, especially Fe disulfides. The disulfides are a principal source for SO_2 emissions during combustion, and in mine wastes, the oxidation of FeS_2 is the principal cause of the development of acidic drainage.

Stach et al. (1982) list more than 40 minerals that have been identified as occurring in coal, and recent observations have expanded the total to more than a hundred. Finkelman (1980a,b) concluded that coals yielding >5 wt% ash have had the bulk of their minerals derived by detrital processes.

Table 1 summarizes and presents an interpretation of the occurrence of the principal nondetrital minerals in coal. The chief detrital minerals are quartz and clay minerals (including K–Al micas), and these minerals commonly form up to 90% of the mineral matter in coals. The bulk of the remainder typically consists of carbonates and pyrite. Renton (1982) observed that most discrete mineral grains observed in coal are about 20 μm in diameter, and few exceed 100 μm . Exceptions are concretions, nodules, and 'balls' that typically contain one or more of pyrite, marcasite [FeS_2], calcite [CaCO_3], and siderite [FeCO_3] and which may be many centimeters in diameter. As well, aggregates of pyrite and marcasite occur within coal and as fracture (cleat) mineralization. The most common cleat-filling minerals are calcite, pyrite, and kaolinite [$\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$] (Renton, 1982). Vassilev et al. (1996) observed that higher rank coals are enriched in elements associated with probable detrital minerals, whereas lower rank coals are enriched in elements associated with probable authigenic minerals and organic material. The magnitude of the concentrations of trace elements in coal is summarized in Table 2. Modern

Table 1 Some minerals observed in coal deposits^a

Mineral group	Inherent	Extraneous source
Clay minerals	Kaolinite, illite–sericite ^b , mixed-layer clays, smectite	Illite–sericite ^b , chlorite
Carbonates	Calcite, siderite, dolomite–ankerite	Calcite, dolomite–ankerite
Sulfides	Pyrite, marcasite, sphalerite, galena, chalcopyrite, pyrrhotite, greigite	Pyrite, marcasite, sphalerite, galena, chalcopyrite
Oxides	Quartz, Fe oxyhydroxides, hematite	Quartz, goethite, lepidocrocite
Phosphates	Apatite, crandallite-group minerals, vivianite	Apatite
Others	Gypsum, halite	Sulfates, chlorides, nitrates

^aInterpreted partly from data in Mackowsky (1968), Renton (1982), Stach et al. (1982), Harvey and Ruch (1986), Birk (1989), Ward (1989), Speight (1994), and Spears (1997). The extraneous source minerals typically form after consolidation has progressed to the state at which the coal can sustain fracturing, and the minerals occur as fracture fillings and in cavities. The minerals in each group are listed in approximate decreasing order of abundance.

^bSericite is a general term for fine-grained, mica-like minerals. Illite has been assigned a specific composition by the International Mineralogical Association (Rieder et al., 1998), but illite and sericite are used here only to designate mica-type minerals. The mineral ankerite has a formula $\text{Fe} > \text{Mg}$, but the name is commonly used for ferroan dolomite [$\text{Ca}(\text{Mg,Fe})(\text{CO}_3)_2$].

Table 2 The magnitude of the trace-element contents (ppm) of coal and coal ash

Element	Coal		Coal ash	
	Average ^a	Range ^b	Average ^c	Range ^d
Antimony	3.0	0.05–10		200?
Arsenic	5.0	0.5–80	500	100–500?
Barium	500	20–1000		300–900
Beryllium	3	0.1–15	45	1–10
Bismuth	5.5		20	2–50
Boron	75	5–400	600	
Cadmium	1.3	0.1–3	5	5?
Chlorine	1000	50–2000		
Chromium	10	0.5–60		100–400
Cobalt	5	0.5–30	300	300
Copper	15	0.5–50		20–200
Fluorine		20–500		
Gallium	7	1–20	100	100?
Germanium	5	0.5–50	500	50–500
Lead	25	2–80	100	5–50?
Lithium	65	1–80		
Manganese	50	5–300		
Mercury	0.01	0.02–1		0.02–0.5?
Molybdenum	5	0.1–10	50	100–200
Nickel	15	0.5–50	700	50–800
Phosphorus	500	10–3000		
Scandium	5	1–10	60	
Selenium	3	0.2–1.4		60?
Silver	0.50		2	1–5
Strontium	500	15–500		80–170?
Thallium		<0.2–1		1?
Thorium		0.5–10		
Tin		1–10		16–200?
Titanium	500	10–2000		
Uranium	1.0	0.5–10	400	
Vanadium	25	2–100		100–1000
Zinc	50	5–300		100–1000?
Zirconium		5–200		100–500?

^aFrom the US National Committee for Geochemistry, as cited in Valković (1983).^bSwaine (1990).^cMason (1958).^dKrauskopf (1955).

technology utilizes coal washing, fluidized bed combustion, activated carbon injection, gasification of coal, flue gas desulfurization, and particulate control devices to improve energy extraction while limiting the gaseous and liquid emissions. The lower temperatures of those processes affect the fate of the individual trace elements that are emitted or are associated with the residues from the consumed coal (Table 2; Clarke, 1993).

11.5.2.2 Base-Metal Deposits

Base metal is a wide-ranging term that refers either to metals inferior in value to those of gold and silver or, alternatively, to metals that are more chemically active than gold, silver, and the platinum metals (AGI, 1957). Accordingly, a review of base-metal mineralogy would encompass much of the world's metal production and geology. Usage of the 'base metal' term in the mining and minerals industry is rather loose, but a

common application is to the nonferrous metals excluding precious metals. These include copper, lead, zinc, nickel, and tin. Kesler (1994), however, grouped nickel with ferroalloy metals along with manganese, chromium, silicon, cobalt, molybdenum, vanadium, tungsten, niobium, and tellurium and copper, lead, zinc, and tin as base metals. Among the base metals, tin is by far the least significant in terms of volumes consumed and monetary value.

World mine production of copper in 2009 exceeded 15 Mt, about a third of which is from Chile. Other large producers are Peru and the United States, followed closely by China, Indonesia, and Australia. The most important ore mineral is chalcopyrite [CuFeS₂] and also significant are bornite [Cu₅FeS₄] and chalcocite [Cu₂S]. These three minerals make up about 90% of the primary copper production. The first two are the primary minerals, whereas chalcocite forms principally by their weathering and subsequent reprecipitation of the solubilized Cu as enriched 'blankets' of chalcocite ore beneath the oxidation zone.

Copper ore is predominantly derived from porphyry copper deposits, with lesser but significant contributions from massive sulfide, skarn, and other types of deposits. The host rocks for porphyry copper deposits are felsic granitoid intrusions, and in skarn deposits, the intrusions penetrate limestone and associated sedimentary-derived assemblages. The deposits are typically large (commonly hundreds of million tonnes) and of low grade (commonly <1% Cu), with successful exploitation dependent mainly on open-pit access and on daily large-tonnage extraction and processing.

Based on the 2009 figures, the world mine production of zinc is about 11 Mt, with almost all of it derived from sphalerite [(Zn,Fe)S], which is also the principal primary source of Cd and several other metals, such as Ge and In. China, Peru, and Australia are the largest producers, but several other countries including Canada, the United States, and India mine significant amounts. About half of the annual consumption is for the manufacture of galvanized products to resist corrosion, primarily in the automotive and construction industries.

Sources of sphalerite in mineral deposits are diverse. Large production is obtained chiefly from skarn (e.g., Antamina, Peru); from volcanogenic massive sulfide deposits (e.g., Kidd Creek and Brunswick No. 12, Canada) in which pyrite is the predominant mineral; from sedimentary-exhalative (SEDEX) deposits (e.g., Broken Hill and Mt. Isa, Australia) in which layers of Pb, Zn, and Fe sulfides occur in fine-grained clastic sedimentary rocks; and from Mississippi Valley-type deposits (e.g., Viburnum Trend, United States) in which sphalerite and galena [PbS] were deposited in large amounts in cavities, breccias, and as replacements of calcareous sedimentary rocks consisting predominantly of limestone.

Whereas sphalerite is the principal mineral source of Zn, galena is the main source of Pb. The annual world mining production of lead reached about 4 Mt in 2009, and the annual consumption is nearing 9 Mt, with the difference made up by recycling. The largest primary producers are China, Australia, and the United States, and the largest consumers are China and the United States. More than 75% of lead consumption is for the manufacture of lead-acid automotive batteries, which also are the principal source of recycled scrap. Unlike Zn, which is an essential biological trace element, Pb has no similar

function and is an important environmental hazard (Kesler, 1994). The processing of Pb and Zn concentrates is almost totally by conventional pyrometallurgical smelting, but the most abundant anthropogenic sources of Pb have been coal combustion and gasoline additives (Kesler, 1994).

Nickel production from primary sources includes Ni sulfide deposits located principally in Canada, Russia, Australia, and South Africa and laterite deposits in New Caledonia, Indonesia, Colombia, Brazil, Cuba, and the Dominican Republic. Russia is the top producer of Ni from the Norilsk–Talnakh Ni sulfide deposits that are considered to be the largest in the world. Canada, Indonesia, and Australia are the other important Ni producers. Stainless steel and alloys account for more than 80% of Ni consumption. Nickel sulfide deposits are associated with mafic and ultramafic complexes where pentlandite is the main mineral source of Ni. Pyrrhotite and chalcopyrite are the other abundant sulfide minerals. Laterite deposits are remnants of highly weathered mafic and ultramafic rocks. Nickel is hosted by silicate minerals and Fe(III) oxyhydroxides such as goethite.

11.5.2.3 Precious-Metal Deposits

The precious-metal group consists of gold, silver, and the platinum-group elements (PGEs). The world's leading producer of Au is China, followed by Australia, the United States, Russia, and South Africa forming nearly half of the total world production figure of 2572 Mt in 2009. Most mining of Au is done specifically for that metal rather than for a polymetallic assemblage, and most Au is produced from auriferous quartz veins also known as lode deposits. However, appreciable amounts (i.e., ~20%) of Au are recovered as a by-product from the processing of base-metal ores, especially Cu deposits. A characteristic feature of all types of deposits is that nearly all of the Au occurs as the native metal, commonly with Ag in solid solution. Another type of economically important Au deposits is known as the refractory Au where Au occurs in solid solution or as nanoparticulate form in arsenical pyrite and/or arsenopyrite. Gold is recovered from such ores by the destruction of the host sulfide through roasting, POX, or bacterial oxidation (BIOX) processes prior to cyanidation. Environmental concerns related to roasting are the high-As emissions and the disposal of the As_2O_3 that precipitates from the condensed gases. The use of pressurized autoclaves to oxidize the host sulfide minerals has increased to decrease As emissions. Another environmental concern has been the use of mercury to recover gold by amalgamation. This practice has been largely discontinued because the effects of mercury poisoning are well known, but a legacy of pollution remains in many areas, and amalgamation on a small scale is still practiced by artisan miners in countries such as Brazil, Guyana, and Indonesia.

Gold in recent years has found increased markets in electrical and electronic applications, but these account for <5% of the annual consumption. About 90% of the annual production is utilized for jewelry and arts purposes.

Peru, China, Mexico, and Chile are the largest producers of silver, whose main usage is in photography, plating, jewelry, and electronic and electrical applications. More than two-thirds of the world's production of silver in 2009 was obtained

as a by-product from smelting of base-metal ores including those from copper–gold deposits. For example, the world's largest silver producer is the metamorphosed, stratabound Cannington deposit in Australia. The deposit is of the Broken Hill type and contains about 44 Mt grading 11.6% Pb, 4.4% Zn, and 538 g t^{-1} Ag (Walters and Bailey, 1998). The dominant sulfide assemblage is galena–sphalerite–pyrrhotite, and the high-Ag content is related mainly to the presence of argenterous galena and freibergite $[(\text{Ag,Cu,Fe})_{12}\text{Sb}_4\text{S}_{13}]$.

Mexico, one of the leading silver producers by country, obtains about half of its output from mines in which silver is the principal ore metal. Many of the mines are epithermal fissure veins, and most host a polymetallic assemblage whose exploitation is economically dependent on the high-Ag values. Although acanthite $[\text{Ag}_2\text{S}]$ and native silver predominate in some veins, in others, much of the Ag occurs in Ag sulfosalts and as Ag substitutions in tetrahedrite $[(\text{Cu,Fe,Ag})_{12}\text{Sb}_4\text{S}_{13}]$ and other minerals.

The platinum-group metals consist of ruthenium, rhodium, palladium, osmium, iridium, and platinum. Each of the metals occurs naturally in its native form, and in economically exploitable deposits, the elements occur overwhelmingly as individual platinum-group mineral (PGM) species. Mutual substitution of the various PGEs is common, but substitutions in other minerals, such as base-metal sulfides, typically occur to only a limited extent. A comprehensive review of PGM and PGE geochemistry is given in Cabri (2002).

The platinum-group metals are generally grouped with gold and silver as precious-metal commodities, but the platinum-group metals have little in common with the other precious metals in terms of their primary geological host-rock associations. The world's largest producer of platinum and rhodium is South Africa, with most of the metal obtained from mines that exploit thin (centimeters rather than meters), PGM-rich layers (averaging $<10 \text{ g t}^{-1}$ PGE) in the Bushveld Complex, a layered mafic intrusion that is also a principal source of chromium and vanadium. Platinum and palladium account for all but a very small percentage of the world's PGE production. Whereas the Bushveld Complex accounts for more than a quarter of the world's palladium production, more than double that amount is obtained as a by-product from Cu–Ni mines in layered intrusive complexes such as those at Sudbury, Canada, and Norilsk–Talnakh, Russia; the latter is the world's largest primary source of palladium. Braggite $[(\text{Pt,Pd})\text{S}]$, cooperite $[\text{PtS}]$, sperrylite $[\text{PtAs}_2]$, michenerite $[\text{PdBiTe}]$, moncheite $[\text{PdTe}_2]$, and Pt–Fe alloys are among the principal sources of PGE.

The principal consumption of PGE is as a catalyst, especially the use of platinum, or the more favored palladium because of its superior high-temperature performance, in catalytic converters in motor vehicles. Among the diverse other chief uses are electrical and electronic applications, jewelry, fabrication of laboratory equipment, and dental repairs.

11.5.2.4 Uranium Deposits

Canada, Australia, and Kazakhstan are the world's largest producers of uranium. Together, they form about 63% of the world's uranium production. All Canadian production is from rich deposits in the Athabasca Basin of Northern Saskatchewan; among those is the McArthur River mine, which

has the world's largest high-grade deposit, estimated at 152 000 t of U from ore grading, 15–18% U. These 'unconformity'-type Saskatchewan deposits, which are also the principal deposit type for Australian uranium production, contain mainly uraninite [UO₂] with associated coffinite [U(SiO₄)_{1-x}(OH)_{4x}] and brannerite [(U,Ca,Y,Ce)(Ti,Fe)₂O₆] (Plant et al., 1999). The chief uses of uranium are in nuclear power plants and weaponry.

11.5.2.5 Diamond Deposits

The world's annual production of natural diamonds, including both the gemstone and industrial types, is about 159 million carats (1 carat = 200 mg). Almost all is derived from kimberlite or its weathered remnants, but Australian production is from the Argyle mine at which the host rock is lamproite. Kimberlites are olivine- and volatile-rich potassic ultrabasic rocks of variable geological age that typically form near-vertical carrot-shaped 'pipes' intruded into Archean cratons. The volatile-rich component is predominantly CO₂ in the carbonate minerals calcite and dolomite, and the texture is characteristically inequigranular,

with large grains (macrocrysts), usually of magnesian olivine [Mg₂SiO₄], in a fine-grained, olivine-rich matrix.

Russia, Botswana, Congo, Australia, and Canada, in decreasing order, account for more than 80% of the carats (both gemstone and industrial) produced in 2008. In terms of gemstone carats, Botswana, Russia, and Canada account for over 70% of the world production, whereas South Africa production is ranked sixth and with less than 0.5% of the world production, Australia is ranked eleventh.

11.5.2.6 Other Deposits

Table 3 summarizes the data on the principal sources and uses of numerous other metals. The listing is not intended to be comprehensive.

11.5.3 Sulfide Oxidation and the Generation of Oxidation Products

A principal environmental concern associated with mining results from the oxidation of sulfide minerals within the waste

Table 3 Principal 'mineral' sources and usage of various metals

	<i>Principal 'mineral' sources</i>	<i>Principal usage</i>
Aluminum	Gibbsite [Al(OH) ₃], böhmite [AlO(OH)]	Transportation, packaging
Antimony	Stibnite [Sb ₂ S ₃], by-product from Pb sulfides	Flame-retardant chemical, hardener for Pb in batteries
Arsenic	By-product as As ₂ O ₃	Wood preservatives
Beryllium	Beryl [Be ₃ Al ₂ Si ₆ O ₁₈] from pegmatite, bertrandite [Be ₄ Si ₂ O ₇ (OH) ₂] from tuff	Be–Cu alloys (telecommunications)
Bismuth	By-product, mainly from galena	Pharmaceuticals, chemicals
Cadmium	By-product from sphalerite	Batteries
Chromium	Chromite [FeCr ₂ O ₄] in mafic–ultramafic intrusions	Stainless steel
Cobalt	Laterites, Ni–Cu sulfide ores, linnaeite [Co ²⁺ Co ³⁺ S ₄] from sedimentary-hosted Cu–Co deposits	Superalloys, alloys with steel
Gallium	By-product from sphalerite	GaAs in electronic devices
Germanium	By-product from sphalerite	Fiber-optic systems
Indium	By-product from sphalerite	Electronics, LCD screens
Iron	Hematite [Fe ₂ O ₃], goethite [FeOOH], magnetite [Fe ²⁺ Fe ₂ ³⁺ O ₄]	Iron and steel
Magnesium	Brines, seawater, magnesite [MgCO ₃]	Al alloys, die casting
Manganese	Mn oxides, hydroxides	Alloys with steel
Mercury	Cinnabar [HgS]	Electrolysis, batteries
Molybdenum	Molybdenite [MoS ₂] from porphyry Mo and Cu deposits	Alloys with iron, steel
Nickel	Laterite, pentlandite [(Fe,Ni) ₉ S ₈] in mafic–ultramafic intrusions	Steel and nonferrous alloys
Niobium	Pyrochlore [(Ca,Na) ₂ Nb ₂ O ₆ (OH,F)] from carbonatites	Alloys with steel, superalloys
Rare earths	Bastnäsite–Ce [(Ce,La)(CO ₃)F] from carbonatites	Chemical catalyst, automotive catalytic converters, glass polishing, ceramics
Rhenium	By-product from molybdenite	Pt–Rh catalysts, superalloys
Scandium	By-product	Al alloys, halide lighting additive
Selenium	By-product from Cu anode slimes	Glass, metallurgical additive, electronics
Silicon	Quartz [SiO ₂]	Additive to steel
Strontium	Celestine [SrSO ₄]	Television faceplate glass, ceramics
Tantalum	Tantalite–columbite [(Fe,Mn,Mg)Ta ₂ O ₆ –(Fe,Mn,Mg)Nb ₂ O ₆] from pegmatites	Electronic components
Tellurium	By-product from refining Cu ores	Steel and copper additive
Thallium	By-product from Cu–Zn–Pb sulfide ores	Semiconductor materials, electronics
Thorium	Monazite [(REE,Th)PO ₄] by-product from heavy-mineral sands	Refractory applications, catalyst
Tin	Cassiterite [SnO ₂] in placers	Plating on cans and containers, solder
Titanium	Ilmenite [FeTiO ₃] from heavy-mineral sands	TiO ₂ pigment
Tungsten	Scheelite [CaWO ₄] from skarns, ferberite [Fe ²⁺ WO ₄] from veins	Tungsten carbide
Vanadium	Magnetite [Fe ²⁺ (Fe ³⁺ ,V ³⁺) ₂ O ₄] in mafic–ultramafic intrusions	Steel additive
Yttrium	By-product from bastnäsite REE production	Phosphors
Zirconium	Zircon [ZrSiO ₄] from heavy-mineral sands	Refractory facings and bricks

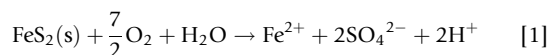
materials and mine workings and the transport and release of oxidation products. The principal sulfide minerals in mine wastes are pyrite and pyrrhotite, but others are susceptible to oxidation, releasing elements such as Al, As, Cd, Co, Cu, Hg, Ni, Pb, and Zn to the water flowing through the mine waste.

11.5.3.1 Sulfide-Mineral Oxidation

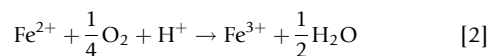
11.5.3.1.1 Pyrite oxidation

Pyrite is the most abundant sulfide mineral in Earth's crust. It is commonly associated with coal, base-metal, and gold deposits. It is also commonly found elsewhere in crustal rocks unrelated to mineral deposits. Pyrite oxidation and the factors affecting the kinetics of oxidation (O_2 , Fe^{3+} , temperature, pH, E_h , and the presence or absence of microorganisms) have been the focus of extensive study because of their importance in both environmental remediation and mineral separation by flotation (Brown and Jurinak, 1989; Buckley and Woods, 1987; Evangelou and Zhang, 1995; Luther, 1987; McKibben and Barnes, 1986; Moses et al., 1987; Nordstrom, 1982; Sasaki et al., 1995; Wiersma and Rimstidt, 1984; Williamson and Rimstidt, 1994). The reviews of pyrite oxidation and the formation of acid mine drainage (AMD) are given by Lawson (1982), Evangelou (1995), Evangelou and Zhang (1995), Nordstrom and Southam (1997), Nordstrom and Alpers (1999a), Rimstidt and Vaughan (2003), and Rosso and Vaughan (2006).

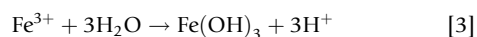
The oxidation of pyrite can occur when the mineral surface is exposed to an oxidant and water either in oxygenated or anoxic systems, depending on the oxidant. The process is complex and can involve chemical, biological, and electrochemical reactions. The chemical oxidation of pyrite can follow a variety of pathways involving surface interactions with dissolved O_2 , Fe^{3+} , and other mineral catalysts (e.g., MnO_2). The oxidation of pyrite by atmospheric oxygen produces one mole of Fe^{2+} , two moles of SO_4^{2-} , and two moles of H^+ for every mole of pyrite oxidized (Nordstrom, 1982):



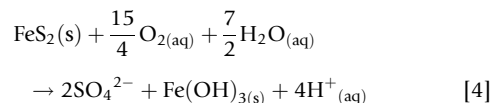
The $Fe(II)$ thus released may be oxidized to $Fe(III)$:



$Fe(III)$ oxyhydroxides such as ferrihydrite (nominally $5Fe_2O_3 \cdot 9H_2O$) may precipitate:



where $Fe(OH)_3$ is a surrogate for ferrihydrite. Adding eqns [1]–[3] yields the overall reaction:



This overall reaction results in the release of four moles of H^+ for each mole of pyrite oxidized.

11.5.3.1.1.1 Chemical oxidation by Fe^{3+} and O_2

Initially, pyrite oxidation involves the adsorption of O_2 and water to the partly protonated pyrite surface by bonding to Fe^{2+} (Fornasiero et al., 1994). Various sulfide species and Fe

oxyhydroxide intermediate products can form on the pyrite surface, depending on the pH. According to Todd et al. (2003), in oxygenated aqueous solutions under acidic conditions ($pH < 4$), an $Fe(III)$ hydroxysulfate is the main oxidation product. As the pH is increased, an $Fe(III)$ oxyhydroxide appears beside an $Fe(III)$ hydroxysulfate, and under alkaline conditions, goethite dominates the pyrite surfaces. Singer and Stumm (1970) suggested that, under acidic conditions, the major oxidant of pyrite is Fe^{3+} , whereas O_2 becomes the predominant oxidant at circumneutral pH because of the diminished solubility of Fe^{3+} . Pyrite oxidation by Fe^{3+} at circumneutral pH has also been observed (Brown and Jurinak, 1989; Evangelou and Zhang, 1995; Moses et al., 1987), but the reaction cannot be sustained without the presence of dissolved O_2 to perpetuate the oxidation to Fe^{3+} . When O_2 is the oxidant under near-neutral pH conditions, one O atom in the sulfate is derived from dissolved O_2 , with the remainder derived from H_2O . Under acidic conditions, all four O atoms in sulfate are derived from H_2O (Reedy et al., 1991). Although both Fe^{3+} and oxygen can bind chemically to the surface, the more rapid oxidation rates for Fe^{3+} compared to those for O_2 are due to a more efficient electron transfer for Fe^{3+} (Luther, 1987). A molecular orbital model proposed by Luther (1987) is consistent with pyrite oxidation data obtained by McKibben and Barnes (1986), Moses et al. (1987), and Wiersma and Rimstidt (1984).

Rate data from the literature for the reaction of pyrite with dissolved O_2 were compiled by Williamson and Rimstidt (1994) to produce a rate law that is applicable for more than four orders of magnitude in O_2 concentration and over a pH range of 2–10:

$$R = 10^{-8.19(\pm 0.04)} \frac{m_{DO}^{0.5(\pm 0.04)}}{m_{H^+}^{0.11(\pm 0.01)}} \quad [5]$$

where R is the rate of pyrite dissolution in units of $\text{mol m}^{-2} \text{s}^{-1}$.

A series of batch and mixed flow reactor experiments were performed at $pH < 3$ to determine the effect of SO_4^{2-} , Cl^- , ionic strength and dissolved O_2 on the rate of pyrite oxidation by Fe^{3+} . Of these, only dissolved O_2 had any appreciable effect on the rate of pyrite oxidation in the presence of Fe^{3+} . Williamson and Rimstidt (1994) combined their experimental results with kinetic data reported from the literature to formulate rate laws that are applicable over a range spanning six orders of magnitude in Fe^{3+} and Fe^{2+} concentrations and for a pH range of 0.5–3.0 when fixed concentrations of dissolved O_2 are present:

$$R = 10^{-6.07(\pm 0.57)} \frac{m_{Fe^{3+}}^{0.93(\pm 0.07)}}{m_{Fe^{2+}}^{0.40(\pm 0.06)}} \quad [6]$$

where R is the rate of pyrite dissolution in units of $\text{mol m}^{-2} \text{s}^{-1}$.

A wide variation in empirical rate laws has been developed to describe pyrite oxidation. The wideness of the range could be due to several factors, among which are the differences in sample preparation, different ratios of surface area to volume, types of pyrite (e.g., low T, high T, framboidal, and massive), and presence of impurities and substitutions in the pyrite or in the solution. The activation energies determined for pyrite oxidation range from 50 kJ mol^{-1} for pH 2–4 to 92 kJ mol^{-1} for pH 6–8, regardless of whether dissolved O_2 or $Fe(III)$ is used as the oxidant (Nicholson, 1994; Wiersma and Rimstidt,

Table 4 Summary of proposed rate expressions for the dissolution of pyrite in solutions containing dissolved oxygen and ferric ion

Source	Rate expression: dissolved oxygen	Rate expression: dissolved iron
Garrels and Thompson (1960)		$r_{\text{FeS}_2} = \frac{k [\text{Fe}^{3+}]}{\sum [\text{Fe}]}$
Mathews and Robins (1972, 1974)	$r_{\text{FeS}_2} = k [\text{O}_2]^{0.81}$	$r_{\text{FeS}_2} = \frac{k [\text{Fe}^{3+}]}{\sum [\text{Fe}] [\text{H}^+]^{0.44}}$
Lowson (1982)		$r_{\text{FeS}_2} = \frac{k [\text{Fe}^{3+}] [\text{Fe}^{2+}]}{\sum [\text{Fe}]}$
McKibben and Barnes (1986)	$r_{\text{FeS}_2} = k [\text{O}_2]^{0.5}$	$r_{\text{FeS}_2} = \frac{k [\text{Fe}^{3+}]^{0.58}}{[\text{H}^+]^{0.5}}$
Williamson and Rimstidt (1994)	$r_{\text{FeS}_2} = k [\text{O}_2]^{0.5} [\text{H}^+]^{-0.11}$	$r_{\text{FeS}_2} = \frac{k [\text{Fe}^{3+}]^{0.3}}{[\text{Fe}^{2+}]^{0.47} [\text{H}^+]^{0.32}}$
Holmes and Crundwell (2000)	$r_{\text{FeS}_2} = k [\text{H}^+]^{-0.18} [\text{O}_2]^{0.5}$	$r_{\text{FeS}_2} = k [\text{H}^+]^{-0.5} \left(\frac{k_{\text{Fe}^{3+}} [\text{Fe}^{3+}]}{k_{\text{Fe}^{2+}} [\text{Fe}^{2+}] k_{\text{FeS}_2} [\text{H}^+]^{-0.5}} \right)^{0.5}$

Source: Holmes PR and Crundwell FK (2000) The kinetics of the oxidation of pyrite by ferric ions and dissolved oxygen: An electrochemical study. *Geochimica et Cosmochimica Acta* 64: 263–274.

1984). **Table 4** provides a summary of the proposed rate expressions for the dissolution of pyrite in solutions containing dissolved O_2 and Fe(III) . The activation energies are observed to be higher for pH values in the range of 6–8 than in the range from 2 to 4. Casey and Sposito (1991) suggested that the proton adsorption/desorption reactions can contribute up to 50 kJ mol^{-1} to the experimental activation energy of dissolution reactions for silicate minerals. The hydrogen ion activity or pH, therefore, may play an important role in the observed activation energy for the oxidation of sulfide minerals. Regardless, the high activation energies observed indicate that the rate-limiting step in pyrite oxidation is related to electron transfer at the pyrite surface.

Holmes and Crundwell (2000) studied the kinetics of pyrite oxidation and reduction independently using electrochemical techniques. The kinetics of the half reactions are related to the overall dissolution reaction assuming no accumulation of charge on the surface. This assumption was used to derive expressions for the mixed potential and rate of dissolution, which agreed with those obtained by McKibben and Barnes (1986) and Williamson and Rimstidt (1994). The results showed that the electrochemical reaction steps occurring at the mineral–solution interface control the rate of dissolution. As summarized by Rimstidt and Vaughan (2003), the steps are (1) cathodic reaction transferring electrons from the pyrite surface to the aqueous oxidant species, (2) electron transport from the anodic to cathodic site, and (3) anodic reaction involving H_2O molecules interacting with S atoms to form a sulfoxy species. Reactivity of pyrite can be influenced by subtle changes in the composition of pyrite because of its semiconductor nature (Rimstidt and Vaughan, 2003). In other words, highly variable minor- and trace-element compositions of pyrite including Au, As, Sb, Co, and Ni (Abratis et al., 2004) and the stoichiometric variability in terms of excess S or its deficiency would influence the electrical properties of pyrite and its reactivity.

Secondary Fe(III) oxyhydroxide coatings that develop at neutral pH values can reduce the rate of pyrite oxidation by limiting the transport of reactants to pyrite surfaces. According to Huminicki and Rimstidt (2009), Fe(III) oxyhydroxide coatings grow in two stages: the formation of Fe oxyhydroxide colloids and their attachment to pyrite surfaces followed by the interstitial precipitation of Fe oxyhydroxide material between the colloidal particles. Both stages play a role in limiting the transport of oxidant to pyrite surfaces.

11.5.3.1.2 Pyrrhotite oxidation

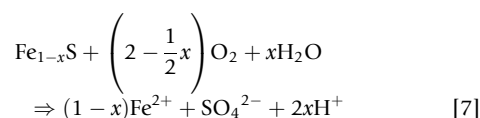
Pyrrhotite is a common Fe sulfide mineral. Although there have been numerous studies of the oxidation of pyrite, fewer studies have focused on pyrrhotite oxidation (Buckley and Woods, 1985; Janzen et al., 2000; Jones et al., 1992; Nicholson and Scharer, 1994; Pratt et al., 1994a,b; Thomas et al., 1998). The pyrrhotite structure is based on hexagonal close packing but is disordered (i.e., NiAs-type structure), giving rise to non-stoichiometric and stoichiometric compositions in which x in the formula Fe_{1-x}S can vary from 0.125 (Fe_7S_8) to 0 (FeS). The Fe vacancies within the structure may be charge-compensated by Fe^{3+} (Vaughan and Craig, 1978) or an approximation thereof. Analyses of cleaved pyrrhotite surfaces under vacuum showed Fe(III) –S interactions on the pyrrhotite surface (Pratt et al., 1994a). The deficiency in Fe within the pyrrhotite structure can result in a symmetry that varies from monoclinic (Fe_7S_8) to hexagonal ($\text{Fe}_{11}\text{S}_{12}$), with the composition progressing to stoichiometric troilite (FeS). Orlova et al. (1988) examined the reaction rates for monoclinic and hexagonal pyrrhotite and concluded that the hexagonal form was the more reactive.

The deficiency of Fe in the pyrrhotite structure may affect the oxidation behavior. Nicholson and Scharer (1994) observed a dependency of activation energy on pH; the energy ranged from 52 to 58 kJ mol^{-1} at pH 2–4 and almost doubled to 100 kJ mol^{-1} at circumneutral pH (i.e., 6). These values are similar to activation energies noted for pyrite, suggesting a

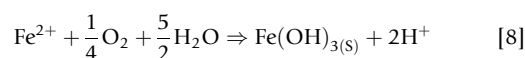
chemical-controlled reaction. Orlova et al. (1988) observed a range of activation energies for both monoclinic and hexagonal varieties ranging from 50 to 46 kJ mol⁻¹, respectively. It was argued that the lower activation energy was a function of the hexagonal crystal structure. Janzen et al. (2000), however, did not observe consistent trends between activation energy and crystal structure.

11.5.3.1.2.1 Chemical oxidation by O₂ and Fe³⁺

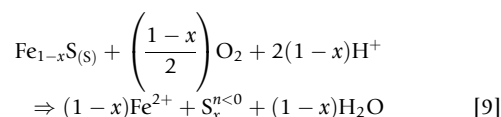
Pyrrhotite dissolution can proceed through oxidative or non-oxidative reactions. Oxidative dissolution can be at least 10³ times slower than nonoxidative reactions (Thomas et al., 1998). Dissolved O₂ and Fe³⁺ can be important oxidants of pyrrhotite. When oxygen is the primary oxidant, the overall reaction may be written as



The production of protons is linked to the mineral stoichiometry. Up to one-quarter mole of the protons produced are derived from the oxidation of one mole of the Fe-deficient form ($x=0.125$), whereas no protons are produced from the stoichiometric form, which is troilite ($x=0$). The release of protons can also result from the oxidation of the dissolved Fe, resulting from the precipitation of ferric hydroxide:



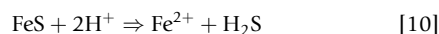
In other circumstances, the oxidation reactions may not proceed to completion. Partial oxidation may result in only a small proportion of the S being transformed to sulfate, with the remainder accumulating as reduced sulfur species (polysulfides and elemental sulfur) at the mineral surface (Janzen et al., 2000):



The rates of oxidation of both pyrite and pyrrhotite at 25 °C and the standard atmospheric oxygen indicate that pyrrhotite can react 20–100 times faster than pyrite. During oxidation of a particle of pyrrhotite, Fe diffuses to the exposed surface, thereby creating a S-enriched inner zone that contains disulfide- and polysulfide-like species (Mycroft et al., 1995).

11.5.3.1.2.2 Nonoxidative mechanism

Nonoxidative dissolution of pyrrhotite occurs in acidic solutions when predominant S²⁻ surface species are exposed. The reaction occurs as



Jones et al. (1992) observed restructuring of sulfur-rich pyrrhotite surfaces in deoxygenated acid, resulting in the development of a surface dominated by a discontinuous layer of a tetragonal intermediate Fe₂S₃ structure. Janzen et al. (2000) showed a significant release of Fe²⁺ from pyrrhotite in acidic solutions in which oxygen was not present. Although Fe²⁺ concentrations increased linearly with time, sulfate values remained unchanged, with sulfur from pyrrhotite dissolution

remaining in a reduced state (S²⁻). Thomas et al. (1998, 2001) proposed a dissolution mechanism that allows two distinct pathways: (1) Fe leaves the surface, with no additional electrons released from the structure, and (2) after a critical accumulation of charge, the reduction of polysulfide to sulfide occurs, resulting in the release of negative charge from the surface in the form of HS⁻. The significant feature of this process is the delay between the release of Fe²⁺ and HS⁻.

11.5.3.1.3 Oxidation of other metal sulfides

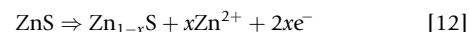
11.5.3.1.3.1 Sphalerite

The oxidation of sphalerite is dependent on a number of factors, among which are the concentration of oxidants, such as dissolved O₂ or Fe(III) in the solution, the temperature, and the pH (Bobeck and Su, 1985; Crundwell, 1988; Olanipekun, 1999; Perez and Dutrizac, 1991; Rimstidt et al., 1994). For sphalerite, Vaughan and Craig (1978) reported a solubility product of $K_{\text{sp}} = 1 \times 10^{-20.6}$ at 25 °C in water. Other researchers have reported similar values (Daskalakis and Helz, 1993).

For sphalerite in dilute Fe(III) solutions, Rimstidt et al. (1994) obtained a dissolution rate of 7.0×10^{-8} mol m⁻² s⁻¹ with a corresponding activation energy of 27 kJ mol⁻¹ over a range of 25–60 °C. The concentration of Fe(III) used was 10⁻³ M, which is similar to dissolved Fe concentrations ($2\text{--}9 \times 10^{-3}$ M) typically observed in acidic mine waters (Lin, 1997). The overall oxidation reaction for pure sphalerite, assuming that all sulfur is oxidized to sulfate, is



The x-ray photoelectron spectroscopy (XPS) examination of oxidized sphalerite showed the development of a surface layer of metal-deficient sulfide (Buckley et al., 1989) whose formation in acid solution is described by

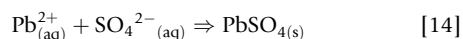
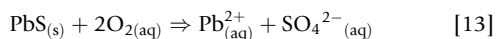


Weisener et al. (2003, 2004) observed increased rates of oxidation and increased acid consumption as a function of the amount of solid-solution Fe in sphalerite [(Zn,Fe)S]. Apparent activation energies of 21–28 kJ mol⁻¹ obtained at 25–85 °C are similar to the values reported by Rimstidt et al. (1994). Weisener et al. (2003, 2004) suggested that the production of polysulfide species results in a lower diffusion gradient at the mineral surface, thus leading to lower reactivity with potential oxidants and to diffusion-limited release of Zn and Fe from the bulk mineral. Elemental sulfur was not observed to limit the reactivity of the mineral surface. The accumulation of polysulfides and S⁰ on the sphalerite surface under oxygenated conditions can affect the acid-neutralization capacity because the polysulfides and S⁰ consume acid when pH is <3. The resulting formation of a S-enriched surface slows the subsequent rate of dissolution of sphalerite in the absence of bacteria. Under these conditions, S⁰ does not passivate the surface (Weisener, 2002). Moncur et al. (2009) showed that the stability of sphalerite in tailings impoundments is generally greater than that of pyrrhotite but less than that of pyrite.

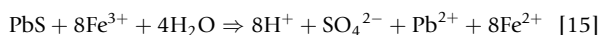
11.5.3.1.3.2 Galena and chalcopyrite

Galena and chalcopyrite are commonly associated with acid-generating minerals, such as pyrite and pyrrhotite. Acid ferric sulfate solutions, generated through the oxidation of Fe sulfides, can enhance the oxidation of Pb- and Cu-bearing

sulfide minerals. The oxidation of galena has been studied by Buckley and Woods (1984a), Tossell and Vaughan (1987), Fornasiero et al. (1994), Kim et al. (1995), Prestidge et al. (1995), Basilio et al. (1996), Kartio et al. (1996, 1998), Chernyshova and Andreev (1997), Jennings et al. (2000), Nowak and Laajalehto (2000), Shapter et al. (2000), and others. The XPS studies showed that S^0 formed when galena was oxidized in a hydrogen peroxide solution and that metal-deficient surfaces resulted from oxidation by dilute acetic acid solutions (Buckley and Woods, 1984a). In natural oxygenated environments, galena will weather to anglesite, which is weakly soluble below pH 6 (Lin, 1997; Shapter et al., 2000):

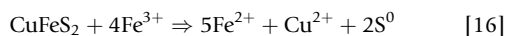


Galena may also be oxidized by $Fe(III)$ under acidic conditions (Rimstidt et al., 1994):

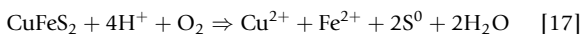


The oxidation of galena in air may result in the formation of lead hydroxide and lead oxide (Buckley and Woods, 1984a; Evans and Raftery, 1982; Laajalehto et al., 1993). The oxidation in aqueous solutions may lead to the formation of lead oxides and lead sulfate surface products (Fornasiero et al., 1994; Kartio et al., 1996; Kim et al., 1995; Nowak and Laajalehto, 2000). In the absence of oxygen, both Pb and sulfide ions are released to the solution in the form of free Pb ions and hydrogen sulfide (Fornasiero et al., 1994). Jennings et al. (2000) showed that galena was not acid-generating when exposed to accelerated oxidation using hydrogen peroxide. This reaction resulted in the accumulation of anglesite on the mineral surface.

An XPS study by Buckley and Woods (1984b) showed that freshly fractured chalcopyrite surfaces exposed to air formed a ferric oxyhydroxide overlayer with a Fe-deficient region composed of CuS_2 . Acid-treated surfaces of fractured chalcopyrite showed an increase in the thickness of the CuS_2 layer and the presence of elemental sulfur. Hackl et al. (1995) suggested that dissolution of chalcopyrite is passivated by a thin ($<1 \mu m$) Cu-rich surface layer, which forms as a result of solid-state changes. The passivating surface layer consists of Cu polysulfide, CuS_n , where $n \geq 2$. Hackl et al. (1995) described the dissolution kinetics as a mixed diffusion and chemical reaction whose rate is controlled by the rate at which the Cu polysulfide is leached. The oxidation of chalcopyrite in the presence of ferric ions under acidic conditions can be expressed as



Hiroyoshi et al. (1997) monitored the oxygen consumption, sulfur formation, total Fe, and $Fe(II)$ concentrations at different pH levels during the oxidation of chalcopyrite. On the basis of the reaction products formed, it was concluded that ferrous ions catalyzed the oxidation by dissolved oxygen in acidic media:



This conclusion agrees with the observation and interpretations made from testing monomineralic chalcopyrite with peroxide solution; acid production was not observed. Chalcopyrite can be acid-consuming via the production of S^0 (Smart et al., 2000).

The dissolution of chalcopyrite also can be influenced strongly by galvanic effects. The presence of pyrite or molybdenite

in association with chalcopyrite can cause accelerated rates of chalcopyrite dissolution (Dutrizac and MacDonald, 1973), whereas the presence of Fe-rich sphalerite and galena can slow the dissolution.

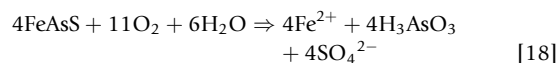
11.5.3.1.3.3 Mercury sulfides

Cinnabar [HgS], the principal ore of mercury, is the most thermodynamically stable form at low temperature (Barnett et al., 2001; Benoit et al., 1999). The presence of trace impurities, such as Zn, Se, and Fe, can impede the conversion of metacinnabar, which is the high-temperature form, to cinnabar. The trace impurities decrease the inversion temperature and thus retard the conversion (Barnett et al., 1997). Therefore, in some environments, the availability of impurities will favor the in situ formation of metacinnabar over cinnabar. The formation of HgS is favored under reducing conditions, in part, because of the high affinity of Hg compounds for S. Cinnabar is kinetically resistant to oxidation and can remain in the soil and tailings impoundments even under oxidizing conditions (Barnett et al., 1997). Although little has been published on its oxidation rates in soils, the observed persistence of HgS in soils at mine sites suggests that its weathering is slow under typical oxidative environments (Barnett et al., 1997; Gray et al., 2000).

11.5.3.1.4 Oxidation of arsenic sulfides

The oxidation of arsenopyrite [$FeAsS$] releases both S and As. Buckley and Walker (1988) studied the oxidation of arsenopyrite in alkaline and in acidic aqueous solutions. In air, the mineral reacted rapidly, and the oxidation of As to $As(III)$ was more rapid than the oxidation of Fe on the same surface. Only a small amount of sulfur oxidation occurred. Under acidic conditions, the mineral formed S-rich surfaces.

Nesbitt et al. (1995) conducted a detailed study of the oxidation of arsenopyrite in oxygenated solutions. Arsenic and S were observed to exist in multiple oxidation states near the pristine surface. After reaction with air-saturated distilled water, $Fe(III)$ oxyhydroxides formed the dominant Fe surface species, and $As(V)$, $As(III)$, and $As(I)$ were as abundant as $As(-I)$ surface species. An appreciable amount of sulfate was observed on the mineral surface. Arsenic was more readily oxidized than S, and similar rates of the oxidation of $As(-I)$ and $Fe(II)^+$ surface species were observed. Nesbitt et al. (1995) concluded that continued diffusion of As to the surface under these conditions can produce large amounts of As^{3+} and As^{5+} , promoting rapid selective leaching of arsenites and arsenates. According to Walker et al. (2006), arsenite forms 60% of the total arsenic releases, while the remainder is made of arsenate and that the molar proportion of arsenic to sulfur is approximately 1:1, suggesting that the dissolution is congruent. Arsenopyrite oxidation involves the following reactions yielding $H_2AsO_4^{2-}$ and $HAsO_4^-$ ions in equal proportions (Walker et al., 2006):



Further acidity is produced with the oxidation of dissolved Fe and its precipitation as a ferric hydroxide as described earlier

for pyrite. Overall, oxidation of one mole of arsenopyrite would result in the formation of 3.5 protons.

Rimstidt et al. (1994) measured the rate of reaction of arsenopyrite under conditions typical of AMD environments. Arsenopyrite was observed to be more reactive than pyrite, chalcopyrite, galena, and sphalerite. Oxidation of arsenopyrite led to the formation of scorodite on the surface. The activation energies for arsenopyrite oxidation varied from 18 kJ mol^{-1} at $0\text{--}25^\circ\text{C}$ to a slightly negative E_a of 6 kJ mol^{-1} at $25\text{--}60^\circ\text{C}$. Rimstidt et al. (1994) attributed the negative activation energies to competition between the dominant reactions that occur at all temperatures and the less vigorous side reactions that contribute to rate-limiting behavior at higher temperatures. Oxidation of arsenopyrite in acidic media is faster than in air, in water, or in alkaline solutions (Corkhill and Vaughan, 2009).

Oxidation of arsenopyrite by dissolved oxygen at 25°C and circumneutral pH (6–7) indicates that the rate of arsenopyrite oxidation is essentially independent of dissolved oxygen and that oxidative dissolution is congruent with respect to As and S (Walker et al., 2006). McKibben et al. (2008) determined the rate of arsenopyrite oxidation to be four orders of magnitude faster than that of Walker et al. (2006). The reason for this discrepancy is due to the use of As release rate by Walker et al. (2006) as opposed to the use of Fe release rate by McKibben et al. (2008) as representing the oxidative dissolution of arsenopyrite. McKibben et al. (2008) indicated the incongruity of arsenopyrite dissolution in acidic solutions. Incongruent dissolution also was evident from the surface characterization studies of Nesbitt et al. (1995) and Nesbitt and Muir (1998), indicating preferential enrichment of As in the oxidized surfaces of arsenopyrite. In other words, As and S releases from the surfaces of arsenopyrite were much slower than the release of Fe; therefore, the rates of As release from arsenopyrite would not represent the oxidative dissolution of arsenopyrite. Similar to pyrite, arsenopyrite oxidation is a three-step electrochemical process involving cathodic reaction, electron transport, and anodic reaction (Corkhill and Vaughan, 2009).

An investigation of the surface composition and the chemical state of three naturally weathered arsenopyrite samples exposed for periods ranging from 14 days to 25 years showed that the arsenopyrite surface has an effective passivating layer that protects the mineral from further oxidation (Nesbitt and Muir, 1998). The same samples were then reacted with mine-waste waters, which caused extensive leaching of the arsenopyrite surface below the oxidized overlayers. The acidic nature of the solution caused dissolution of the previously accumulated ferric arsenite and arsenate salts. In mine tailings, Paktunc et al. (2004) reported that arsenopyrite grains are often surrounded by secondary minerals, such as scorodite, arseniosiderite, and goethite and suggested that the relict arsenopyrite would be protected from further oxidative dissolution by the rims limiting the reaction to diffusion of reactants and products. Furthermore, As releases would be controlled by the solubility and stability of the secondary products. Similar occurrences of relict arsenopyrite enveloped by secondary products were reported by Salzsauler et al. (2005) and Moncur et al. (2009).

Pyrite can contain significant amounts of arsenic. Concentrations of up to 10 wt% (Fleet et al., 1993), 19 wt% (Reich et al., 2005), and 12 wt% (Paktunc et al., 2006) have been reported to occur in association with gold deposits. Micro x-ray

absorption fine structure (XAFS) spectra collected from a range of arsenical pyrite grains having different morphologies and arsenic concentrations from 0.03 to 4.4 wt% are characterized by the dominant edge aligned with As^{1-} valence state similar to that of arsenopyrite and that of As substitutes for S in the structure (Paktunc, 2008). Based on XAFS results, it appears that As occurs in clusters within the pyrite structure (Paktunc, 2008; Savage et al., 2000), and such clustering of As atoms in the structure causes about 2.6% expansion of the unit cell, making pyrite more susceptible to oxidative dissolution (Savage et al., 2000). Arsenic substitutes for S in the pyrite structure forming As–S dianion groups and the presence of arsenic in pyrite would make it more reactive and accelerate its dissolution (Abratis et al., 2004; Blanchard et al., 2007).

Oxidation of arsenic sulfides including orpiment and realgar produces arsenite as the dominant arsenic species and several intermediate sulfur species (Lengke and Tempel, 2003). Oxidation rates increase with pH, dissolved oxygen concentrations, and temperature. In comparison with pyrite, oxidation rates of orpiment and realgar are lower at pH 2.5–9, whereas those of amorphous arsenic sulfides are higher at neutral to alkaline pH (Lengke and Tempel, 2003). Under acidic conditions, arsenopyrite oxidizes four to five orders of magnitude faster than orpiment and realgar and three to four orders of magnitude faster than arsenical pyrite (McKibben et al., 2008).

11.5.3.2 Bacteria and Sulfide-Mineral Oxidation

11.5.3.2.1 Mechanisms of sulfide-mineral dissolution and the role of microorganisms

The mechanism(s) by which acidophilic microorganisms catalyze the dissolution of sulfide minerals has/have long been the subject of conjecture and debate. Much of the current consensus on this issue, described in the succeeding texts, comes from the insights provided by research groups headed by Wolfgang Sand and Helmut Tributsch over the past 10 years or so.

Sulfide minerals can be divided into those, such as sphalerite (ZnS) and chalcocite (Cu_2S), that are acid-soluble and those, such as pyrite (FeS_2) and tungstenite (WS_2), that are acid-insoluble. As their name suggests, the first of these are susceptible to acid dissolution by bacteria such as *Acidithiobacillus thiooxidans* that oxidize elemental sulfur and reduced inorganic sulfur compounds (RISCs) to sulfuric acid. In contrast, acid-insoluble sulfide minerals are stable in dilute solutions of sulfuric acid but are subject to oxidative dissolution. In acidic liquors, ferric iron is a more important oxidant of sulfide minerals than molecular oxygen, and therefore, bacteria, such as *Leptospirillum* spp., that oxidize ferrous iron to ferric assume the primary role in sulfide-mineral destruction. Even with acid-soluble sulfides, rates of mineral dissolution in acidic media are greatly accelerated when ferric iron is available (Table 6).

Two different mechanisms (the ‘thiosulfate’ and ‘polysulfide’ mechanisms) have been proposed for the biologically accelerated dissolution of sulfide minerals in acidic environments (Schippers and Sand, 1999; Figure 1). Acid dissolution of susceptible sulfide minerals is initiated by the mineral being attacked by protons (hydronium ions), releasing the metal component and generating hydrogen sulfide (eqn [21]):



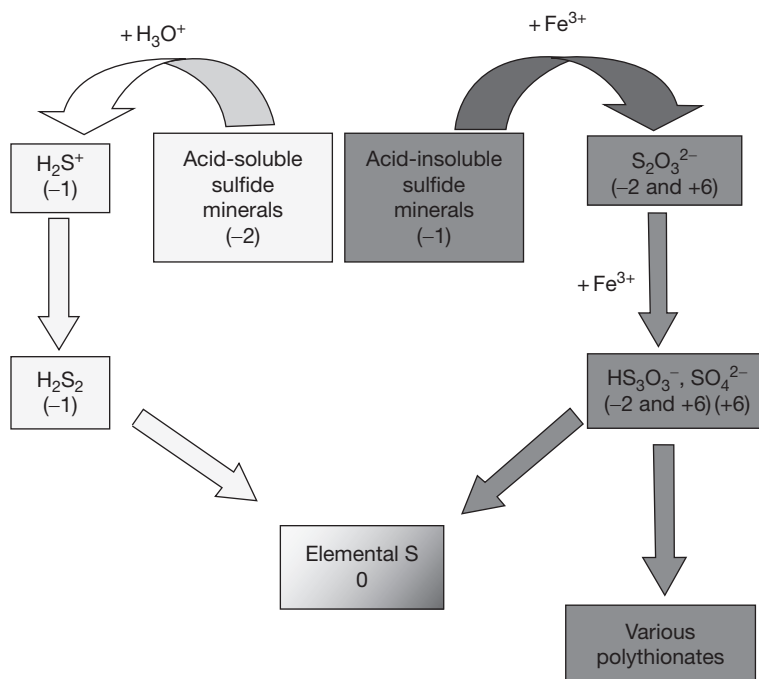
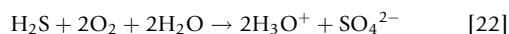
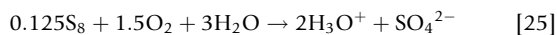
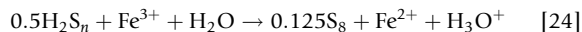
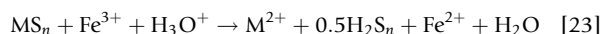


Figure 1 Oxidation pathways for acid-soluble and acid-insoluble sulfide minerals.

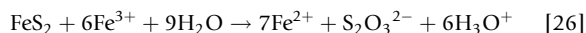
The hydrogen sulfide formed is oxidized, by acidophilic microorganisms, to sulfuric acid, allowing the process to continue (eqn [22]):



Acid-soluble sulfides may also be attacked by ferric iron (eqn [23]), in which case, the first free sulfur compound is thought to be an unstable sulfide cation (H_2S^+) that first dimerizes (to H_2S_2) and then oxidizes via various polysulfides (from which the mechanism derives its name) to elemental sulfur (eqn [24]), which is also oxidized by some acidophilic bacteria and archaea to sulfuric acid (eqn [25]), allowing the cycle of acid dissolution of the sulfide minerals to continue (eqn [21]):

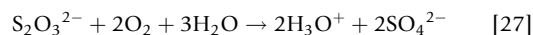


In acidic solutions, acid-insoluble minerals are oxidized by soluble ferric iron (eqn [26]), with a total of six successive one-electron transfer steps required to break the sulfur-metal bonds in minerals such as pyrite:



Since the initial sulfur product released from the degrading mineral is thiosulfate, this form of oxidative sulfide-mineral dissolution has been described as the 'thiosulfate mechanism' (Rohwerder et al., 2003). Thiosulfate is unstable in acidic

solutions (more so when ferric iron is present) and oxidizes via tetrathionate and other sulfur oxyanions, ultimately to sulfate (eqn [27]):



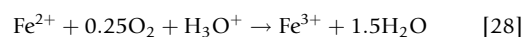
Many acidophilic bacteria have been shown to attach to, and to form biofilms on, sulfide minerals (Rohwerder et al., 2003). The attachment may be (semi-)permanent and specific to certain sulfide minerals, though different species of acidophiles, and even strains within a species, have been shown to display significant differences in how rapidly and permanently they attach to mineral (pyrite, schwertmannite, and glass) surfaces (Ghauri et al., 2007). How acidophilic bacteria attach to sulfide-mineral surfaces is now quite well understood. In the case of *Acidithiobacillus ferrooxidans*, an acidophile that can oxidize both ferrous iron and reduced sulfur, ferric iron is complexed by two uronic acid residues in the bacterial glycocalyx, and the resulting net positive charge facilitates attachment to the negatively charged pyrite (Rohwerder et al., 2003). Once attached, further production of exopolymeric substances (EPS) by the bacteria enhances the development of biofilms on the mineral surfaces, within which conditions (pH, dissolved oxygen concentrations) may differ from the bulk solution phase and be more conducive for mineral dissolution. The biofilm formation and mineral attack by *Leptospirillum ferrooxidans*, an iron-oxidizing acidophile that is unable to oxidize reduced sulfur, appear to be different to that of *At. ferrooxidans*. When it is grown on pyrite, the EPS produced by *L. ferrooxidans* is dotted with small particles, which Tributsch and Rojas-Chapana (2007) identified as colloidal, microcrystalline pyrite. The oxidation of these microscopic grains appears to support the growth of biofilm communities of *L. ferrooxidans*. The pitting

caused by this process can develop into deep holes within sulfide minerals, a process described as 'electrochemical machining.' Colloidal sulfur was detected in the EPS of *At. ferrooxidans* attached to pyrite, and Tributsch and Rojas-Chanapa (2007) provided evidence for the role of thiol groups (e.g., in cysteine) being involved in the breakup of interfacial chemical complexes by *At. ferrooxidans*.

However, it has long been recognized that physical contact between bacteria and sulfide minerals is not a prerequisite for accelerated dissolution of the latter to occur. Historically, two scenarios were described to account for mineral dissolution catalyzed by either bacteria that were attached to sulfide minerals (the 'direct' mechanism) or planktonic-phase cells that generate the oxidant (ferric iron) in the solution, which subsequently diffuses and oxidizes the mineral (the 'indirect' mechanism). It is now recognized that ferric iron mediates oxidative dissolution of acid-insoluble sulfide minerals in both scenarios. Ferric iron is tightly associated with the glycocalyx of attached cells and is recycled with biofilms of iron-oxidizing acidophiles. The terms 'contact' and 'noncontact' leaching, proposed by Hallberg and Johnson (2001), are now used widely to describe that which is mediated by attached and free-swimming bacteria and archaea, respectively.

One consequence of mineral dissolution being possible by noncontact leaching is that, while the hydronium ions and ferric iron necessary for leaching may be generated in situ, they can also originate from an external location, which may be distant from the minerals themselves. For example, acidic ferric iron-rich solutions formed in the aerated surface zones of tailings lagoons can percolate to lower anoxic zones and cause oxidative dissolution of pyrite and other potentially reactive minerals, which are often considered to be secure when stored in oxygen-free environments. However, although oxidation of pyrite by ferric iron can occur in anoxic environments (eqn [28], which is valid for both aerobic and anoxic environments,

as oxygen is not involved), regeneration of ferric iron is an oxygen-requiring reaction (eqn [28]):



Given the different roles of acidophilic bacteria and archaea in the dissolution of sulfide minerals, a division of different physiological groups based on their roles has been proposed (Johnson, 2010). The primary mineral degraders are those autotrophic, mixotrophic, and heterotrophic species that catalyze the oxidation of ferrous iron to ferric iron (Section 11.5.3.2.2), thereby generating the oxidant that initiates the breakdown of pyrite and most other metal sulfides. Secondary acidophiles are sulfur-oxidizing bacteria and archaea. The sulfuric acid that they generate maintains conditions that favor the activities of other acidophiles, and extreme acidity (pH <2.5) causes ferric iron to remain in the solution. Lastly, tertiary acidophiles are those obligate and facultative heterotrophic bacteria and archaea that degrade organic carbon, some of which inhibit the growth of some iron and sulfur oxidizers. Most acidic mineral-oxidizing environments, including biomining sites and those involved in the genesis of mine-impacted waters, are colonized by microbial consortia comprising different species of all three of these groups (Rawlings and Johnson, 2007). A schematic representation of the roles of primary, secondary, and tertiary acidophiles in the dissolution of pyrite is shown in Figure 2.

11.5.3.2.2 Biodiversity of iron- and sulfur-oxidizing acidophilic microorganisms

Mine-impacted waters can be colonized by a wide diversity of microorganisms. Eukaryotic life-forms observed in, and sometimes isolated from, mine-impacted waters include species of acidophilic and acid-tolerant microalgae, fungi and yeasts, protozoa, and some multicellular animals such as rotifers

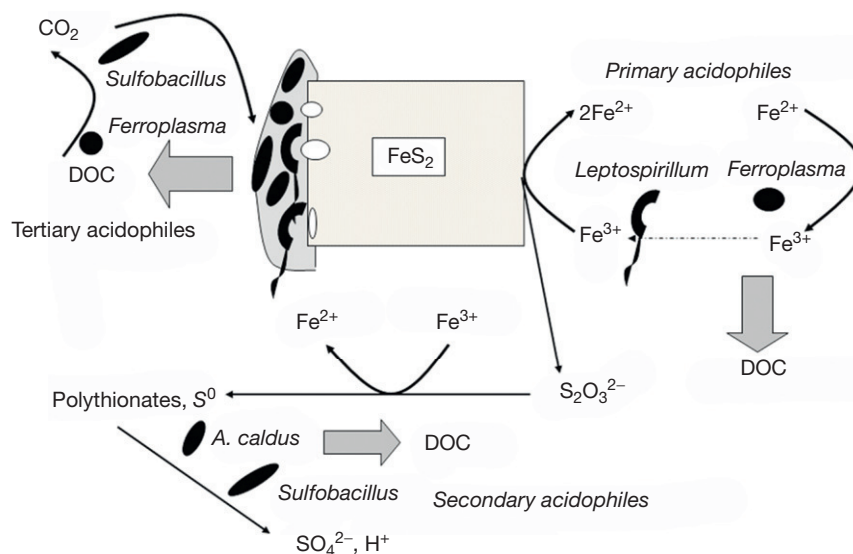


Figure 2 Microbial consortium involved in the oxidative dissolution of pyrite. Iron-oxidizing autotrophs constitute the primary mineral degraders, the secondary sulfuric acid-generating bacteria, the secondary-level subgroup, and the dissolved organic carbon (DOC)-degrading acidophiles, the tertiary-level subgroup. Reproduced with permission from Johnson DB (2010) The biogeochemistry of biomining. In: Barton L and Mandl M (eds.) *Geomicrobiology: Molecular and Environmental Perspective*, pp. 401–426. Dordrecht: Springer.

(Johnson, 2007). However, prokaryotic microorganisms (archaea and particularly bacteria) are found in greater numbers of individuals and of species than eukaryotes, and these have been the focus of a large body of applied and fundamental research.

Bacteria with contrasting physiological characteristics can thrive in mine-impacted waters. These species differ in their requirements for oxygen (obligate aerobes, and facultative and obligate anaerobes), methods of carbon assimilation (obligate and facultative autotrophs, and heterotrophs), and the nature of their electron donors (organic or inorganic). The common characteristic of all prokaryotes that are active in acidic mine waters is that they are metabolically active in low-pH environments, which are highly toxic to most other life-forms. Acidophiles can be divided into three groups, depending on their pH for optima for growth. Extreme acidophiles grow best at pH < 3, moderate acidophiles at pH 3–5, while acid-tolerant species have pH optima at above pH 5 but are able to grow at pH 3. Additional stress factors, such as elevated osmotic potentials and metal concentrations in mine-impacted waters, further restrict their indigenous biodiversities. Acidophiles also vary in their response to temperature. Thermophilic species include those that grow optimally at 40–60 °C (moderate thermophiles) or at 60–80 °C (extreme thermophiles), while thermotolerant acidophiles grow optimally at <40 °C but can also grow at >45 °C. Most acidophiles are mesophilic (growing between about 15 and 40 °C), while a subgroup of these that have been less studied (psychrotolerant species) can grow at <5 °C. Given the temperatures of the majority of mine-impacted waters, mesophilic and psychrotolerant species might be anticipated to be abundant in many of these environments, though there are notable exceptions, such as within the former Richmond Mine at Iron Mountain, California (Nordstrom et al., 2000).

The main protagonists involved in the formation of acidic mine waters, for reasons described in Section 11.5.3.2.1, are

acidophilic bacteria and archaea that can catalyze the dissimilatory oxidation of ferrous iron or reduced forms of sulfur (Table 5). Other prokaryotes (and sometimes even the same species, under different environmental conditions) can, however, catalyze the dissimilatory reduction of ferric iron or elemental sulfur (or sulfur oxyanions) and thereby allow the cycling of these elements, which are among the most abundant of all in mine-impacted waters, at mine water/sediment interfaces (Johnson and Hallberg, 2009a,b; Johnson et al., 2012).

Acidophilic bacteria and archaea that oxidize iron. Some bacteria and archaea can catalyze the dissimilatory oxidation of ferrous iron but appear not to also oxidize reduced sulfur. The most widely studied of these are *Leptospirillum* spp., a species of the deep-branching *Nitrospira* phylum. These are obligately acidophilic and autotrophic bacteria that are well known for their key roles in biomining operations and in the formation of acid rock drainage. Three species of *Leptospirillum* have been validated, but the sole strain of one of these (*Leptospirillum thermoferrooxidans*) has been lost. *L. ferrooxidans* and *Leptospirillum ferriphilum* can coexist in mine-impacted waters, though the higher temperature for growth of most strains of the latter means that they are less commonly encountered in most mine waters. A third species, *Leptospirillum ferrodiazotrophum* awaits validation, though it appears to have many similar traits to *L. ferrooxidans*. *Leptospirillum* spp. generally grow more slowly than iron-oxidizing *Acidithiobacillus* spp., though their tolerance to ferric iron and affinity for ferrous iron tend to be significantly greater. In terms of mine water microbiology, these differences often result in *Acidithiobacillus* spp. being the numerically dominant iron oxidizers in freshly discharged, ferrous iron-rich waters, and *Leptospirillum* becoming increasingly significant downstream as ferrous iron concentrations fall and those of soluble ferric iron increase.

Two other iron-oxidizing species of bacteria that do not oxidize sulfur are known to be important in mine waters. One of these has been described recently (and also awaits

Table 5 Characteristics of iron- and sulfur-oxidizing prokaryotes that have been found in mine-impacted waters

	Oxidation of Fe ²⁺	S	Carbon assimilation ^a	pH range (approx.)	T-response	Other traits
<i>Acidiferrobacter thiooxydans</i>	+	+	A	1.2 to >3	T	Reduction of Fe ³⁺
<i>Acidimicrobium ferrooxidans</i>	+	–	A/H	1.3 to >3	T	Reduction of Fe ³⁺
<i>Acidiphilium acidophilum</i>	–	+	A/H	1.5 to 6.5	M	Reduction of Fe ³⁺
<i>Acidithiobacillus caldus</i>	–	+	A	1 to 4	T	
<i>Acidithiobacillus ferrivorans</i>	+	+	A	1.9 to >3	P	Reduction of Fe ³⁺
<i>Acidithiobacillus ferrooxidans</i>	+	+	A	1.3 to >3	M	Reduction of Fe ³⁺
<i>Acidithiobacillus thiooxydans</i>	–	+	A	0.5 to 5.5	M	Some strains are halotolerant
<i>Ferrimicrobium acidiphilum</i>	+	–	H	1.4 to >3	M	Reduction of Fe ³⁺
<i>Ferroplasma</i> spp.	+	–	H	0.8 to >2	T	Reduction of Fe ³⁺
<i>Ferrovum myxofaciens</i>	+	–	A	2 to >4	P	Copious production of exopolymeric substances
<i>L. ferrooxidans</i>	+	–	A	1.3 to >2	M	
<i>L. ferriphilum</i>	+	–	A	0.8 to >2	T	
<i>Sulfobacillus</i> spp.	+	+	A/H	1 to >3	T	Reduction of Fe ³⁺
<i>Thiobacillus prosperus</i>	+	+	A	1 to 4.5	M	Halophilic
<i>Thiomonas</i> spp.	–	+	A/H	3 to 8	M	Some species oxidize As(III)
					(T, one species)	

^aCarbon assimilation: A = autotroph, H = heterotroph; T-response: P = psychrotolerant, M = mesophilic, and T = thermotolerant or moderately thermophilic.

validation), though, paradoxically, it is the most readily observed in mine-impacted waters, due to its tendency to form macroscopic growths. *Ferrovum myxofaciens* is a *Betaproteobacterium* that shares several key characteristics with *Leptospirillum* spp. (Table 5), though it is far less acid-tolerant (the lower pH for its growth being ~ 2.0). Its most notable trait, however, is its propensity for generating copious amounts of EPS that enmesh the cells and cause them to grow in gelatinous biofilms or, more characteristically, in flowing mine-impacted waters, as 'acid streamers' (Kimura et al., 2011). Although streamer growths can be formed by other acidophiles (Johnson, 2009), those in lower-temperature ($<30^\circ\text{C}$) mine-impacted waters of pH 2–4 are often dominated by *Fv. myxofaciens*.

In contrast with both *Leptospirillum* spp. and *Fv. myxofaciens*, *Ferrimicrobium* (*Fm.*) *acidiphilum* (a Gram-positive *Actinobacterium*) is an obligate heterotroph, obtaining energy from the oxidation of ferrous iron, and carbon from an organic source (such as yeast extract) rather than carbon dioxide. However, a coculture of *Fm. acidiphilum* and the sulfur-oxidizing autotroph *At. thiooxidans* can degrade pyrite in the absence of an added organic carbon source by establishing a syntrophic relationship involving carbon transfer and access to reduced sulfur (polythionates, etc.). *Acidimicrobium ferrooxidans* is also a Gram-positive *Actinobacterium*, but in contrast to mesophilic *Ferrimicrobium*, this iron oxidizer is a moderate thermophile and therefore of more limited distribution in most mine-impacted waters. The thermotolerant iron-oxidizing archaeon, *Ferroplasma*, is also considered by most researchers to be obligately heterotrophic. Its occurrence in acidic mine waters has not

yet been widely documented (except at Iron Mountain), though related euryarchaea have been detected in low-temperature mine waters (e.g., Kimura et al., 2011).

11.5.3.2.2.1 Acidophilic bacteria and archaea that oxidize sulfur

Some acidophilic bacteria and archaea can utilize the energy available from oxidizing various forms of reduced sulfur (oxidation states from -2 to $+4$) ultimately to sulfate (oxidation state $+6$) but do not oxidize iron (Dopson and Johnson, 2012). Elemental sulfur and polythionate RISCs are typical substrates for these prokaryotes. Some (mostly archaea) are thermophiles and are generally not found in mine waters. Prokaryotes that oxidize sulfur at lower temperatures ($5\text{--}30^\circ\text{C}$) are exclusively bacteria and include *At. thiooxidans*, which was the first acidophile to be described in 1919. *Acidithiobacillus caldus* shares many physiological traits with *At. thiooxidans*, with the notable exception that it is a moderate thermophile rather than a mesophile (Table 5).

Several species of *Thiomonas* have been described (e.g., Slyemi et al., 2011), though some of these have been combined into a single species on the basis of comparative 16S rRNA gene analysis (Battaglia-Brunet et al., 2011). *Thiomonas* spp. are moderate, rather than extreme, acidophiles and are generally not found in extremely acidic and metal-rich mine waters. They can grow autotrophically or as heterotrophs using organic carbon. All *Thiomonas* spp. can use the energy from oxidizing reduced sulfur to support their growth, though in closed environments (e.g., in laboratory shake-flask cultures), this oxidation reaction can cause problems as the consequent

Table 6 Comparison of sulfide-mineral oxidation rates from abiotic, microbial, and field measurements

Process	Rates of oxidation pH < 2, 25 °C			Source
	Abiotic	Microbial	Field	
Pyrite oxidation by $[\text{Fe}^{2+}]$	$3 \times 10^{-12} \text{ mol l}^{-1} \text{ s}^{-1}$	$3 \times 10^{-7} \text{ mol l}^{-1} \text{ s}^{-1}$	$5 \times 10^{-7} \text{ mol l}^{-1} \text{ s}^{-1}$	Singer and Stumm (1968, 1970) and Nordstrom (1985) McKibben and Barnes (1986) and Rimstidt et al. (1994)
Pyrite oxidation by $[\text{Fe}^{3+}]$	1×10^{-3} to $2 \times 10^{-8} \text{ mol m}^{-2} \text{ s}^{-1}$ $2.7 \times 10^{-7} \text{ mol m}^{-2} \text{ s}^{-1}$			
Pyrite oxidation by [DO]	0.3×10^{-9} to $3 \times 10^{-9} \text{ mol m}^{-2} \text{ s}^{-1}$ $1.2 \times 10^{-9} \text{ mol m}^{-2} \text{ s}^{-1}$ $0.5 \times 10^{-9} \text{ mol m}^{-2} \text{ s}^{-1}$	$8 \times 10^{-8} \text{ mol m}^{-2} \text{ s}^{-1}$		
Pyrite nonoxidative dissolution	$1.9 \times 10^{-10} \text{ mol m}^{-2} \text{ s}^{-1}$			Weisener (2002) Ritchie (1994) Jerz and Rimstidt (2000)
Oxidation of waste dump			$0.03 \times 10^{-8} \text{ mol m}^{-2} \text{ s}^{-1}$	
Oxidation in high relative humidity	$\sim 10^{-8} \text{ mol m}^{-2} \text{ s}^{-1}$			
Pyrrhotite oxidation by [DO]	$1.3 \times 10^{-8} \text{ mol m}^{-2} \text{ s}^{-1}$ $4 \times 10^{-9} \text{ mol m}^{-2} \text{ s}^{-1}$			Nicholson and Scharer (1994) and Janzen et al. (2000) Janzen et al. (2000) Janzen et al. (2000) and Thomas et al. (1998, 2000)
Pyrrhotite oxidation by $[\text{Fe}^{3+}]$	$3.5 \times 10^{-8} \text{ mol m}^{-2} \text{ s}^{-1}$			
Pyrrhotite nonoxidative dissolution	$5 \times 10^{-10} \text{ mol m}^{-2} \text{ s}^{-1}$ $6 \times 10^{-7} \text{ mol m}^{-2} \text{ s}^{-1}$			
Sphalerite oxidation by $[\text{Fe}^{3+}]$	3×10^{-7} to $7 \times 10^{-7} \text{ mol m}^{-2} \text{ s}^{-1}$			Rimstidt et al. (1994)
Sphalerite oxidation by [DO]	1×10^{-8} to $3 \times 10^{-8} \text{ mol m}^{-2} \text{ s}^{-1}$			
Chalcopyrite oxidation by $[\text{Fe}^{3+}]$	$9.6 \times 10^{-9} \text{ mol m}^{-2} \text{ s}^{-1}$			Rimstidt et al. (1994)
Galena oxidation by $[\text{Fe}^{3+}]$	$1.6 \times 10^{-6} \text{ mol m}^{-2} \text{ s}^{-1}$			Rimstidt et al. (1994)

decrease in pH can be sufficient to cause their rapid mortality. Some species of *Thiomonas* (e.g., *Thiomonas intermedia* and *Thiomonas arsenitoxydans*) can also obtain energy from oxidizing arsenic (III) to arsenic (V) and therefore have potential for remediating As-contaminated mine waters (As(V) rapidly combines with schwertmannite and other ferric precipitates in mine waters, whereas As(III) is more bioavailable). Claims that some *Thiomonas* spp. can directly oxidize ferrous iron have been questioned, as pH perturbations in solid and liquid cultures of these bacteria can give rise to spontaneous chemical oxidation of iron.

Acidiphilium acidophilum is another classified acidophilic sulfur-oxidizing bacterium that is somewhat unusual in that, although it is capable of growing as a heterotroph (a trait that characterizes all other known species of *Acidiphilium*), it can also grow autotrophically using reduced sulfur as an energy source, which is unique among species of this genus. It was originally considered to be a species of *Thiobacillus* (*Thiobacillus acidophilus*), though it was later reclassified as an *Acidiphilium* sp. on the basis of 16S rRNA gene sequencing. Interestingly, other species of *Acidiphilium* (which are widely distributed in mine waters) can also accelerate the oxidation of reduced sulfur though only when provided with organic carbon.

11.5.3.2.2.2 Acidophilic bacteria and archaea that oxidize both iron and sulfur

Some acidophilic bacteria that are commonly encountered in acidic mine waters are capable of accelerating the dissimilatory oxidation of both ferrous iron and reduced sulfur. Of these, the Gram-negative species *Acidithiobacillus* spp., *Acidiferrobacter thiooxydans*, and *Thiobacillus prosperus* are obligate autotrophs, while the Gram-positive species can use organic compounds as a carbon source and also fix carbon dioxide.

At. ferrooxidans is, by far, the most well studied of all acidophiles and was first isolated and characterized in the late 1940s. It is capable of growing autotrophically by oxidation of reduced sulfur or hydrogen, as well as ferrous iron, and can grow in anaerobic (using ferric iron as an electron acceptor, in place of oxygen) and aerobic environments. It is very frequently encountered in acidic mine waters, though its faster growth than many other iron oxidizers (such as *L. ferrooxidans*) means that it tends to dominate enrichment cultures of mine waters that contain ferrous iron, often leading to erroneous conclusions of its relative abundance and significance in these environments. It is now recognized that bacteria that have been classified as *At. ferrooxidans* comprise at least four phylogenetic types (species) on the basis of multigene sequence analysis (Amouric et al., 2011). One of these phylotypes had previously been recognized as the novel species, *At. ferrivorans*. This iron/sulfur oxidizer has many physiological traits in common with *At. ferrooxidans*, though an important difference is that it is a psychrotolerant bacterium (growing at 4 °C and above), and appears to be particularly well distributed in low-temperature mine waters. *At. ferrivorans* is also less acidophilic than *At. ferrooxidans*, and it grows poorly (if at all) on hydrogen. Amouric et al. (2011) also provided evidence of different biochemical pathways for iron oxidation in iron-oxidizing acidithiobacilli, including *At. ferrooxidans* and *At. ferrivorans*.

At least two other species of obligately autotrophic iron- and sulfur-oxidizing acidophilic bacteria are known, and

although these are also *Proteobacteria*, they are only distantly related to *Acidithiobacillus* spp. *Af. thiooxydans* was initially named *Thiobacillus ferrooxidans* strain m-1, though fairly soon after its isolation (in the early 1980s), it was apparent that these bacteria were phylogenetically distant from other (bona fide) strains of *At. ferrooxidans* (Harrison, 1982). The original strain was isolated from coal spoil refuse in Missouri, United States, and there have been relatively few reports of its occurrence in other acidic waters (described in Hallberg et al. 2011b). *T. prosperus*, on the other hand, still awaits official classification over 20 years since first being isolated from the island of Vulcano (Italy), though closely related strains have been isolated more recently by Davis-Belmar et al. (2008). The key trait of *T. prosperus*-like acidophiles is their requirement for salt (sodium chloride), a compound that tends to inhibit the growth of more well-known iron oxidizers, including *At. ferrooxidans*, at relatively small concentrations. *T. prosperus*-like acidophiles have, to date, been isolated from acidic marine-impacted waters, though it might be anticipated that they could also thrive in saline mine waters. Interestingly, growth of both *Af. thiooxydans* and *T. prosperus* on ferrous iron has been reported to be stimulated by the inclusion of reduced sulfur, such as tetrathionate, in growth media.

A number of species of Gram-positive iron- and sulfur-oxidizing acidophilic bacteria have been described, the majority belonging to the spore-forming *Firmicutes* phylum. While most of these are thermotolerant or moderately thermophilic, and would not be anticipated to thrive in most mine waters, some unclassified species (possibly belonging to novel genera) have been detected in low-temperature mine waters (e.g., Kimura et al., 2011).

11.5.4 Acid-Neutralization Mechanisms at Mine Sites

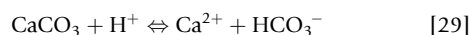
The oxidation of sulfide minerals in mine waste-rock piles and tailings impoundments generates acidic waters containing high concentrations of SO_4 , Fe(II) , and other metals. The water affected by sulfide oxidation is displaced into underlying or adjacent geological materials, or it is discharged directly to the adjacent surface-water flow system. Geochemical reactions with the gangue minerals result in progressive increases in the pore-water pH and the attenuation of some dissolved metals. A sequence of geochemical reactions occurring in the mine wastes and in the underlying aquifers results in profound changes in the concentrations of dissolved constituents and in the mineralogy and physical properties of the mine waste and aquifer materials. Low-pH conditions promote the dissolution of many metal-bearing solids and the desorption of metals from solid surfaces. Increases in pH through acid-neutralization reactions can lead to pronounced declines in the concentrations of dissolved metals released from mine wastes.

11.5.4.1 Mechanisms of Acid Neutralization

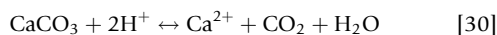
The acid generated through sulfide oxidation reacts with the nonsulfide gangue minerals within the mine wastes. The most significant pH-buffering reactions in mine settings are the dissolution of carbonate minerals, aluminum hydroxide and ferric oxyhydroxide minerals, and aluminosilicate minerals.

11.5.4.1.1 Carbonate-mineral dissolution

The most abundant carbonate minerals in mine wastes are calcite [CaCO_3], dolomite [$\text{CaMg}(\text{CO}_3)_2$], ankerite [$\text{Ca}(\text{Fe}, \text{Mg})(\text{CO}_3)_2$], siderite [FeCO_3], or mixtures thereof. The dissolution of calcite can be described as



or, at low pH, as



The dissolution of carbonate minerals has the potential to raise the pH of the pore water to near neutral. The dissolution of carbonate minerals releases Ca, Mg, Mn, Fe, and other cations that are as solid-solution substitutions or as impurities and increases the alkalinity of the water. At many sites, the mass of carbonate minerals contained in the mine wastes exceeds that of the sulfide minerals, and the rapid dissolution of carbonate minerals is sufficient to maintain neutral pH conditions through the mines or their waste piles. Neutral pH conditions have been observed in tailings impoundments derived from processing carbonate-bearing vein-type gold deposits (Blowes, 1990; McCreadie et al., 2000) and in open-pits derived from gold mining (Shevenell et al., 1999).

Many mines and mine wastes derived from massive sulfide ore deposits have a sulfide content in excess of the carbonate-neutralization capacity. At these sites, the available carbonate content is completely consumed, with the most soluble and most reactive carbonate minerals depleted first. Calcite is depleted initially, followed by dolomite–ankerite and siderite (Blowes and Ptacek, 1994). As the carbonate content of the waste dissolves, the pH is buffered to near neutral. The depletion of carbonate minerals through consumption by acid-neutralization reactions has been documented for a number of settings, including base-metal mine-tailings impoundments (Blowes and Jambor, 1990; Dubrovsky, 1986; Dubrovsky et al., 1984; Gunsinger et al., 2006a,b; Johnson et al., 2000; Moncur et al., 2005), aquifers impacted by drainage from mine wastes (Morin et al., 1988), and coal-mine spoils (Cravotta, 1994). In these studies, the greatest depletion occurs at the advancing acid front. An abrupt increase in solid-phase carbonate concentration is observed, suggesting that the dissolution of carbonate minerals is rapid. Measurements of the carbonate content of mine wastes, in front of and behind of the carbonate dissolution front, indicate that carbonate depletion occurs over a small distance along the path of groundwater flow and that carbonate minerals are absent behind the acid front (Dubrovsky et al., 1984; Johnson et al., 2000). Detailed laboratory column studies conducted by Jurjovec et al. (2002, 2003) confirmed that the depletion of carbonate minerals in response to an advancing acid front is rapid. Jurjovec et al. (2002, 2003) observed stable pH conditions, while alkalinity and cations were produced through carbonate dissolution, followed by a pronounced drop in pH after carbonate minerals were depleted (Figure 3).

The solubility products for the dissolution of calcite, dolomite–ankerite, and siderite vary, with calcite having the highest solubility product and siderite the lowest (Table 7). Field observations suggest that dissolution of the most soluble of these minerals will proceed first, followed by dissolution of

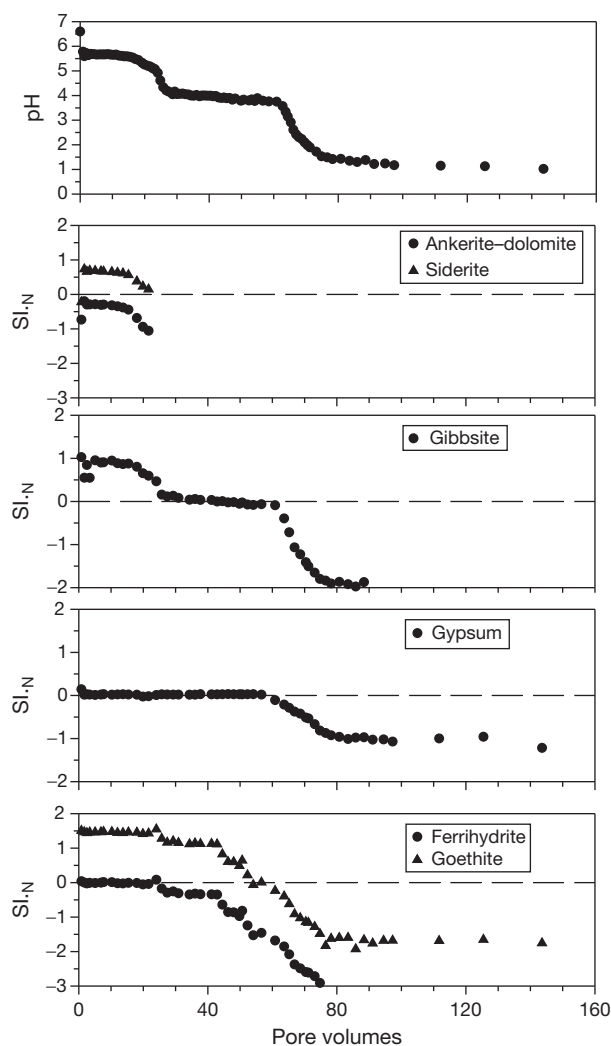


Figure 3 pH-buffering sequence observed in mine tailings: results of laboratory column experiment with 0.1 N H_2SO_4 input solution (reproduced from Jurjovec J, Ptacek CJ, and Blowes DW (2002) Acid neutralization mechanisms and metal release in mine tailings: A laboratory column experiment. *Geochimica et Cosmochimica Acta* 66: 1511–1523). SI_N is the saturation index normalized per mole of reactant.

the next most soluble mineral. As Fe sulfide minerals oxidize, $\text{Fe}(\text{II})$ can be released to anoxic pore waters and can react with the HCO_3^- produced through calcite dissolution, thereby forming siderite. Observations from field sites, coupled with geochemical modeling results, suggest that secondary siderite does form through this reaction (e.g., Blowes and Jambor, 1990; Ho et al., 1984; Moncur et al., 2005; Morin et al., 1988; Ptacek and Blowes, 1994). Al et al. (2000) identified secondary siderite on calcite surfaces, consistent with theoretical predictions. In laboratory studies conducted by Jurjovec et al. (2002, 2003), the dissolution of various carbonate minerals, including calcite, dolomite–ankerite, and siderite, could not be distinguished. Under the near-neutral pH conditions maintained by carbonate-mineral dissolution, the precipitation of crystalline and amorphous metal hydroxides is favored. Cations released through sulfide oxidation and the dissolution

Table 7 Solubility products for selected phases that form at mine-drainage sites

Mineral	Reaction	log <i>k</i>
<i>Oxides and hydroxides</i>		
Goethite	$\text{FeOOH}_{(s)} + 3\text{H}^+_{(aq)} \leftrightarrow \text{Fe}^{3+}_{(aq)} + 2\text{H}_2\text{O}_{(l)}$	-1.0
Ferrihydrite	$\text{Fe}(\text{OH})_{3(s)} + 3\text{H}^+_{(aq)} \leftrightarrow \text{Fe}^{3+}_{(aq)} + 3\text{H}_2\text{O}_{(l)}$	3.0–5.0
Gibbsite	$\text{Al}(\text{OH})_{3(s)} + 3\text{H}^+_{(aq)} \leftrightarrow \text{Al}^{3+}_{(aq)} + 3\text{H}_2\text{O}_{(l)}$	7.94
Boehmite	$\text{AlOOH}_{(s)} + 3\text{H}^+ \leftrightarrow \text{Al}^{3+}_{(aq)} + 2\text{H}_2\text{O}_{(l)}$	7.83
<i>Sulfates</i>		
Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}_{(s)} \leftrightarrow \text{Ca}^{2+}_{(aq)} + \text{SO}_4^{2-}_{(aq)} + 2\text{H}_2\text{O}_{(l)}$	-4.58
Celestine	$\text{SrSO}_{4(s)} \leftrightarrow \text{Sr}^{2+}_{(aq)} + \text{SO}_4^{2-}_{(aq)}$	-6.62
Barite	$\text{BaSO}_4 \leftrightarrow \text{Ba}^{2+}_{(aq)} + \text{SO}_4^{2-}_{(aq)}$	-0.97
<i>Carbonates</i>		
Calcite	$\text{CaCO}_{3(s)} \leftrightarrow \text{Ca}^{2+}_{(aq)} + \text{CO}_3^{2-}_{(aq)}$	-8.48
Dolomite	$\text{CaMg}(\text{CO}_3)_{2(s)} \leftrightarrow \text{Ca}^{2+}_{(aq)} + \text{Mg}^{2+}_{(aq)} + 2\text{CO}_3^{2-}_{(aq)}$	-17.09

Source: Nordstrom DK and Munoz JL (1994) *Geochemical Thermodynamics*. Boston, MA: Blackwell Scientific.

of carbonate and aluminosilicate minerals can lead to the formation of secondary hydroxide phases that accumulate in the waste solids. These phases may subsequently dissolve as additional acid is produced through sulfide-mineral oxidation reactions. The dissolution of these secondary minerals contributes to acid neutralization.

11.5.4.1.2 Dissolution of hydroxide minerals

After the carbonate minerals are depleted, the pH falls until equilibrium with the most soluble secondary hydroxide mineral is attained. An abrupt increase in the concentration of dissolved Al has been observed to coincide with carbonate depletion at several mine-tailings impoundments. These observations suggest that the initial hydroxide mineral to dissolve is an aluminum hydroxide. Although a discrete Al-bearing phase has not been isolated, it is generally assumed to be amorphous Al hydroxide $[\text{Al}(\text{OH})_3]$ or gibbsite [crystalline $\text{Al}(\text{OH})_3$]. Dissolution of this Al-bearing phase buffers the pH in the region of 4.0–4.5 (Blowes and Jambor, 1990; Dubrovsky, 1986; Gungsinger et al., 2006b; Johnson et al., 2000; Moncur et al., 2005). Although repeated attempts have been made, the presence of these Al-bearing phases has not been confirmed through mineralogical studies. After the secondary Al phases present in the tailings are depleted, the pH falls until equilibrium with an Fe oxyhydroxide mineral, typically ferrihydrite (nominally $9\text{Fe}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$) or goethite ($\alpha\text{-FeOOH}$), has been attained. Dissolution of ferric oxyhydroxide minerals typically maintains pH values in the range of 2.5–3.5.

11.5.4.1.3 Dissolution of silicate minerals

Gangue minerals associated with sulfide-rich tailings include various aluminosilicate minerals such as chlorite, smectite, biotite, muscovite, plagioclase, and amphibole and other silicate minerals such as olivine and pyroxene. Their contribution to acid-neutralization reactions has been examined in a number of field studies and through controlled laboratory experiments. During the period of carbonate- and hydroxide-mineral dissolution, aluminosilicate minerals may also dissolve, thus

consuming H^+ and releasing H_4SiO_4 , Al^{3+} , and other cations, including K, Ca, Mg, and Mn. Silicate dissolution is generally not rapid enough to buffer the pore water to a specific pH. Such reactions, however, do consume H^+ and contribute to the overall acid-neutralization potential of mine wastes. For instance, biotite has high neutralization capacity in that the dissolution of one mole of biotite (e.g., $\text{KMg}_{1.5}\text{Fe}_{1.5}\text{Al Si}_3\text{O}_{10}(\text{OH})_2$) can consume between 7 and 10 H^+ depending upon the type of dissolution. In addition to H^+ consumption, Al and other metals released from aluminosilicate minerals may accumulate in secondary products such as amorphous $\text{Al}(\text{OH})_3$ or gibbsite that act as secondary pH buffers. As with other phases contributing to acid neutralization, the mass and composition of silicates vary from site to site and within an individual site.

Detailed field studies conducted at the Heath Steele tailings impoundment, New Brunswick, Canada, illustrate the role of aluminosilicate-mineral dissolution in contributing to acid neutralization. The tailings contain in excess of 85 wt% sulfide minerals, where partial oxidation has led to the production of low-pH pore waters ($\text{pH} < 0.5$) and the extensive dissolution of aluminosilicate minerals. Blowes (1990) noted increased concentrations of Al and Si (up to 2500 and 200 mg l^{-1} , respectively) due to the dissolution of the aluminosilicate minerals. In the deeper tailings, a diverse silicate gangue was observed, including chlorite, muscovite, quartz, and an Fe-bearing amphibole. In the upper 30 cm of the tailings, a less diverse silicate gangue, extensively depleted in chlorite and amphibole, was observed. The order of depletion of the aluminosilicate minerals followed closely the order predicted on the basis of the zero point of charge of the aluminosilicate mineral.

Johnson et al. (2000) documented the depletion of aluminosilicates at the Nickel Rim mine-tailings impoundment. There solid-phase Al and biotite are depleted in the upper oxidized portion of the tailings, whereas biotite is abundant in the underlying unoxidized tailings. At the Sherridon tailings impoundment, Manitoba, Moncur et al. (2003, 2005) observed extensive depletion of biotite, chlorite, and possibly, smectite in the upper 1 m of the tailings. At this site, there is evidence of secondary replacement of other aluminosilicates, namely, plagioclase, cordierite, and amphibole. At Sherridon, the first appearance of biotite occurs at 72 cm, which coincides with the first appearance of secondary sulfides (marcasite). Chlorite and smectite first appear at a 90 cm depth, indicating the greater tendency for dissolution. In a controlled laboratory experiment using a 0.1 M H_2SO_4 solution, Jurjovec et al. (2002) observed extensive depletion of chlorite and minor depletion of biotite as a result of acid leaching of the tailings.

Dissolution rates of biotite and iron-rich chlorite at acidic pH (Nagy, 1995; Ritchie, 1994) are slow in comparison with calcite but several orders of magnitude faster than K-feldspars and albite (Lasaga et al., 1994; Stillings and Brantley, 1995). In addition, biotite and chlorite can be considered as slightly more reactive than muscovite. Dissolution rates of olivine at acidic pH (i.e., 3.98×10^{-11} – 1.02×10^{-12} $\text{mol}^{-1} \text{cm}^{-2} \text{s}^{-1}$) as reported by Wogelius and Walther (1991) and Sverjensky (1992) are comparable to the oxidative dissolution rates of pyrite. Other minerals with relatively fast dissolution rates under acidic conditions that are comparable with the oxidation rates of pyrite include Ca end-member of plagioclase (anorthite) and wollastonite. In addition, Fe-rich pyroxenes

and amphiboles may also be considered as having potentials to neutralize acidity. The presence of Fe^{2+} or other cations that may oxidize reduces the structural stability; therefore, the rates for ferromagnesian minerals may be higher depending upon their Mg to Fe ratios.

11.5.4.1.4 Development of pH-buffering sequences

The sequence of pH-buffering reactions observed within tailings impoundments results in a progressive increase in the pore-water pH along the groundwater flow path. These changes in pH occur as long zones of relatively uniform pH, which are dominated by a single pH-buffering reaction, separated by fronts or relatively sharp changes in pH as a buffering phase is depleted (Figure 3). Field data from the Nickel Rim tailings impoundment, as were presented by Johnson et al. (2000), illustrate a pH-buffering sequence. At the Nickel Rim site, the pH of the pore water at the base of the impoundment is near neutral, varying from 6.5 to 7.0. The geochemical calculations suggest that the pore water in this zone approaches or attains equilibrium with respect to calcite. Overlying the calcite-buffered zone, calcite is absent from the tailings, and the pH is buffered by the dissolution of siderite and possibly by the dissolution of dolomite. The complete depletion of carbonate minerals is accompanied by a sharp decline in pH from 5.8 to 4.5. Immediately above the depth of carbonate-mineral depletion, the pH is relatively uniform, varying from 4.0 to 4.5, the dissolved Al concentrations increase sharply, and the pore water approaches or attains equilibrium with respect to gibbsite. Near the surface of the impoundment, the pH decreases sharply from 4.0 to 3.5, then more gradually from 3.5 to approximately 2.5. Throughout this decrease in pH, the pore water is undersaturated with respect to ferrihydrite and approaches equilibrium with respect to goethite.

11.5.5 Geochemistry and Mineralogy of Secondary Minerals

In mineral processing and mine-waste settings, the secondary minerals are those that form during and after processing and after disposal of the wastes. Thus, the mined material may contain both the primary, fresh minerals and the oxidation products that were formed through geological processes (e.g., paleoweathering). In rare cases, a mineral deposit will consist wholly of oxidized material, but in the acid-drainage setting, all of these minerals are deemed to constitute the primary assemblage because they were formed prior to processing of ores and disposal as wastes. After final discharge or even during temporary storage of wastes and low-grade ores, crystallization of secondary minerals occurs in situ principally as a result of oxidation of the primary sulfide assemblage. However, upon removal from its disposal site, as occurs during a sampling program, a waste may form additional minerals as the material dries; such products, which are predominantly water-soluble salts that crystallize during evaporation of the pore water, are classified as 'tertiary' minerals (Jambor, 1994; Jambor and Blowes, 1998).

11.5.5.1 Soluble Sulfates: Iron Minerals

Iron sulfates form under low-pH conditions and are generally highly soluble. The most common secondary Fe sulfate minerals associated with oxidized mine wastes are of the type $\text{Fe}^{2+}\text{SO}_4 \cdot n\text{H}_2\text{O}$, wherein n ranges from 7 to 1. The minerals of the most frequent occurrence and abundance are melanterite ($n=7$), rozenite ($n=4$), and szomolnokite ($n=1$); the pentahydrate (siderotil) and hexahydrate (ferrohexahydrite) have also been reported but are much less widespread and occur sparingly. The divalent salts seem to form in the earliest stages of sulfide-mineral oxidation, but they are observed more commonly as superficial evanescent coatings that appear during dry periods and are susceptible to solubilization during rainfall events. Melanterite has been reported from several localities as a precipitate from greenish pools of Fe-rich solutions that drained from pyritiferous deposits, and the mineral has also been noted to form a hardpan layer in pyrite-rich tailings at the Heath Steele Mines, New Brunswick, Canada (Blowes et al., 1992), in the main hardpan of the Sherridon tailings in Manitoba (Moncur et al., 2005, 2009), and in high-sulfide mine wastes at the Berikul mine site in Russia (Sidenko et al., 2005). The tetrahydrate, rozenite, more typically forms as a dehydration product of melanterite.

Although the soluble salts of ferric iron may crystallize on a microscale in the earliest stages of Fe sulfide oxidation, in field settings, the paragenesis indicates that the progression is from the early formation of the divalent salts of Fe to minerals containing both ferrous and ferric Fe and thence to the trivalent-Fe minerals (Jambor et al., 2000a,b). For minerals devoid of other cations, the most commonly occurring of the mixed-valence type is copiapite [$\text{Fe}^{2+}\text{Fe}_4^{3+}(\text{SO}_4)_6(\text{OH})_2 \cdot 20\text{H}_2\text{O}$], and less common is römerite [$\text{Fe}^{2+}\text{Fe}_2^{3+}(\text{SO}_4)_4 \cdot 14\text{H}_2\text{O}$]. Among the trivalent-Fe minerals, the most common are coquimbite [$\text{Fe}_2^{3+}(\text{SO}_4)_3 \cdot 9\text{H}_2\text{O}$], ferricopiapite [$\text{Fe}_{2/3}^{3+}\text{Fe}_4^{3+}(\text{SO}_4)_6(\text{OH})_2 \cdot 20\text{H}_2\text{O}$], and rhomboclase [$(\text{H}_5\text{O}_2)^{1+}\text{Fe}^{3+}(\text{SO}_4)_2 \cdot 2\text{H}_2\text{O}$]. Also common is fibroferrite [$\text{Fe}^{3+}(\text{SO}_4)(\text{OH}) \cdot 5\text{H}_2\text{O}$], which, however, is relatively insoluble except in acidic conditions.

A notable locality for acid drainage is Iron Mountain, California, where oxidation of pyritiferous massive sulfide deposits has resulted in the formation of many of the aforementioned soluble salts in the underground workings. The precipitation order of the salts at Iron Mountain follows the divalent to trivalent trend (Nordstrom and Alpers, 1999b). These secondary iron sulfates form and evolve during dry periods through dehydration, oxidation, and neutralization reactions, and they are dissolved during wet periods by releasing metals and acidity into local environments.

11.5.5.2 Soluble Sulfates: Other Elements

Other than the soluble sulfates of Fe, only those of Mg and Al are of moderately common occurrence in association with acid drainage. The Mg salts are chiefly of the type $\text{MgSO}_4 \cdot n\text{H}_2\text{O}$, with epsomite [$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$] and hexahydrite [$\text{MgSO}_4 \cdot 6\text{H}_2\text{O}$] the predominant minerals. Occurrences of the pentahydrate (pentahydrite) and the tetrahydrate (starkeyite), which are known from a few localities, could be categorized as rare, and the monohydrate (kieserite), although identified as a weathering product of pyritiferous shales, has not yet been observed specifically in association with acid drainage.

Whereas the oxidation of pyrite provides the principal source for the Fe analogues of the simple Mg sulfates, the chief source of the latter is dolomite $[\text{CaMg}(\text{CO}_3)_2]$. Dissolution of dolomite attenuates the acidity generated by pyrite oxidation, thereby releasing Ca and Mg to form gypsum $[\text{CaSO}_4 \cdot 2\text{H}_2\text{O}]$, simple Mg salts, and more complex ones such as magnesiocopiapite $[\text{MgFe}_4^{3+}(\text{SO}_4)_6(\text{OH})_2 \cdot 20\text{H}_2\text{O}]$. Thus, both gypsum and Mg sulfates may form without the development of acid drainage if dolomite is sufficiently abundant to neutralize the acidity that is generated by oxidation of the associated sulfides.

The formation of soluble salts containing Al is paragenetically later than that of the simple Mg salts because Al is sourced from aluminosilicates, the dissolution of which is typically slow and requires low-pH conditions to accelerate the process. The most common soluble sulfates of Al are members of the halotrichite–pickeringite series $[\text{Fe}^{2+}\text{Al}_2(\text{SO}_4)_4 \cdot 22\text{H}_2\text{O} - \text{MgAl}_2(\text{SO}_4)_4 \cdot 22\text{H}_2\text{O}]$ and of less widespread occurrence are aluminocopiapite $[\text{Al}_{2/3}\text{Fe}_4^{3+}(\text{SO}_4)_6(\text{OH})_2 \cdot 20\text{H}_2\text{O}]$ and alunogen $[\text{Al}_2(\text{SO}_4)_3 \cdot 17\text{H}_2\text{O}]$.

Various soluble salts, such as kalinite $[\text{KAl}(\text{SO}_4)_2 \cdot 11\text{H}_2\text{O}]$, blödite $[\text{Na}_2\text{Mg}(\text{SO}_4)_2 \cdot 4\text{H}_2\text{O}]$, and bilinite $[\text{Fe}^{2+}\text{Fe}_2^{3+}(\text{SO}_4)_4 \cdot 22\text{H}_2\text{O}]$, occur sparingly but may be locally prominent. The most common of the Cu salts is chalcantite $[\text{CuSO}_4 \cdot 5\text{H}_2\text{O}]$. Both goslarite $[\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}]$ and gunningite $[\text{ZnSO}_4 \cdot \text{H}_2\text{O}]$ have been reported as secondary minerals in sphalerite-bearing wastes, but the salts occur sparingly.

11.5.5.3 Less Soluble Sulfate Minerals

Gypsum, which has a moderate solubility in water, is the most abundant secondary mineral in sulfide-rich tailings due to high-sulfate concentrations resulting from sulfide oxidation and high-Ca concentrations arising from lime and/or limestone use during metallurgical processing and tailings neutralization. Jarosite $[\text{KFe}_3(\text{SO}_4)_2(\text{OH})_6]$, which is relatively insoluble (Table 7), is the other abundant and widespread secondary mineral associated with acid drainage. Bassanite $[\text{2CaSO}_4 \cdot \text{H}_2\text{O}]$ has also been identified as a secondary mineral in tailings impoundments, but the quantities are minute and are insignificant relative to those of gypsum. Jarosite is, by definition, K-dominant, but as a mine-drainage precipitate, it invariably is oxonium-bearing, commonly with $\text{K} > \text{H}_3\text{O} > \text{Na}$. In the early stages of acid generation, the bulk of the K at most localities is derived by incongruent dissolution of trioctahedral micas. Various other members of the jarosite supergroup occur in acid-drainage settings; other members detected are natrojarosite $[\text{NaFe}_3(\text{SO}_4)_2(\text{OH})_6]$, hydronium jarosite $[(\text{H}_3\text{O})\text{Fe}_3(\text{SO}_4)_2(\text{OH})_6]$, plumbojarosite $[\text{PbFe}_6(\text{SO}_4)_4(\text{OH})_{12}]$, alunite $[\text{KAl}_3(\text{SO}_4)_2(\text{OH})_6]$, and As-bearing species, but all are of rare occurrence relative to jarosite. Jarosite is also common in weathered zones of sulfide ore deposits and in acid sulfate soils. Furthermore, jarosite is an important compound in hydrometallurgy because its formation is widely utilized to control Fe, sulfate, alkalis, and impurity elements. Precipitation of jarosite occurs from acidic ($\text{pH} < 3$) solutions rich in $\text{Fe}(\text{III})$, SO_4 , and alkalis such as K and Na at elevated temperatures (Dutrizac and Kaiman, 1976; Majzlan and Myneni, 2005). Dissolution of jarosite produces acidity.

Among the Al sulfate salts, jurbanite $[\text{Al}(\text{SO}_4)(\text{OH}) \cdot 5\text{H}_2\text{O}]$ is commonly used in geochemical modeling, but the mineral itself is of extremely rare occurrence; the most common Al hydroxysulfates associated with acid drainage correspond to crystalline and amorphous felsöbányaite (basaluminite) $[\text{Al}_4(\text{SO}_4)(\text{OH})_{10} \cdot 4\text{H}_2\text{O}]$ and hydrobasaluminite $[\text{Al}_4(\text{SO}_4)(\text{OH})_{10} \cdot 15\text{H}_2\text{O}]$.

A poorly crystalline hydroxysulfate, schwertmannite (Bigham et al., 1990), having the ideal formula $\text{Fe}_8\text{O}_8(\text{OH})_6 \cdot \text{SO}_4 \cdot n\text{H}_2\text{O}$, may be the most common direct precipitate of Fe from acidic effluents at pH 2–4 (Bigham and Nordstrom, 2000). The composition deviates from the ideal formula in terms of the proportions of Fe, OH, and SO_4 . Although schwertmannite occurs in acidic waters and soils and commonly the initial precipitate from AMD, it is difficult to identify because of its poor crystallinity and the almost invariable presence of associated Fe oxyhydroxides and jarosite, but numerous occurrences have been reported in the literature. Schwertmannite typically occurs as ‘hedgehog’-type rounded aggregates with filamentous or needlelike whiskers (Bigham and Nordstrom, 2000; Loan et al., 2004) or spheroidal aggregates (Peretyazhko et al., 2009; Regenspurg et al., 2004) that are typically less than a micrometer. Its structure is described by Bigham et al. (1990) as having some similarities to that of akaganeite $[\beta\text{-FeO}(\text{OH}, \text{Cl})]$ where double chains of FeO octahedra share corners to produce square tunnels extending parallel to the *c*-axis. Transmission electron microscopy (TEM) and nanodiffraction studies of Loan et al. (2004) did not find any evidence for an akaganeite-like structure. Instead, the TEM observations suggested that the schwertmannite whiskers possess structures similar to maghemite and highly disordered ferrihydrite. Recently, Fernandez-Martinez et al. (2010) proposed a structural model similar to the original akaganeite-like structure having two sulfate molecules per unit cell where one sulfate occurs as an inner-sphere and the other as an outer-sphere complex within the akaganeite tunnel structure. Schwertmannite is metastable, and it transforms to goethite (Bigham et al., 1996; Peretyazhko et al., 2009; Schwertmann and Carlson, 2005) and to jarosite at low pH (Majzlan and Myneni, 2005). Its transformation rate to goethite increases with an increase in pH (Regenspurg et al., 2004) and temperature (Schwertmann and Carlson, 2005), whereas it decreases with increases in the concentrations of sulfate and dissolved organic carbon (Knorr and Blodau, 2007). Detailed studies of the oxidation of pyrite and pyrrhotite indicate that schwertmannite is not an early formed mineral, and the occurrences suggest an origin in which the requisite Fe and SO_4 have undergone solubilization, transportation, and subsequent precipitation rather than maintaining an intimate association with precursor sulfides. Regenspurg et al. (2004) describe schwertmannite as the first mineral that forms from oxidation and hydrolysis of $\text{Fe}(\text{II})\text{SO}_4$ solutions at pH 5–6.

The principal ore mineral of Pb is galena $[\text{PbS}]$, and the predominant sulfates derived from it are anglesite $[\text{PbSO}_4]$ and plumbojarosite. Partial substitution of Pb for the alkali-site cations also occurs in other members of the jarosite supergroup. An unusual sink for Cu and Al is the mineral hydrowoodwardite $[\text{Cu}_{1-x}\text{Al}_x(\text{OH})_2(\text{SO}_4)_{x/2}(\text{H}_2\text{O})_n]$, recently discovered as a supergene product at several old mines in Saxony, Germany (Witzke, 1999). The same mineral was

almost simultaneously reported to occur as bluish coatings on weathered waste rocks and in stream sediments at two former mines in Northern Italy (Dinelli et al., 1998).

11.5.5.4 Metal Oxides and Hydroxides

The oxidation of Fe sulfides invariably leads to the formation of goethite [α -FeOOH] both in intimate association with the sulfides and in distal precipitates. Other oxyhydroxide minerals identified as secondary precipitates are crocote [γ -FeOOH] and akaganéite [β -FeO(OH,Cl)], but occurrences are generally insignificant relative to the abundance and distribution of goethite, which is thermodynamically the most stable iron oxyhydroxide at low temperatures. However, lepidocrocite has been observed to be important at some sites; for example, Roussel et al. (2000) determined the mineral to be the principal component of suspended particulate matter in drainage from an As-rich tailings impoundment in France, and Bowell and Bruce (1995) observed lepidocrocite to be a significant component of ochres that precipitated from low-pH (<5) waters at the Levant Mine in Cornwall, England. In general, Fe(III) oxyhydroxides are the other abundant secondary minerals forming from slightly acidic to neutral drainages in sulfidic mine tailings.

Ferrihydrite is a poorly crystalline mineral of widespread occurrence in soils and secondary alteration products of hydrothermal ore deposits, mining wastes, AMD settings, and iron-bearing sediments. In addition, it has a considerable importance in mineral processing and extractive metallurgy (Jambor and Dutrizac, 1998). Ferrihydrite is commonly referred to as 'amorphous Fe(OH)₃,' poorly crystalline iron oxyhydroxide, or colloidal ferric hydroxide. Its particle size is defined by Waychunas et al. (1996) as being in the 0.8–1.5 nm range and in the 2–6 nm range by Janney et al. (2000). There are several varieties of ferrihydrite identified by the number of broad peaks in the x-ray diffraction (XRD) patterns such as the two-line and six-line varieties. The two-line variety is less crystalline and smaller in size in comparison to the six-line variety (Drits et al., 1993; Manceau and Drits, 1993). Ferrihydrite is considered to be a mixture rather than a single compound (Drits et al., 1993; Marchand and Rancourt, 2009). Recently, Michel et al. (2007, 2010) proposed a new structural model for ferrihydrite based on synchrotron x-ray total scattering data. The structure is analogous to akdalaite (Al₁₀O₁₄(OH)₂) having 20% iron in tetrahedral coordination and possessing a low density of defects. Iron octahedra and tetrahedra share corners and edges, and unlike the conventional model (Drits et al., 1993), there are no face-sharing octahedra. The new model received criticism due to its shortcomings in describing XRD, x-ray absorption spectroscopy, and Mössbauer data and not being in accordance with the fundamental crystal chemical principles (Hiemstra and van Riemsdijk, 2009; Manceau, 2009, 2010, 2011; Rancourt and Meunier, 2008).

Mixing of acidic waters with oxygenated freshwaters and neutralization of acidic mine effluents result in the formation of Fe(III) oxyhydroxides. These Fe(III) oxyhydroxide compounds and minerals begin as small clusters of Fe(O,OH,OH₂)₆ octahedra that evolve into larger polymeric units with time, eventually reaching colloidal sizes (Combes et al., 1989).

Precipitation of ferrihydrite is favored when oxidation of Fe²⁺ and hydrolysis occur rapidly from supersaturated solutions. Due to their nanoparticle or colloidal sizes, Fe(III) oxyhydroxides have high surface areas; therefore, they can play important roles in terms of immobilizing heavy metals and toxic elements through sorption.

Ferrihydrite is metastable and transforms to hematite or goethite with aging (Combes et al., 1989; Waychunas et al., 2005) or to hematite through internal structural arrangements combined with dehydration (Schwertmann et al., 2004). The transformation from ferrihydrite to hematite occurs at neutral pH. In contrast, transformation of ferrihydrite to goethite requires dissolution and recrystallization because face-sharing FeO octahedra are not present in the goethite structure. Instead, the goethite structure consists of edge-sharing and corner-sharing FeO octahedra. Because of the energy barriers involving breakage of the Fe–O bonds linking octahedra through a face-sharing arrangement, ferrihydrite–goethite transformation requires disintegration of the structure through dissolution (Combes et al., 1990).

The presence of ferrihydrite in precipitates from mine-waste effluents is commonly reported, but unequivocal identifications of its presence in mine-waste solids, such as tailings, have been fewer. For the latter, because the crystallinity is poor and the mineral forms only a small proportion of the total waste solids, recourse is generally made to selective dissolution whereby the fraction that is more soluble in certain reagents is inferred to represent ferrihydrite rather than goethite, which is more dissolution-resistant. In addition, the likelihood of finding ferrihydrite in tailings is slim due to its metastability.

Although hematite [α -Fe₂O₃] is a common product of the oxidation of sulfide-bearing mineral deposits, only a few unequivocal occurrences are known in mine-drainage settings (e.g., Boulet and Larocque, 1996; Hochella et al., 1999). Various Fe oxides and oxyhydroxides, including hematite, maghemite [γ -Fe₂O₃], and ilmenite [Fe²⁺TiO₃], have been observed as microcrystalline aggregates that have formed within acidophilic microorganisms in sediments affected by effluents from tailings at Elliot Lake, Ontario, Canada (Mann and Fyfe, 1989). Minerals of the spinel group, which includes magnetite [Fe²⁺Fe³⁺O₄], have been detected, but most associations are on a minute scale. However, the presence of extensive accumulations of magnetite derived by the anoxic reduction of transported Fe in mining-impacted lake-bottom sediments has been reported by Cummings et al. (2000).

Manganese oxides–oxyhydroxides which are known to be potent scavengers of trace metals in both aquatic and terrestrial environments (Villalobos et al., 2005) have been observed at a few acid-drainage localities. For example, Benvenuti et al. (2000) observed pyrolusite [MnO₂] and pyrochroite [Mn(OH)₂] in weathered jig tailings and waste rock at a site in Tuscany, Italy. At the Clark Fork River Superfund site, Hochella et al. (2005) reported the occurrence of a vernadite-like mineral (nominally MnO₂·*n*H₂O) having high sorption capacity due to its exceptionally high surface area. It occurs as a mixture with ferrihydrite and carries significant amounts of heavy metals such as Pb, Cu, Zn, and As. Hochella et al. (1999) identified hydrohetaerolite [Zn₂Mn₄³⁺O₈·H₂O] among the secondary minerals in stream sediments at an acid-drainage

site in Montana, and nsutite [$\gamma\text{-MnO}_2$], birnessite [$\text{Na}_4\text{Mn}_{14}\text{O}_{27}\cdot 9\text{H}_2\text{O}$], and minerals in the ranci  ite-takanelite series [$(\text{Ca}, \text{Mn}^{2+})\text{Mn}_4^{4+}\text{O}_9\cdot 3\text{H}_2\text{O}-(\text{Mn}^{2+}, \text{Ca})\text{Mn}_4^{4+}\text{O}_9\cdot \text{H}_2\text{O}$] were among the various Mn minerals detected in precipitates in the streambed of Pinal Creek, Arizona (Lind and Hem, 1993). At Pinal Creek, the acidic groundwater plume reacted with calcite-bearing alluvium, initially forming ‘amorphous Fe oxide’ as pH increased, and subsequently precipitating the Mn-bearing assemblage farther downstream, where the pH of the stream water was near neutral. Similar relationships were observed by Hudson-Edwards et al. (1996), who identified cesarolite [$\text{PbH}_2\text{Mn}_3\text{O}_8$], coronadite [$\text{Pb}(\text{Mn}^{4+}, \text{Mn}^{2+})_8\text{O}_{16}$], woodruffite [$(\text{Zn}, \text{Mn}^{2+}\text{Mn}_3^{4+}\text{O}_7\cdot 1\text{-}2\text{H}_2\text{O})$], and hydrohetaerolite in contaminated stream sediments of the Tyne River catchment, northeastern England, and noted the disappearance of these minerals in downstream areas in which the pH of the water was lower.

The presence of plumboferrite [$\sim\text{PbFe}_4\text{O}_7$] in the Tyne River sediments was reported by Hudson-Edwards et al. (1996), and the mineral has also been identified as a secondary product in tailings (Morin et al., 1999). Although Al hydroxides play a prominent role in geochemical modeling, few occurrences of such material have been described. Bove et al. (2000) reported the presence of an amorphous Al hydroxide precipitate in a stream channel that drains an acid sulfate hydrothermal system near Silverton, Colorado, and Bove et al. (2000) reported the occurrence of a poorly crystalline Al hydroxide in particulate matter that was filtered from seeps below an abandoned waste-rock dump in New Mexico; however, detailed characterizations were not published. Diaspore [AlOOH] has been reported to occur as a secondary mineral in a waste-rock dump in Tuscany, Italy (Benvenuti et al., 1997). In the study by Hochella et al. (1999), a nearly pure amorphous Si–Al oxyhydroxide (Si:Al = 2:1) rather than $\text{Al}(\text{OH})_3$ was identified as the major sink for Al in the local drainage system.

The occurrence of secondary $\text{SiO}_2\cdot n\text{H}_2\text{O}$ has been observed in several tailings impoundments, predominantly as amorphous hydrated pseudomorphs after biotite. Cristobalite [SiO_2] has also been detected in the pseudomorphs. Partial replacement of various other minerals by amorphous silica has also been observed.

An extensive suite of supergene minerals, including chlorides, sulfates, and arsenates, occurs at Levant Mine, Cornwall, England, where ingress of seawater has influenced the variety of minerals in the suite. In stagnant pools of mine drainage at the mine, cuprite [Cu_2O] and native copper have precipitated within the ochres (Bowell and Bruce, 1995).

11.5.5.5 Carbonate Minerals

Like the precipitation of secondary $\text{Al}(\text{OH})_3$, that of secondary siderite [FeCO_3] is widely employed in geochemical models. Secondary siderite and associated Fe oxyhydroxides were determined to have formed as coatings, up to 1000-nm thick, on ankerite–dolomite in the Kidd Creek tailings impoundment at Timmins, Ontario, Canada (Al et al., 2000). At Elliot Lake, Ontario, nodules of siderite, each <10 μm across, and calcite in aggregates, up to 400 μm across, occur as secondary minerals in a small tailings impoundment (Paktunc and Dav  , 2002). The precipitation of the carbonate minerals at the Elliot

Lake site was interpreted to result from high alkalinity related to the presence of decaying organic debris.

At the Matchless pyritiferous deposit in Namibia, huntite [$\text{CaMg}_3(\text{CO}_3)_4$] occurs within high-pH zones in an oxidized tailings impoundment (Dill et al., 2002). The high-pH zones are sandwiched between low-pH layers, with phyllosilicate-rich layers acting as aquitards that impede vertical movement and homogenization of the pore waters.

Wastes from Pb-rich mineral deposits typically form angle-site in sulfate-dominant environments, but in limestone-dominated host rocks and in gangue containing abundant carbonates, both cerussite [PbCO_3] and hydrocerussite [$\text{Pb}_3(\text{CO}_3)_2(\text{OH})_2$] have been reported as secondary minerals in mining-related wastes. Several carbonate and hydroxycarbonate minerals of Cu and Zn were reported by Hudson-Edwards et al. (1996) as secondary products in stream sediments in the Tyne Basin, England.

11.5.5.6 Arsenic Oxides

Common arsenic oxide minerals include arsenates or As^{5+} species such as scorodite, arseniosiderite, and pharmacosiderite and arsenites or As^{3+} species such as claudetite (As_2O_3) and arsenolite (As_2O_3). Arsenates are common in mine wastes and mine-drainage settings, mainly because of the common and widespread occurrence of arsenopyrite and arsenian pyrite as gangue minerals. Once discarded in tailings or as constituents of waste rocks, arsenopyrite and arsenian pyrite oxidize to form secondary arsenates as discussed earlier and as indicated from the occurrence of scorodite around relict arsenopyrite particles (Moncur et al., 2005, 2009; Paktunc et al., 2004; Salzsauler et al., 2005). Arsenate minerals are also common in the oxidized (paleoweathered) portions of some gold deposits (Paktunc, unpublished data) and impacted soils (Ashley and Lottermoser, 1999). The common secondary arsenates in mine wastes include scorodite, amorphous ferric arsenate, arseniosiderite, yukonite, and pharmacosiderite (Filippi et al., 2007; Foster et al., 1998a; Morin and Calas, 2006; Paktunc et al., 2003, 2004, 2008; Walker et al., 2009). Among the secondary arsenates, scorodite is the most abundant variety due to its higher stability in the near-surface environments (Harvey et al., 2006; Paktunc and Bruggeman, 2010).

By far, the most important sink for arsenate is sorption to Fe oxyhydroxides. Examples include Fuller et al. (1993), Nriagu (1994), Kimball et al. (1995), Manceau (1995), Waychunas et al. (1996), Lutzenkirchen and Lovgren (1998), Manning et al. (1998), Raven et al. (1998), Webster et al. (1998), Foster et al. (1998b), Ding et al. (2000), Jain and Loeppert (2000), and Farquhar et al. (2002). Fe oxyhydroxides have the capacity to adsorb significant amounts of arsenate as bridging or bidentate–binuclear type complexes (Fuller et al., 1993; Paktunc et al., 2004, 2008; Waychunas et al., 1996). Sorption of As on clays is reported by Manning and Goldberg (1996), Foster et al. (1998a), Lin and Puls (2000), and Garcia-Sanchez et al. (2002) and on schwertmannite by Carlson et al. (2002).

Arsenopyrite and arsenian pyrite are the common carriers of gold in refractory gold deposits. In order to liberate the contained gold, these sulfide minerals are decomposed through accelerated oxidation at high temperature and/or pressure. Roasting, POX, and BIOX are the proven technologies in the

pretreatment of refractory gold ores. The POX of sulfide concentrate from refractory gold ores in autoclaves produces residue containing various arsenates including scorodite, sulfoarsenates, sulfates, and As-bearing hematite and jarosite. Arsenates and sulfoarsenates include compounds referred to as type 1 and type 2 (Swash and Monhemius, 1994) and phase 3 and phase 4 (Dutrizac and Jambor, 2003). Phase 3 has a highly variable composition ($\text{Fe}_{0.9-1.3}\text{As}_{0.3-0.6}\text{S}_{0.4-0.7}\text{O}_4(\text{OH})_{0.3-3.3}$), and its As-rich end-member is similar to type 2 ($\text{Fe}_{1.2-1.3}\text{As}_{0.6-0.8}\text{S}_{0.2-0.4}\text{O}_4(\text{OH})_{2.6-3.5}$) in terms of both the long- and short-range structures (Paktunc, 2009). Spherical Fe sulfarsenates from the Red Lake autoclave reported by McCreadie et al. (1998) is similar in composition to phase 3. Jarosite can accommodate limited arsenate in the structure through substitution of sulfate (Dutrizac and Jambor, 2000; Paktunc and Dutrizac, 2003; Savage et al., 2000).

During roasting pretreatment of the refractory gold ores, the host sulfide minerals are progressively transformed to pyrrhotite, magnetite, and hematite (Swash and Ellis, 1986) and to maghemite and hematite (Paktunc et al., 2006) as the sulfur is driven off as SO_2 from the sulfide minerals. Arsenic that occurs in -1 valence state in arsenian pyrite and arsenopyrite is volatilized as As_2O_3 and in some cases oxidized to As_2O_5 under the highly oxidizing conditions of the roaster. Depending upon the roasting conditions and the efficiency of the process, some of this As is trapped in maghemite and hematite, which report to the cyanidation circuit for gold extraction (Paktunc et al., 2006). These high-temperature Fe oxide oxidation products typically appear as porous and concentrically layered grains. McCreadie et al. (2000) reported that hematite and maghemite occurring as porous grains measuring 20–50 μm across are the As-rich phases in the residue from the Red Lake roaster. These grains have inclusions of a variety of gangue minerals and also have a high-As content, which varies widely between and within grains but is not correlated with the type of iron oxide. No discrete As phases were identified on the surfaces of the grains, and the As is not part of the crystal structure. The As was presumably incorporated during the roasting process. Arsenic is released due to bacterially mediated reduction of As-rich hematite and maghemite in the roaster tailings at the Campbell Mine (McCreadie et al., 2000). Foster et al. (1998a) reported that arsenic occurs in the roasted sulfide ore of the Spencerville mine in California as As^{5+} species in jarosite and as sorbed complexes on hematite and ferric oxyhydroxide grains. Micro x-ray absorption near edge structure studies indicated that arsenic occurs as both As^{3+} and As^{5+} species in maghemite and hematite (Paktunc, 2008). Walker et al. (2005) reported similar speciation data on the Fe oxide particles in the tailings of the Giant Gold Mine roaster. It appears that As^{3+} species are preferentially enriched in maghemite (Paktunc, 2008, 2009). Although modern roasters allow effective capture of As from off-gases as As_2O_3 (arsenolite) through condensation and collection in bags, complete removal of As from roaster circuits remains challenging, and there are major environmental issues with the disposal of pure As_2O_3 wastes stored in bags. For example, more than 100 000 tonnes of As_2O_3 are temporarily stored in the underground workings of the Giant Gold Mine at Yellowknife and permanent disposal is a major environmental concern.

Trivalent arsenic oxide species also occur as secondary oxidation products after arsenopyrite and arsenian pyrite in mine-drainage settings, although they are rare in comparison to the As^{5+} species. Reported occurrences of As^{3+} oxides in mine wastes and impacted soil/sediments include arsenolite (Ashley and Lottermoser, 1999), picroparmacolite [$\text{Ca}_4\text{Mg}(\text{AsO}_3\text{OH})_2(\text{AsO}_4)_2 \cdot 11\text{H}_2\text{O}$] (Juillot et al., 1999), and tooeileite [$\text{Fe}_6(\text{AsO}_3)_4(\text{SO}_4)(\text{OH})_4 \cdot 4\text{H}_2\text{O}$] (Morin et al., 2003).

11.5.5.7 Phosphates

Secondary phosphates occur rarely in acid-drainage settings. Pyromorphite [$\text{Pb}_5(\text{PO}_4)_3\text{Cl}$] has been identified in tailings (Morin et al., 1999) and in mining-related stream sediments (Hudson-Edwards et al., 1996). Brushite [$\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$] was reported by Dill et al. (2002) to be present in the oxidized tailings of the Matchless Mine, Namibia, but the absence of an adequate source of PO_4 in the primary mineral assemblage indicated that the anion was most likely derived from a flotation reagent.

11.5.5.8 Secondary Sulfides

The principal secondary sulfides that have been observed in oxidized mine wastes are marcasite [FeS_2] and covellite [CuS], which have distinctly different origins. The initial oxidation of pyrrhotite results in the solid-state diffusion of Fe toward grain boundaries at which the Fe is oxidized (Pratt and Nesbitt, 1997; Pratt et al., 1994a,b). Upon further Fe migration and the consequent enhanced S enrichment of the original pyrrhotite, replacement by marcasite is affected. Hence, pseudomorphism of the pyrrhotite by marcasite and other oxidation products, including native sulfur, is common if leaching rates are not rapid.

In contrast to the spatial restriction of marcasite to replacement of pyrrhotite, most covellite in mine wastes results from redeposition of solubilized Cu that is typically derived from primary chalcopyrite. Sorption of Cu on Fe oxyhydroxides is common, but redeposition as a sulfide occurs where reductive conditions are present. Such conditions seem to be available locally on a microscale in proximity to altering pyrrhotite, but on a broader scale, the formation of covellite is predominant at the interface between the oxidized and reduced zones of a waste body, thus emulating the supergene enrichment process that takes place in sulfide deposits, especially those of porphyry copper. In mine wastes, other Cu sulfides that resemble covellite in reflected light may also be present, but small grain size has impeded specific identification.

Occurrences of secondary pyrite with framboidal and massive morphologies were reported in the Elliot Lake uranium mine tailings (Paktunc and Davé, 2002). The geochemical conditions in the tailings that promote the formation of pyrite at depths of greater than 0.75 m are described as neutral to weakly alkaline and reducing assisted by microbial activity.

In permeable reactive barriers (PRBs) (described in Section 11.5.5.9), chalcopyrite, bornite [Cu_5FeS_4], greigite [$\text{Fe}^{2+}\text{Fe}_2^{3+}\text{S}_4$], and abundant framboidal pyrite have been observed as secondary precipitates within the reactive medium emplaced to treat waters laden with heavy metals and sulfate derived from oxidized tailings and sulfide concentrates

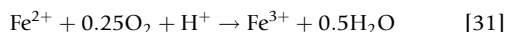
(Jambor et al., 2005). Organic carbon is added to the barrier mixture to promote sulfate reduction. The occurrence of mackinawite [Fe_9S_8] at one of the barriers was described by Herbert et al. (1998).

Nanometer-scale secondary sulfides associated with sulfide-reducing bacteria were reported by Fortin and Beveridge (1997) to have formed within the pyritiferous tailings at Kidd Creek, Timmins, Ontario. The sulfides were identified as mackinawite, amorphous FeS , and pyrite. A bacterially associated secondary Ag_2S phase, possibly acanthite, was observed by Davis (1997) in the Kidd Creek tailings.

11.5.5.9 Role of Microorganisms in the Formation and Dissolution of Secondary Minerals

Microorganisms have a major impact on the behavior and cycling of metals in the environment. Their role influences metal behavior through mediation of a variety of opposing mechanisms, including assimilation/adsorption and mineralization, precipitation and dissolution, oxidation and reduction, and methylation and dealkylation (Johnson, 2006). Sometimes, these processes are interrelated; for example, redox changes can result in the release of metal ions that are more or less soluble under prevailing environmental conditions, leading to their spontaneous precipitation or dissolution. Whereas the dissolution of pyrite and other sulfide minerals is central to the genesis of AMD, other microbially mediated processes can occur downstream at the point of discharge of water from a mine, leading to the formation of new (secondary) mineral phases, and these, in turn, may also be subjected to dissolution. While most of these processes involve redox transformations, other aspects of microbial metabolism (e.g., oxygenic photosynthesis by acidophilic and acid-tolerant microalgae, which results in increased alkalinity in mine waters) can also cause or contribute to the formation of secondary minerals.

At its point of discharge, AMD frequently contains little or no dissolved oxygen, and the dominant (or even exclusive) form of soluble iron present is Fe(II) (e.g., Rowe et al., 2007). Oxygenation of AMD streams by diffusion, mass transport, and oxygenic photosynthesis creates conditions favorable to the diverse species of bacteria and archaea that can obtain energy from the dissimilatory oxidation of ferrous iron. Ferrous iron oxidation is a proton-consuming reaction (eqn [31]), which causes, at least in the short term, the pH to increase



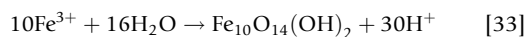
However, ferric iron is far less soluble under most physiological conditions than is ferrous and reacts with hydroxyl ions (i.e., the reaction is pH-dependent) to form a variety of amorphous (e.g., ferric hydroxide, eqn [32]) and mineral phases:



This reaction (referred to as iron 'hydrolysis') causes the pH to decrease, and because two protons are 'produced' for each iron atom, which oxidizes and hydrolyzes, the pH of AMD is often found to be lower downstream than at its point of discharge, as a result of these combined reactions.

In iron-rich mine waters, the dominant ferric iron mineral that forms is very much determined by the pH. In relatively

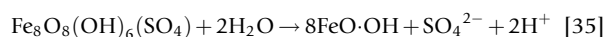
high-pH waters, ferrihydrite is a frequently encountered secondary ferric iron mineral (eqn [33]):



However, at lower pH (~ 2.3 – 4), schwertmannite is often the dominant secondary ferric precipitate found in AMD (eqn [34]):



Both of these are hydrous oxide minerals, but the formation of schwertmannite generates less acidity than does ferrihydrite (2.75 protons for each ferric ion precipitated, compared to 3) and also results in coprecipitation of sulfate, a useful attribute with regard to mine water remediation. However, schwertmannite is metastable with regard to goethite (Schwertmann and Cornell, 2000) and releases sulfate when it is transformed to the latter (eqn [35]):



Ferrihydrite is also a metastable mineral and transforms to goethite and/or hematite (Schwertmann and Cornell, 2000).

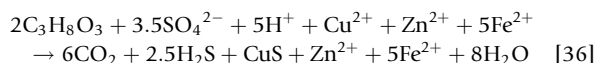
Below $\text{pH} \sim 2.3$, the solubility of ferric iron is greatly increased and extremely acidic, iron-rich mine waters can assume colors ranging from light orange to burgundy red. However, a third type of secondary ferric minerals can precipitate under such extreme conditions. Jarosites are a group of basic hydrous sulfate ferric iron minerals. Jarosite also is a principle waste product formed during metallurgical mill processing, where it can be confined to large repositories (Al et al., 1994; Dutrizac, 1999; Dutrizac and Jambor, 2000; Welham et al., 2000). Structurally, jarosite is a member of the jarosite–alunite group of minerals with a general formula of $\text{DG}_3(\text{TO}_4)_2(\text{OH}, \text{H}_2\text{O})_6$. The D sites can be occupied by monovalent (i.e., K, Na, Ag, Tl, and H_3O), divalent (i.e., Ca, Sr, Ba, and Pb), trivalent (Bi and rare earth element (REE)), and quadrivalent (Th and U) ions; G sites in this case can undergo solid-solution substitution with Al, Fe^{3+} , and T sites by S^{6+} , As $^{5+}$, or P $^{5+}$ with lesser amounts of Cr and/or Si (Stoffregen et al., 2000). Jarosites are most commonly encountered in low-pH, high-ferric iron/sulfate-containing liquors that also contain significant concentrations of basic metal cations, such as process waters at bio-mining sites as well as mine-waste sites and mine waters (e.g., at Iron Mountain, California; Nordstrom et al., 2000). Due to its low solubility in water, jarosite will scavenge metals, allowing it to serve as an excellent medium for trace-metal attenuation. Jarosite thus is important with respect to the fate and transport of metals in AMD environments (Accornero et al., 2005; Lee et al., 2005).

Some microorganisms appear to have a role in forming secondary ferric iron minerals in AMD beyond that of generating ferric iron (which is primarily a microbiological process in acidic ($\text{pH} \sim 4$) waters). Filaments of ferrihydrite, deposited as trailing stalks attached to bacterial cells, are produced by one of the longest known of all bacteria, *Gallionella ferruginea*, a neutrophilic bacterium not commonly encountered in AMD. However, there is evidence that the EPS produced in abundance by the acidophilic iron oxidizer, *Fv. myxofaciens*, can serve as a site of nucleation for schwertmannite precipitates. *Ferrovum*-like bacteria have been identified as the dominant bacteria present in a pilot-scale system used for remediating

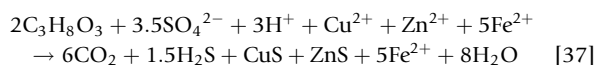
iron-rich groundwater in Germany, where the major precipitate formed is schwertmannite (Heinzel et al., 2009).

Metal sulfides can also form as secondary minerals in mine waters, though these are less frequently encountered than secondary ferric minerals. Many metals frequently found in elevated concentration in AMD from metal mines (e.g., copper and zinc) form highly insoluble sulfide phases that can precipitate (in the case of copper) even in extremely acidic mine waters. The prerequisite here is a source of sulfide (often denoted as S^{2-} , whereas H_2S is actually the dominant form below pH 6.8). Sulfide is produced as a waste product in several biological processes, the most important of which involve the dissimilatory reduction of sulfate, elemental sulfur, and polythionates by sulfate- and sulfur-reducing bacteria (Jameson et al., 2010). These bacteria are obligate or facultative anaerobes that only produce hydrogen sulfide in anoxic conditions, such as in sediments or within thick microbial biofilms and mats in mine streams, where sulfate, etc. (as opposed to oxygen), is used as a terminal electron acceptor. An electron donor (energy source) is also required, which can be organic (e.g., small molecular weight alcohols, such as ethanol) or inorganic (e.g., hydrogen) in nature. Since AMD streams are characterized by having very elevated concentrations of dissolved sulfate, there is great potential in these waters for bacterial sulfide formation. The availability of a suitable electron donor is usually the limiting factor for bacterial sulfidogenesis in AMD streams and sediments (mine waters are generally oligotrophic with regard to dissolved organic carbon), though colonization by acidophilic algae can lead to large inputs of organic carbon in mine waters, which in turn can support massive streamer/mat growths where sulfidogenic bacteria proliferate (e.g., Rowe et al., 2007).

Bacterial sulfate reduction is a proton-consuming reaction, whereas the formation of metal sulfides from the reaction between the dissolved metals and hydrogen sulfide generates protons. Whether the net reaction is pH-neutral or proton-consuming depends on the concentration of a metal(s) in the AMD that can form a sulfide phase under the prevailing pH conditions. Equations [36] and [37] show a hypothetical example for mine water sediment containing 1 mM of both Cu and Zn and 5 mM of Fe(II), where the electron donor is glycerol. At pH 2, only CuS forms (due to its much lower solubility product than those of ZnS and FeS), and the net reaction is alkali-generating (eqn [36]):



Such a scenario has been described in AMD draining a copper mine in Spain (Cantareras) where only CuS was identified in the anaerobic microbial mat growth, even though the mine water also contained appreciable concentrations of Zn and Fe (Rowe et al., 2007). At pH 4, both Zn and Cu sulfides form, but not FeS (even though there still may be free sulfide available), as its solubility product is too large. The net reaction is now less proton-consuming (eqn [37]):



The secondary minerals that form in AMD streams and sediments are themselves potentially subject to microbial

dissolution. Secondary sulfides, such as the primary sulfides, are stable as long as conditions remain anoxic, but when sediments are disturbed by high-streamflow rates, etc., and exposed to oxygen, they are susceptible to the same processes of microbial oxidative dissolution as described in Section 11.5.3.2.1 In contrast, secondary ferric iron minerals (jarosites, schwertmannite, and ferrihydrite) are subject to reductive dissolution in oxygen-depleted (or limiting) environments. Some species of bacteria and archaea are able to use ferric iron as an electron acceptor in anaerobic respiration (Lovley, 1993), and ferric iron reduction also can occur during fermentative metabolism. While dissimilatory ferric iron reduction has been most intensively studied with neutrophilic bacteria such as *Geobacter* spp., the ability to reduce iron appears to be particularly widespread among heterotrophic and autotrophic acidophiles (Coupland and Johnson, 2008; Johnson and Hallberg, 2009a,b). The reasons for this capacity are probably the elevated concentrations of iron that characterize AMD and related environments and also the greatly enhanced solubility of ferric iron at extremely low pH, making it more accessible as an electron acceptor.

Heterotrophic acidophilic bacteria, such as *Acidiphilium* spp. and *Ferrimicrobium acidiphilum*, use organic electron donors to fuel ferric iron reduction (Bridge and Johnson, 2000; Johnson et al., 2009), while autotrophic acidophiles, such as *Acidithiobacillus* spp., use reduced forms of sulfur and, occasionally, hydrogen (Hallberg et al., 2010; Ohmura et al., 2002). Mixotrophic acidophiles, such as *Sulfobacillus* spp. and *Am. ferrooxidans*, can use both organic and inorganic electron donors, and since these, such as *Acidithiobacillus* spp., are also iron oxidizers, they are capable of iron cycling when grown as pure cultures (Bridge and Johnson, 1998).

Bridge and Johnson (2000) compared rates of reductive dissolution of ferric iron minerals by *Acidiphilium* sp. SJH, using glycerol as an electron donor. They found that the iron mineral reduction rates followed the pattern amorphous $Fe(OH)_3 > \text{magnetite} > \text{goethite} = \text{natrojarosite} > \text{akageneite} > (\text{potassium}) \text{ jarosite}$ and that hematite was not reduced under the experimental conditions used. The rates of mineral dissolution were also faster at lower pH and that contact between the bacteria and the mineral phase was not necessary for reduction of the latter. They concluded that *Acidiphilium* sp. SJH caused the reductive dissolution of ferric iron minerals by an indirect mechanism in which bacterial reduction of soluble ferric iron caused a shift in equilibrium between the mineral phase and soluble ferric iron (which was affected by pH), thereby causing accelerated dissolution of the minerals. The optimum ferric iron reduction by *Acidiphilium* sp. SJH was observed under microaerobic rather than strictly anoxic conditions. Reductive dissolution of amorphous ferric hydroxide, goethite, and jarosite has also been reported for moderately thermophilic acidophiles (Bridge and Johnson, 1998), and recently, Hallberg et al. (2011a) have reported that reductive dissolution of goethite in a limonitic nickel laterite ore, using elemental sulfur as an electron donor, resulted in extraction of >80% of nickel, most of which was intimately associated with the goethite fraction.

In mine-impacted sediments, jarosite may decompose to form a variety of secondary mineral phases, for example, schwertmannite, iron oxides (usually goethite), and hydroxides

(Baron and Palmer, 1996; Gasharova et al., 2005; Smith et al., 2006; Welham et al., 2000). Depending on the geochemical conditions, these secondary minerals may also sequester metals such as Pb, Zn, Ni, Cr, As, and Ag either through adsorption or incorporation into their crystalline structure (Hudson-Edwards et al., 1999; Paktunc and Dutrizac, 2003; Sidenko and Sherriff, 2005).

Dissolution of jarosite (Bridge and Johnson, 2000) and schwertmannite (Coupland and Johnson, 2008; Gramp et al., 2009; Kusel et al., 1999) through the action of acidophilic iron and sulfate-reducing bacteria (SRB) under acidic conditions (pH < 2.5) has been demonstrated. The bacteria, in this case, accelerate the dissolution of these ferric iron hydroxysulfate minerals by an indirect mechanism, such that the bacterial reduction of the dissolved Fe(III) results in a shift in equilibrium between solid Fe(III) phases, promoting dissolution of the mineral. As noted earlier, soluble ferric iron derived from acid dissolution of the ferric iron minerals appears to be the form of iron used by iron-reducing acidophiles. In contrast, concentrations of dissolved Fe(III) in equilibrium with the solid would be too low to support such an indirect mechanism at circumneutral pH. However, dissimilatory metal-reducing bacteria have been shown to thrive under circumneutral conditions, although most commonly through direct mechanisms utilizing metal sites (i.e., iron) as a terminal electron acceptor from the mineral itself (Crowe et al., 2007; Hansel et al., 2003; Nevin and Loveley, 2002; Weisener et al., 2011). In this respect, many questions remain regarding the long-term stability of metal-substituted iron hydroxysulfate minerals under prolonged reducing conditions and perhaps moderate pH settings associated with AMD settings either proximal or downstream.

Jones et al. (2006) investigated the microbial susceptibility of K jarosite. In this study, the dissolution character was controlled by a function of temperature and particle size and differences in experimental conditions (media, organisms, number of cells, etc.). These observations appear to be the case for other metal-substituted jarosite minerals. Jones et al. (2006) and Weisener et al. (2008) both show similar trends in sulfate release rates during reductive dissolution compared to ferrous iron for two different jarosite compounds containing different D-substituted elements (i.e., K vs. Ag). The effect of these trace metals relative to iron and sulfate is often not clear, especially when the range of redox-sensitive metals (i.e., As, Cr, Se, Pb, Tl, and Ag) that can occupy lattice positions in these hydroxy sulfate phases is considered. In the latter case, Ag was observed to form intracellularly within bacteria in part due to its very low solubility, coupled with the potential for metal-reducing bacteria (i.e., *Shewanella putrefaciens* and *Geobacter metallireducens*) to use Ag along with ferric iron as a terminal electron acceptor. Similar trends have been observed for Pb-substituted iron hydroxy sulfate minerals (Smeaton et al., 2009). However, in this case, Pb was observed to precipitate within the cytoplasmic membrane, in contrast to the cell wall, of the same bacterial species. Smeaton et al. (2009) note that, within such systems, key genetic differences related to how specific organisms regulate toxicity may be a deciding factor. These differences would certainly have a strong influence on potential pathways that would regulate metals in these AMD environments.

Other investigations have focused on the biogeochemical cycling of solid-phase Fe and S mineral phases often observed

in these systems with the aim of understanding the oxidative and reductive pathways of hydroxysulfate end-members and their mineral counterparts (Gramp et al., 2009). The authors note that the reduction of Fe(III) hydroxysulfate occurs via solution-phase transformations, resulting in the dissolution and remobilization of minor trace metals occupying the D location. However, the precise initial mechanism associated with the biological reduction of sulfate is undetermined, but it is hypothesized that electron transfer in sulfate reducers to the surfaces of the mineral and the reduction of low concentrations of sulfate in the solution result in a shift in the equilibrium toward Fe(III) hydroxysulfate solubilization (Gramp et al., 2009).

In reduced aqueous environments, the presence of As in the solution can be a function of both biotic and abiotic mechanisms. The release of As to groundwater from anthropogenic sources is problematic in some areas proximal to mining operations (Williams, 2001) and can be strongly influenced by the microbial communities found within the mine wastes. The prevalent and notably most toxic As species in these systems are the inorganic forms: arsenate [As(V) as HAsO_4^{2-} or H_2AsO_4^-] and arsenite [As(III) as H_3AsO_3 or H_2AsO_3^-], which act to disrupt cellular processes by posing as an analogue of phosphate or by binding to sulfhydryl groups, respectively (Oremland and Stolz, 2003). The geochemical controls on the fate of As in mine environments are complicated by a number of interacting biotic and abiotic processes, and thus the limits on its mobilization are often difficult to isolate (Smedley and Kinniburgh, 2002). In general, As provenance is most strongly associated with that of Fe. Iron oxides and oxyhydroxides provide sorption sites for As (Bowell, 1994) along with many clay minerals (Violante and Pigna, 2002). Extensive studies of As sorption have demonstrated that pH and redox conditions are the major factors affecting As sorption onto mineral phases (Burton et al., 2009; Casiot et al., 2005). However, factors such as surface area, mineralogy, and the presence of competing ions also may control As sorption (Lim et al., 2008). Following the onset of reducing conditions, adsorbed and coordinated arsenate can be remobilized during the microbial Fe(III) to Fe(II) transformations of Fe oxides (e.g., Cummings et al., 1999). Furthermore, As-bearing phases, such as scorodite [$\text{FeAsO}_4 \cdot 2\text{H}_2\text{O}$], can be reduced by a variety of dissimilatory iron-reducing bacteria (Babechuk et al., 2009; Cummings et al., 1999; Newman et al., 1997; Papassiopi et al., 2003), leading to the release of Fe(II) and As(V). In some cases, dissimilatory release of As is thought to be a prerequisite for subsequent arsenate reduction (Cummings et al., 1999). The transformation of arsenic between pentavalent and trivalent states also can occur directly through microbial oxidation and reduction (Lee et al., 2010). Recent studies focusing on arsenic-rich mine tailings from the Ketza River mine, Canada, suggest that the stability of these mineral phases is dependent upon the increasing Fe/As ratio (Paktunc et al., 2003, 2004). Leaching tests performed by the authors have shown that the tailings material has significant As concentrations associated with Fe(III) oxyhydroxides. The adsorbed As associated with the Fe(III) oxyhydroxides was less soluble compared to other crystalline minerals such as arseniosiderite [$\text{Ca}_2\text{Fe}_3(\text{AsO}_4)_3\text{O}_2 \cdot 3\text{H}_2\text{O}$] (Paktunc et al., 2004).

The susceptibility of As-bearing Ca-Fe hydroxide minerals collected from tailings cores at the Ketza River mine (Yukon,

Canada) was investigated by Weisener et al. (2011). In this study, samples were collected from exposed and saturated tailings within the impoundment area. The principle concern was to establish the materials' behavior when exposed to reducing conditions in the presence of bacteria. The rates of Fe and As reduction using two strains of dissimilatory As- (*S. putrefaciens* ANA3) and Fe-reducing (*S. putrefaciens* 200R) bacteria were measured. Coatings containing Ca, Fe, and As in the tailings surfaces appeared to have an inhibitory effect on the bacteria slowing the rates of reduction for Fe and As. In part, this reduction in dissolution rate is likely due to surface area characteristics. However, when the carbonate coatings were removed through acid treatment and reinoculated with the bacteria strains, a significant increase in the rate of Fe and As reduction occurred. While, simultaneous reduction of both As(v) and Fe(III) was observed in this study in both cases, As(III) respiration rates showed a significant increase in the untreated material compared to Fe(II) and was likely due to favorable thermodynamic conditions utilizing different electron receptor half reactions, for example, As(III/v) reduction versus Fe(II/III).

11.5.6 AMD in Mines and Mine Wastes

11.5.6.1 Underground Workings

Access to deeply buried ore bodies is attained through underground mining. Workings are excavated to provide access to the ore body (Figure 4) and to mine the ore. These operations typically involve the removal of rock with no or little valuable metal content. Mining and economic limitations frequently result in large amounts of residual sulfide minerals being left adjacent to the excavated stopes. The extent of oxidation of these residual sulfides is primarily dependent on the surface area of exposed sulfide minerals and the duration of exposure

to oxygen (Morin and Hutt, 1997). After a mine is decommissioned, the weathering products of these exposed sulfides may become a source of acidity, sulfate, and dissolved metals.

Many underground mines are allowed to flood shortly after mining ceases, thus limiting the extent of sulfide-mineral oxidation. In areas where the water table is deep or where adits have been excavated to drain the mine workings, extensive oxidation of the wall rock may persist for years or decades after mining ends. The extent of sulfide oxidation and the quality of the water derived from a mine site are dependent on the duration of exposure to atmospheric O₂, the local geological conditions, and the methods of mining used.

Nordstrom and Alpers (1999b) and Nordstrom et al. (2000) described the generation of extremely acidic water at the Iron Mountain Mine near Redding, California, where a series of tunnels had been excavated to allow access to deep portions of the ore body. The excavation of drainage tunnels from these sites had the effect of exposing large volumes of sulfide minerals, principally pyrite with lesser amounts of chalcopyrite and sphalerite, to oxidation. Large accumulations of secondary sulfate solids, such as roemerite, rhomboclase, and Zn-Cu-bearing melanterite, have been observed as efflorescences on the mine walls and as stalagmites and stalactites (Nordstrom and Alpers, 1999b). Water dripping from the mine walls and in pools contained extremely high concentrations of dissolved solids (up to 760 000 mg l⁻¹ SO₄, 86 mg l⁻¹ Fe, and very high concentrations of other metals) and had a pH as low as -3.6 (Nordstrom et al., 2000; Table 8). During dry periods, secondary minerals form within the mine workings and the dissolved metals in pools become concentrated from evaporative effects. After rainfall events, flushing of the accumulated oxidation products results in effluents that have very high-metal concentrations. The concentrations of dissolved metals associated with the Iron Mountain site are much higher

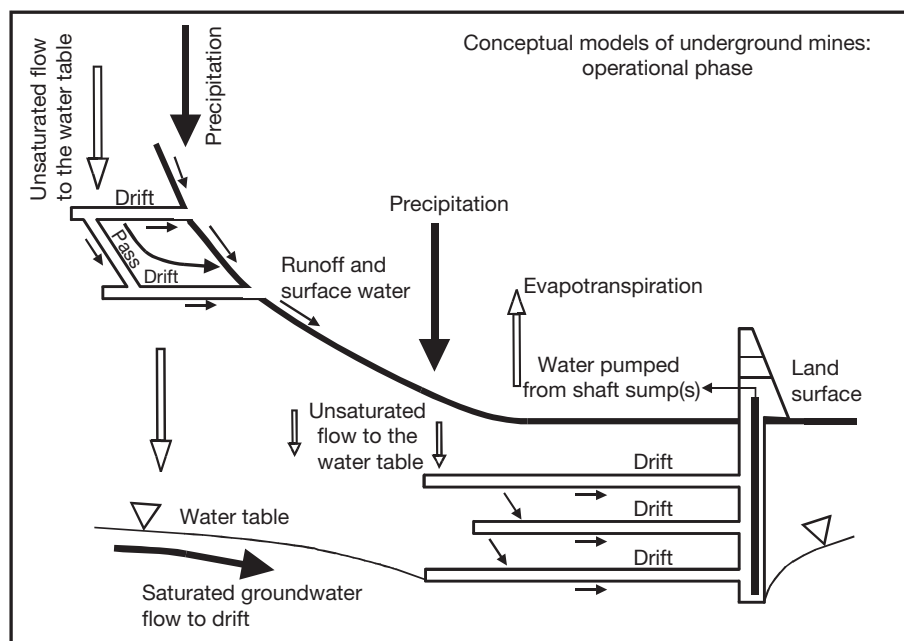


Figure 4 Schematic diagram showing underground mine workings. Modified from Morin KA and Hutt NM (1997) *Environmental Geochemistry of Minesite Drainage: Practical Theory and Case Studies*. Vancouver, BC: MDAG Publishing.

Table 8 Example of water chemistry observed in mine workings

	<i>Königstein Mine^a Groundwater in fourth aquifer</i>	<i>Königstein Mine^a Unsaturated unleached blocks</i>	<i>Königstein Mine^a Unsaturated leached blocks</i>	<i>Königstein Mine^a Flooded leached blocks</i>	<i>Richmond Mine Portal^b</i>	<i>Carlton Mine Tunnel^c</i>	<i>Roosevelt Mine Tunnel^d</i>
pH (SU)	5.99	1.88	2.92	1.88	0.5–1.0	7.81	7.69
E_h (mV)	307	747	807	651	–	–	–
TDS (mg l ⁻¹)	1736	12 322	3827	13 296	–	–	–
SO ₄ (mg l ⁻¹)	33	8220	2090	8800	20 000–108 000	1292	849
As (mg l ⁻¹)	–	–	–	–	34–59	–	–
Cd (mg l ⁻¹)	–	–	–	–	4–19	–	–
Cr _T (mg l ⁻¹)	<0.002	0.97	0.072	1.34	–	–	–
Cu (mg l ⁻¹)	–	–	–	–	120–650	–	–
Fe _T (mg l ⁻¹)	1.51	1171	15.32	1570	13 000–19 000	0.006	0.012
Pb (mg l ⁻¹)	0.013	2.1	0.010	1.43	–	–	–
Zn (mg l ⁻¹)	<0.01	132	24	164	700–2600	0.044	0.22
Ra (Bq kg ⁻¹)	104 ± 7	0.520 ± 0.047	0.0073 ± 0.0016	2.74 ± 0.24	–	–	–
Th (Bq kg ⁻¹)	n.a.	1333 ± 100	49 ± 6	1051 ± 95	–	–	–
U (mg l ⁻¹)	<0.02	12.3	18.1	50	–	–	–

^aKönigstein Mine, Germany (Biehler and Falck, 1999).^bRichmond Mine portal, California, United States (Nordstrom et al., 2000).^cCarlton Mine Tunnel, Colorado, United States (Eary et al., 2003).^dRoosevelt Mine Tunnel, Colorado, United States (Eary et al., 2003).

than those observed in drainage from other base metal and gold mines, such as the Carlton Mine and the Roosevelt Mine in Colorado (Table 8).

At the Wismut Königstein mine, near Dresden, Germany, acidic solutions were used in an in situ leaching procedure to extract uranium from a sandstone aquifer. During mining, zones of the mine were isolated, and acidic leaching solutions were percolated through blocks of aquifer material. The residual sulfuric acid was collected at the base of the block. The blocks were blasted to enhance permeability. In addition to the acid used in the leaching process, acidity was released by the oxidation of sulfide minerals contained in the sandstone (Biehler and Falck, 1999). The concentrations of dissolved constituents vary in the vicinity of the mine (Table 8). Low concentrations of sulfate and dissolved metals are observed in the groundwater. Higher concentrations are observed in the mined blocks, with the maximum concentrations observed in blocks that had been leached and flooded. Wismut plans to decommission the Königstein mine. The fate of dissolved metals and radionuclides in the flooding water is an important factor in the development of decommissioning plans (Bain et al., 2001).

The effects of Au mining on groundwater quality was assessed at the Cripple Creek Mining District, Cripple Creek, Colorado, using field measurements of water quality, mineralogical analyses, and geochemical modeling techniques (Eary et al., 2003). The underground workings were dewatered using a series of drainage adits. These adits have not been plugged, allowing the workings to continue to be exposed to atmospheric O₂. The Au ore occurs as disseminated native Au and is generally associated with pyrite. Oxidation reactions have produced low-pH conditions (pH < 3), with elevated concentrations of SO₄ (>2000 mg l⁻¹), Fe (>200 mg l⁻¹), and other dissolved metals in the shallow groundwater. Deeper drainage waters are near neutral in pH (> 7) and have low concentrations

of dissolved metals. The upper host rocks were depleted in carbonate minerals, suggesting the increase in pH was a result of carbonate-mineral dissolution reactions. The removal of Fe was attributed to the formation of ferrihydrite, Mn to the formation of Mn oxyhydroxides and rhodochrosite, and Zn to the formation of Zn silicates and/or Zn substitutions in calcite. The predictions suggest that the neutralizing capacity of the rock will result in continued improvements in water quality for the foreseeable future (Eary et al., 2003).

11.5.6.2 Open-Pits

Open-pits are excavated to extract ore from near-surface ore bodies. Many open-pits are adjacent or connected to underground workings. Open-pit mining has increased substantially over the past two decades as improved metallurgical techniques, and the increased scale of mechanical operations has made it possible to extract metals from low-grade ores (Miller et al., 1996). These improvements have made large-scale open-pit mining of low-grade Cu and Au ores economically viable. Open-pits are excavated through a series of benches that provide access to the ore and maintain the stability of the pit walls. In areas where the water table is high, it may be necessary, during the mining operation, to dewater the pits and surrounding materials by using pumping or diversion techniques (Figure 5).

The excavation of the open-pit and dewatering expose portions of the ore body and country rock, which were previously buried or submerged, to atmospheric oxygen. The oxidation of sulfide minerals within the wall rock results in the release of acidity, sulfate, dissolved metals, and other trace elements (Davis and Ashenberg, 1989). The extent of sulfide oxidation is dependent on the surface area of the exposed sulfide minerals and the duration of mining. In upper portions of the pit, where benches may be pushed back only occasionally, sulfide

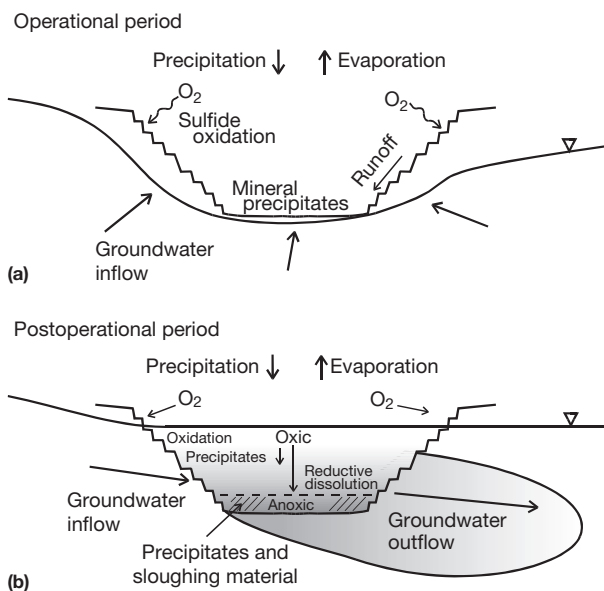


Figure 5 Schematic diagram of an open-pit mine during and after operation. Modified from Eary and Castendyk (1999).

minerals may oxidize for a prolonged period of time. The quantity of sulfide minerals exposed is dependent on their abundance and on the surface area of the pit walls. The pit-wall surface area is considerably greater than would be estimated from an ideal three-dimensional model because of irregularities in the pit walls, the presence of fractures and faults intersecting the pit walls, and the accumulation of rubble zones (Morin and Hutt, 1997). The reactive surface of a pit wall has been estimated to be 27–161 times larger than the ideal three-dimensional surface area (MEND, 1995). The total reactive rock-surface area for an open-pit can be large. For example, the reactive rock-surface area at the Island Copper Mine, near Port Hardy, British Columbia, Canada, was estimated to be 244 000 000 m², with a 161:1 ratio of reactive to modeled planar surfaces (MEND, 1995). Many pits are back-filled with sulfide-bearing waste rock, tailings, and other materials, some of which may have oxidized extensively prior to placement in the pit. The presence of these backfill materials may also contribute to the generation of low-quality effluent.

After mining is complete, pumping is terminated and the pits are allowed to flood. The principal components of water gain are precipitation, groundwater inflow, and surface-water inflow. The final elevation of the pit lake will depend on the local hydrological conditions and the geological properties of the surrounding rocks. The principal components of water loss are evaporation and groundwater and surface-water outflow. A water balance based on these principal components can be constructed to estimate the relative magnitude of each component and to assist in the prediction of pit-water quality (Eary and Castendyk, 2009). For the Spenceville Mine at Spenceville, California, Levy et al. (1997) estimated that the principal inflow components were surface runoff > precipitation ≥ groundwater inflow and the principal outflow components were direct evaporation > groundwater outflow as the pit reflooded. On this basis, it was concluded that 2 years had been required to reflood the pit. After the pit had filled, the

outflow of surface water commenced and the magnitude of the groundwater and surface-water outflow components increased. In contrast, the Berkeley Pit at Butte, Montana, filled over a period of three decades and is anticipated to continue to fill for several more years (Gammons and Duaiame, 2006), thus allowing prolonged exposure of sulfide minerals on the pit walls and the possible accumulation of secondary minerals. These secondary minerals may dissolve into precipitation runoff or into inflowing groundwater or during pit refilling. After the pit floods, further oxidation of sulfide minerals by atmospheric oxygen below the pit-lake surface will be limited because of the low solubility of O₂ in water. The sulfide minerals above the surface of the pit lake will continue to oxidize long after mining has ceased. In addition, oxidation of sulfide minerals with Fe(III) as an electron acceptor may continue within the flooded zone of the pit (Gammons and Duaiame, 2006).

The chemistry of pit lakes depends on a number of factors, among which are the geology of the pit walls and surrounding geological materials, the duration of sulfide oxidation, and the local hydrologic conditions (Table 9; Miller et al., 1996; Shevenell et al., 1999). Some pit lakes have near-neutral pH and contain low concentrations of dissolved constituents. These lakes may be suitable for recreational use (Shevenell et al., 1999). Other lakes, however, are acidic and contain high concentrations of dissolved metals and may be hazardous to waterfowl (Miller et al., 1996; Morin and Hutt, 1997). Miller et al. (1996) and Shevenell et al. (1999) compared the water chemistry of several pit lakes in the Western United States. Pit lakes associated with sulfide ores and carbonate-deficient rocks generated acidic water containing high concentrations of dissolved metals, whereas pits excavated in country rock containing abundant carbonates were neutral in pH and contained low concentrations of dissolved metals. Some elements, including As, Se, and Hg, were present at high concentrations under the neutral pH conditions that are predominant in carbonate-rich terrains. Eary and Castendyk (2009) summarized the typical host-rock composition, acid–base characteristics, and the expected pit-lake water quality associated with major ore-deposit types.

The relative depth, which is the ratio of the maximum pit depth to the mean diameter of the pit lake expressed as percentage, typically is greater than for natural lakes. The large relative depth typical of pit lakes, combined with the high concentrations of dissolved solids in some pit-lake waters, results in a stable stratification; hence, seasonal turnover and metal cycling are limited. For example, Ramstedt et al. (2003) observed a halocline, redoxcline, and thermocline at different depths in the Udden pit lake, northern Sweden. The limited mixing that persists in some pit lakes can result in the development of a permanently anoxic zone at the base of a lake. Sanchez-España et al. (2008) examined 22 mine pit lakes of the Iberian Pyrite Belt in Spain and noted that many of the lakes exhibited a thermal and chemical stratification with a well-defined chemocline, separating an anoxic Fe(II)-rich monimolimnion from a well-mixed, oxygenated upper layer. The stratification isolates the anoxic zone from the atmosphere, limiting Fe(II) oxidation and providing an environment suitable for anaerobic bacteria, including Fe-reducing bacteria and SRB (Sanchez-España et al., 2008). In lakes derived from metal mining, concentrations of organic carbon may be low enough to limit the extent of Fe and sulfate reduction, whereas open-pits derived from coal mines may have sufficient organic

Table 9 Example of water chemistry observed in existing pit lakes

Constituent	EPA drinking water standard	Berkeley Pit Butte, MT, 10/16/87	Robinson District		Yerington Pit Yerington, NV, 1991	Getchell Mine			Cortez Pit Cortez, NZ 1992–93
			Liberty Pit 1993 (0.5 m)	Kimbley Pit 1993 (0.5 m)		South Pit ^a 4/28/82	Center Pit ^a 4/28/92	North Pit ^a 4/28/92	
pH ^b	6.5–8.5 (s)	2.8	3.21	7.61	8.45	5.96	5.27	7.67	8.07
TDS ^c	100 (s)		6240	3580	631	2110	2140	2420	432
Cl	250 (s)	9	48.9	286	36	34.4	30.2	25.7	24.4
F	1.4–2.4		18.5	3.01	1.4	2.4	2.4	1.6	2.4
NO ₃ as N	10		<.04	<0.02	0.67	0.01	0.01	0.01	0.207
SO ₄	250 (s)	5740	3700	1800	270	1380	1410	1570	90.2
As	0.5	0.05	<0.005	<0.005	0.003	0.009	0.008	0.38	0.038
Ba	1		<0.002	<0.002	0.034				0.06
Cd	0.01	1.3	0.647	<0.005	<.001	<.005	<0.005	<0.005	
Cr	0.05		0.107	0.059	<0.005	<.02	<0.02	<0.02	
Cu	1 (s)	156	37.1	0.06	0.16	0.04	0.04	<0.005	
Fe	0.3 (s)	386	62.2	<0.05	0.01	0.8	2.1	0.16	0.134
Pb	0.05		<0.005	<0.005	<0.005	<.05	<0.05	<.05	0.0043
Mn	0.05 (s)	95	116	0.17	0.32	1.8	4.3	0.13	0.0017
Hg	0.002		<0.0002	<0.0002	<0.0005	<0.2	<0.2	<0.02	0.00046
Se	0.01		<0.005	<.005	0.13	<.002	<0.002	0.003	
Ag	0.05		0.022	0.021	<.01	<.005	<0.005	<0.005	
Zn	5 (s)	280	52.1	1.81	0.01	0.33	0.4	0.02	0.002
Ca		462	506	605	93	401	438	530	45.4
Mg		201	344	156	15	79.3	61.9	72.7	18.1
K		10	63	11.4	6.9	7.85	6.36	10.82	11.7
Na		72	53.6	95.3	76	56.3	40	38.1	68.6
Total alk.			0	189	24				

^aFiltered samples; composites of samples from various depths.^bAmounts are in mg l⁻¹; pH is given in pH units.^cTotal dissolved solids.Source: Eary LE (1996) Geochemical and equilibrium trends in mine pit lakes. *Applied Geochemistry* 14: 963–987.

carbon to promote extensive sulfate reduction (Pfeiffer et al., 2002; Ramstedt et al., 2003). In the oxic portion of the water column, oxidation of ferrous iron and precipitation of ferric oxyhydroxide solids may occur. These solids can scavenge trace metals and can settle through the water column. The upper, oxic portion of a lake may therefore contain lower concentrations of dissolved constituents than the deeper waters. The distribution of dissolved constituents in the pit will affect the quality of water discharged from the pit into the adjacent groundwater system and surface-water outflow. For example, very high concentrations of dissolved Zn have been observed in groundwater adjacent to the Brunswick No. 6 open-pit at Bathurst, New Brunswick (Morin and Hutt, 1997). Eary (1999) provides a summary of secondary minerals that may form in open-pits. Recently, the Acid Drainage Technology Initiative, metal mining sector (ADTI-MMS) developed a comprehensive guidebook for assessing the characteristics, fate, and remediation of pit lakes that result from metal and coal mining (Eary and Castendyk, 2009).

11.5.6.3 Waste-Rock Piles

Waste rock, the large volume of rock excavated to access the ore and low-grade ore that cannot be processed economically, is commonly deposited in large piles that are typically 30–100 m high but range up to 500 m in height and several square

kilometers in area (Figure 6). Waste-rock piles may be deposited in successive lifts or benches, or waste rock may be end dumped from the top of the pile, enhancing sorting and segregation according to particle size. The selection of a disposal technique will depend on the site conditions, economic considerations, and environmental policy.

Waste-rock materials vary in grain size from fine-grained sand and gravel-sized materials to large blocks up to several meters in diameter (Ritchie, 1994). The hydrology of waste-rock piles is an active area of research (Gelinas et al., 1992; MEND, 1994; Nichol et al., 2002; Ritchie, 1994). Measurements made on waste-rock piles and in laboratory columns suggest that the coarse nature of waste rock leads to a relatively large gas-filled porosity and high permeability. Ritchie (1994) summarized the characteristics of waste-rock piles (Table 10). This summary indicates that the travel time for vertical transport of water from the surface to the base of a 15 m high waste-rock pile is approximately 3 years. These calculations assume that water transport through the waste rock can be described in the same manner as water flow through an unsaturated soil. Recent studies of waste materials at a mine site in Northern Saskatchewan suggest that more than one flow regime may be present in waste-rock piles, with a portion of the water moving along rapid flow paths (Nichol et al., 2002). At many locations, waste-rock piles are constructed on permeable geological materials (Ritchie,

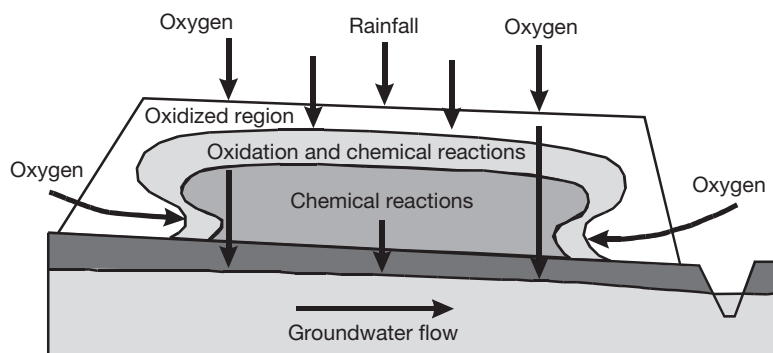


Figure 6 Schematic diagram of a waste-rock dump. Reproduced with permission from Ritchie AIM (1994) Sulfide oxidation mechanisms: Controls and rates of oxygen transport. In: Blowes DW and Jambor JL (eds.) *The Environmental Geochemistry of Sulfide Mine-Wastes*, vol. 22, pp. 201–246. Nepean, ON: Mineralogical Association of Canada.

Table 10 Physical properties of a typical dump of mine waste

Symbol	Definition	Value	Units
L	Dump height	15	m
A	Dump area	25	ha
ρ_r	Bulk density of dump material	1500	kg m^{-3}
ρ_{rs}	Sulfur density as pyrite	30 (2%)	kg m^{-3}
Q_w	Infiltration rate	0.5	m year^{-1}
ε_g	Porosity of the dumped material	0.4	
ε_w	Water-filled porosity at specified infiltration	0.1	
K_s	Saturated hydraulic conductivity of dump	10	m day^{-1}
D	Oxygen diffusion coefficient in the dump	5×10^{-6}	$\text{m}^2 \text{s}^{-1}$
C_o	Oxygen concentration in air	0.265	kg m^{-3}
ε	Mass of oxygen consumed per unit mass sulfur oxidized	1.75	
S^*	Mass of oxygen consumed per unit volume and unit time by the dump material	1×10^{-8}	$\text{kg m}^{-1} \text{s}^{-1}$
ρ_c	Carbonate density	0.6 (0.04%)	kg m^3

Source: Ritchie AIM (1994) Sulfide oxidation mechanisms: Controls and rates of oxygen transport. In: Blowes DW and Jambor JL (eds.) *The Environmental Geochemistry of Sulfide Mine-Wastes*, vol. 22, pp. 201–246. Mineralogical Association of Canada.

1994). At these locations, pore water affected by geochemical reactions within the waste-rock pile may be displaced into the underlying geological materials. Travel through the base of a waste-rock pile, 25 ha in area and undersaturated flow conditions, is anticipated to require 5 years or more (Ritchie, 1994). Under conditions that prevail in most waste-rock piles, the supply of oxygen limits the rate and extent of sulfide-mineral oxidation (Ritchie, 1994). Initially, the oxygen that is contained in a waste-rock pile upon deposition is consumed. This oxygen is gradually replenished by the oxygen from the surface of the pile via four gas-transport mechanisms: diffusion, advection, convection, and barometric pumping, with the last probably the least significant (Figure 6).

The rate of oxygen diffusion is proportional to the diffusivity of the waste-rock pile. Although the diffusivity of waste rock

is high because of the low moisture content of the waste rock, diffusive transport of oxygen is sufficiently slow that this process limits the rate of sulfide oxidation. Advective transport of oxygen results from changes in gas pressure between the waste-rock pile and the atmosphere. Wind has the potential to drive oxygen deeper into the pile than would occur under diffusive transport mechanisms alone (Amos et al., 2009; Ritchie, 1994; Figure 6). The oxidation of pyrite is exothermic, and the progressive accumulation of heat can result in high temperatures in a waste-rock pile undergoing intense oxidation. Temperatures in excess of 60 °C have been measured in oxidizing waste-rock piles. These increased temperatures induce the convective transport of atmospheric oxygen into the pile (Ritchie, 1994, 2003; Figure 7).

Convective transport of oxygen results in the penetration of oxygen deep into the waste-rock pile, accelerating the rate of oxidation of sulfide minerals and reinforcing the development of the convection cell (Cathles, 1979, 1994). The accelerated rate of oxidation near the pile margins both increases the short-term release of contaminants and decreases the duration of sulfide oxidation at those sites. Measurements of temperature and gas-phase oxygen concentrations in a waste-rock pile at the Rum Jungle mine site in Australia illustrate the regimes of gas transport within the pile (Harries and Ritchie, 1985). On the surface of the pile, away from the pile margins, the temperature remains relatively uniform and the concentration of oxygen in the pore gas decreases from atmospheric levels (20.9%) to <1% within 10 m. Gas transport in this region of the pile is dominated by diffusion. At its western margin, temperatures of up to 50 °C were observed within the pile. High O_2 concentrations coincide with these high temperatures, indicating that convective transport of gas is drawing oxygen-rich air from the margin of the pile to approximately 150 m into the interior of the pile. The convective transport of O_2 in this region accelerates the rate of sulfide oxidation at the pile margin relative to the pile surface.

Although seemingly dramatic in its effects, the convective transport of atmospheric gases is of relatively limited importance because a zone of only about 100 m inward from the pile margins is affected. Based on model calculations, Ritchie (1994) estimated that approximately 150 years would be required to oxidize all of the pyrite in a typical waste-rock pile containing 2 wt% pyrite. The duration of oxidation is longer if

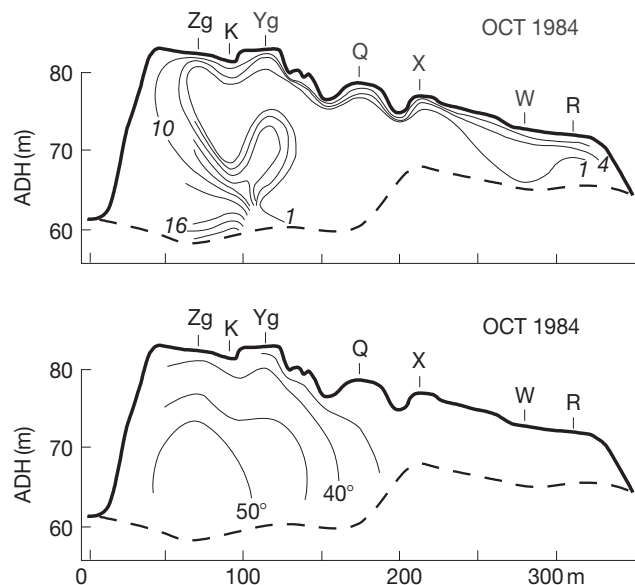


Figure 7 Measurements of temperature and oxygen concentration in the Rum Jungle Mine waste-rock pile. Reproduced from Harries JR and Ritchie AIM (1985) Pore gas composition in waste rock dumps undergoing pyritic oxidation. *Soil Science* 140: 143–152.

the pile has a greater sulfide content. The rapid oxidation of sulfide minerals in waste-rock piles can generate low-pH conditions and release very high concentrations of dissolved constituents to the pore water that flows through the waste-rock pile. Because of the difficulty associated with obtaining water samples from unsaturated waste rock, few measurements of water chemistry from within waste-rock piles are available. Measurements made within the unsaturated zone at the Mine Doyon waste-rock pile show low-pH waters containing very high concentrations of dissolved SO_4 , Fe, and Al (Table 11; Gelinas et al., 1992; MEND, 1994).

The transport and release of dissolved constituents from the oxidation zone are controlled by the physical hydrogeologic processes that control water flow through the unsaturated pile. These processes include diffusion, flow through the porous matrix, and preferential flow through more permeable pathways. The interaction between these flow and transport processes is complex because of waste pile heterogeneity (Nichol et al., 2005), structure, and geometry (Lamontagne et al., 1999; Lefebvre et al., 2001). Transient flow conditions can result in spatial variations in flow patterns and flow intensities, which lead to spatial variability in flushing rates and, thus, variability in drainage solute and metal loadings (Nichol et al., 2005; Velbel, 1993).

11.5.6.4 Coal-Mine Spoils

One of the most serious environmental concerns associated with coal mining is the production of AMD. Coal mining exposes sulfur-bearing minerals to atmospheric oxygen and water. Pyrite is the principal source of acid production in coal spoils (Rose and Cravotta, 1998). Concerns associated with acidic coal-mine drainage include sedimentation of chemical

Table 11 Concentrations of dissolved constituents in samples collected in lysimeters installed into the Mine Doyon waste-rock pile

Sample	92-1 L3A	92-1 L4A	92-1 L5A	92-2 L3A	92-2 L4A	92-2 L5A
Depth (m)	1.67	2.42	4.05	1.67	2.54	4.07
pH (SU)	6.81	6.81	6.97	1.77	2.03	1.9
Cond. (μS)	1242	1625	2315	21185	17588	22532
E_h (mV)	249	250	226	514	484	432
Fe_T (mg l^{-1})	0	0	0	16614	2878	7888
SO_4 (mg l^{-1})	629	1550	0	63029	43210	40750
Al (mg l^{-1})	—	—	—	2324	2412	2634

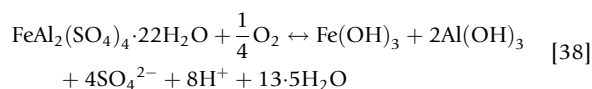
MEND (1994) Monitoring and modeling of acid mine drainage from waste rock dumps. *Natural Resources Canada MEND Report 1.14.2g*.

precipitates, soil erosion, and loss of aquatic habitats in contact with waters with high-metal loads (Williams et al., 2002). A bimodal distribution of coal-mine drainage has been observed, with acidic (pH 3–5) and near-neutral (pH 5–7) pH values (Brady et al., 1997). Figure 8 illustrates the range of pH values observed for both bituminous and anthracite coals of the Eastern United States. Acidic drainage associated with coal spoils develops where carbonate minerals, such as calcite and dolomite, or other calcareous strata that could neutralize acid production are insufficient to neutralize the acidity released by sulfide oxidation. Coal-mine effluents can range in composition from acidic to alkaline, depending on the host-rock geology. Effluents can have pH values as low as 2 and high concentrations of SO_4 , Fe, Mn, and Al, along with common elements such as Ca, Na, K, and Mg. The latter constituents are commonly elevated due to aggressive acidic dissolution of carbonate, oxide, and aluminosilicate minerals along flow paths that are down-gradient from the sources of oxidizing pyrite (Cravotta, 1994).

In cases where neutral or alkaline mine drainage predominates, problems may arise because of elevated concentrations of SO_4 , Fe, Mn, and other solutes that are derived from sulfide oxidation or from reactions with carbonate or aluminosilicate minerals. Dissolved Fe and Al may precipitate as the pH increases, and these precipitates can act as substrates for adsorption and coprecipitation (Brake et al., 2001; Foos, 1997; Stumm and Sulzberger, 1992). The dissolution of siderite, which is commonly associated with coal spoils, followed by the precipitation of Fe(III) oxyhydroxides generates no net alkalinity (Rose and Cravotta, 1998).

High-sulfate concentrations are not dependent on pH conditions and can pose a significant problem in both acidic and alkaline conditions (Rose and Cravotta, 1998). The formation of hydrous sulfate minerals in coal spoils can be significant sources of 'stored acidity' (Alpers et al., 1994). This stored acidity can be released when the soluble secondary minerals are dissolved during periods of recharge or runoff and when Fe or Al components of the minerals undergo hydrolysis (Rose and Cravotta, 1998). An example of this effect is the dissolution of halotrichite and coquimbite, respectively:

Halotrichite



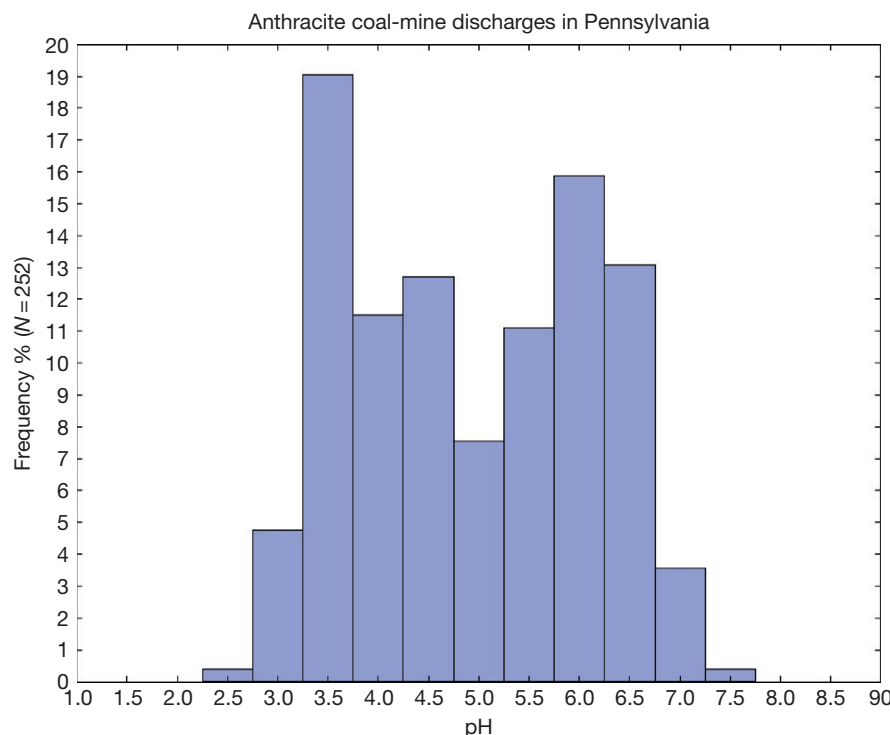
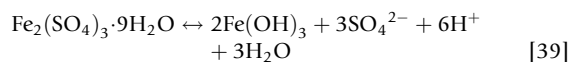


Figure 8 The bimodal distribution of pH in coal-mine drainage, where approximately half of the discharges from bituminous and anthracite coal mines are acidic with a pH less than 5 (<http://pa.water.usgs.gov/projects/amd/index.html>).

Coquimbite



The storage and release of acidity by these mechanisms can cause considerable temporal variability in water quality and can cause acid drainage to continue even after pyrite oxidation has ceased.

11.5.6.5 Tailings Impoundments

Mine tailings are the finely ground residue from ore extraction. The grain size of the tailings depends on the nature of the ore and the milling process. Size measurements (Robertson, 1994) of tailings from four mines in Ontario, Canada, indicated the tailings materials to be predominantly silt and fine to medium sand with <10% clay content. Tailings are transported from the mill and are discharged into an impoundment as a slurry containing ~30 wt% solids. The method of deposition affects the distribution of tailings particles within the impoundment. Discharge commonly takes place at elevated perimeter dams; hence, there is potential for extensive hydraulic sorting, with coarser fractions settling near the discharge point and the finer fractions settling in distal portions of the impoundment (Robertson, 1994). At some sites, tailings are thickened to >60 wt% solids prior to deposition. Thickening the tailings allows a more rapid settling of the solids, which therefore reduces the potential for hydraulic sorting, resulting in a more uniform grain-sized distribution than is observed in conventional tailings areas (Al, 1996; Robinsky, 1978).

During tailings disposal, water is continuously added to the impoundment, and the water table remains near the impoundment surface. After tailings deposition ceases, precipitation becomes the dominant source of recharge to the impoundment. The water table falls to an equilibrium position controlled by the rate of precipitation, the rate of evapotranspiration, and the hydraulic properties of the tailings and the underlying materials (Blowes and Jambor, 1990; Dubrovsky et al., 1984).

The fine-grain size of mine tailings results in a high moisture-retaining potential for these materials, which is a situation distinctly different from that in waste-rock piles. Whereas waste-rock piles commonly have a large open and free-draining porosity, mine tailings drain slowly, maintaining a large residual moisture content under gravity drainage. Measured moisture contents of conventional tailings impoundments vary from 10% to 100% saturation (Blowes, 1990; Smyth, 1981). The residual moisture content of thickened tailings is greater than that observed for conventional tailings (Al and Blowes, 1996; Robinsky et al., 1991). The high residual moisture content of mine tailings results in a low gas-filled porosity and in rapid changes in hydraulic gradient in response to precipitation (Al and Blowes, 1996; Blowes and Gillham, 1988).

Precipitation that falls on the impoundment surface migrates downward and laterally through the tailings impoundment into underlying geological materials (Figure 9). Groundwater velocities in tailings impoundments are relatively low. Coggans et al. (1998) estimated that the groundwater vertical velocity ranged from 0.2 to 1.0 m a⁻¹ at the Inco Ltd. Copper Cliff Central Tailings area in Sudbury, Ontario, whereas horizontal velocities were on the order of 10–15 m a⁻¹. At the Nickel Rim tailings impoundment, also

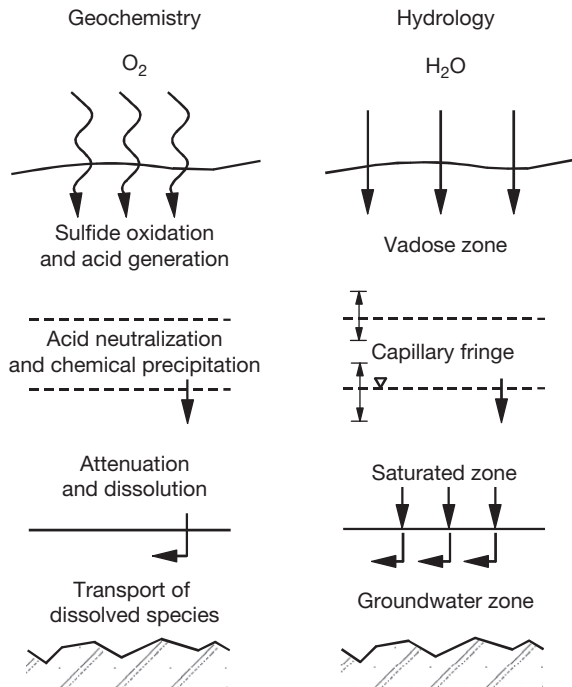


Figure 9 Schematic diagram showing the hydrology and geochemistry of a decommissioned mine-tailings impoundment. Reproduced with permission from Blowes DW and Ptacek CJ (1994) Acid-neutralization mechanisms in inactive mine tailings. In: Blowes DW and Jambor JL (eds.) *The Environmental Geochemistry of Sulfide Mine-Wastes*, vol. 22, pp. 271–292. Nepean, ON: Mineralogical Association of Canada.

near Sudbury, [Johnson et al. \(2000\)](#) estimated groundwater vertical velocities ranged from 0.1 to 0.5 m a⁻¹ and horizontal velocities ranged from 1 to 16 m a⁻¹.

The surface area of tailings impoundments varies from less than 10 ha to several square kilometers, and the thicknesses of the tailings deposits vary from a few meters to more than 50 m. The relatively low groundwater velocities and the large areal extent of tailings impoundments result in long time intervals between the time of groundwater infiltration and the time of groundwater discharge to an underlying aquifer or to the surface water environment ([Figure 10](#)). These long travel times result in the delay of measurable environmental degradation at the groundwater discharge point until long into the life of the impoundment. The severity of the negative environmental effects associated with tailings impoundments may not be evident until long after mine closure and decommissioning of the impoundments. The subsequent prevention and remediation of low-quality drainage waters are more difficult than during the active mining. The long travel distances and low groundwater velocities result not only in the potential for prolonged release of contaminants from the tailings impoundment but also in large long-term treatment costs. For example, [Coggans \(1992\)](#) combined estimates of the rate of sulfide oxidation with estimates of groundwater velocity at the Inco Copper Cliff Central Tailings area in Sudbury, Ontario, and predicted (1) that the peak release of sulfide-oxidation products will occur approximately 50 years after the impoundment is decommissioned and (2) that high concentrations of oxidation products will persist for approximately 400 years thereafter.

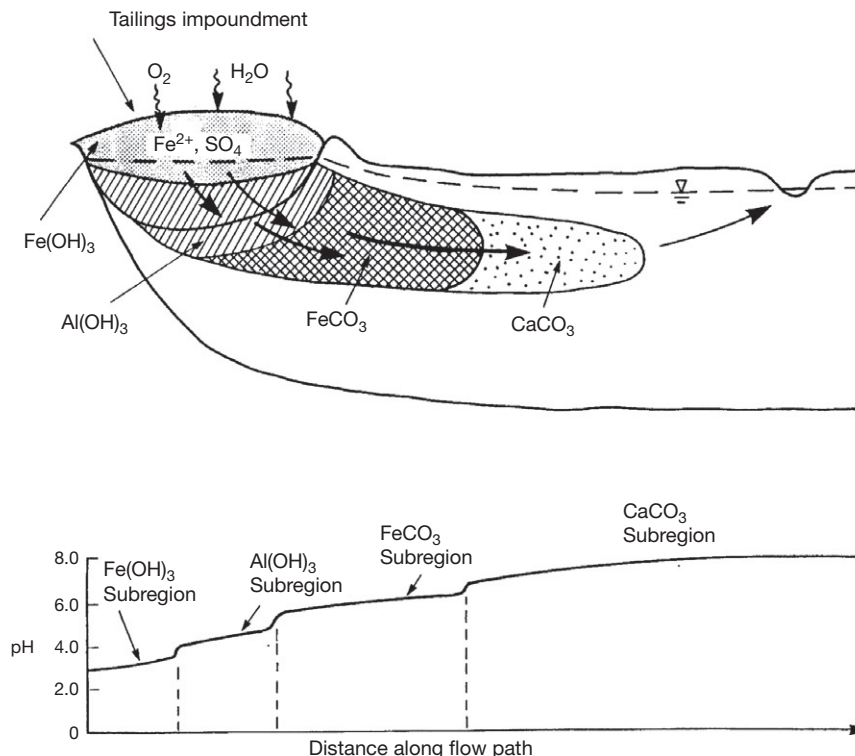


Figure 10 Schematic diagram of tailings impoundment and underlying aquifer, and pH-buffering regions. Reproduced with permission from Blowes DW and Ptacek CJ (1994) Acid-neutralization mechanisms in inactive mine tailings. In: Blowes DW and Jambor JL (eds.) *The Environmental Geochemistry of Sulfide Mine-Wastes*, vol. 22, pp. 271–292. Nepean, ON: Mineralogical Association of Canada.

In most tailings impoundments, gaseous diffusion is the most significant mechanism for oxygen transport. The rate of diffusion of oxygen gas is dependent on the concentration gradient and the diffusion coefficient of the tailings material. The diffusion coefficient of tailings is dependent on the air-filled porosity of the tailings; the coefficient increases as the air-filled porosity increases, and the coefficient decreases as the moisture content increases. Several empirical relationships have been developed to describe the dependence of the gas diffusion coefficient on the tailings moisture content (e.g., Reardon and Moddle, 1985). These relationships indicate a maximum diffusion coefficient at low moisture contents, with a gradual decline in diffusion coefficient as moisture content increases to about 70% saturation, followed by a more rapid decline as the moisture content increases further. The relationship between moisture content and diffusion coefficient results in rapid oxygen diffusion in the shallow portion of the vadose zone of a tailings impoundment, where the moisture content is low. The rapid diffusion of oxygen in this zone replenishes oxygen consumed by the oxidation of sulfide minerals. As the sulfide minerals in the shallow portion of the tailings are depleted, the rate of sulfide oxidation decreases due to the longer diffusion distance and the higher moisture content of the deeper tailings.

In many tailings impoundments, a variety of sulfide minerals is present. Jambor (1994) reported a general sequence of sulfide-mineral reactivity observed in several tailings impoundments, from the most readily attacked to the most resistant, to be pyrrhotite > galena–sphalerite > pyrite–arsenopyrite > chalcopyrite > magnetite. Blowes and Jambor (1990) observed systematic variations in sulfide-mineral alteration versus depth at the Waite Amulet tailings impoundment, Rouyn-Noranda, Quebec. On the basis of the observations, the degree of alteration was classified into a numerical scale as shown in Table 12. The sulfide alteration index indicates the relative degree of alteration of sulfides. Because pyrrhotite is the sulfide mineral most susceptible to alteration, the extent of its replacement forms the basis for the alteration index. When plotted versus depth on a vertical axis, the alteration index correlated well with geochemical parameters measured in

adjacent drill holes and with gas-phase O_2 and CO_2 concentrations (Figure 11; Blowes and Jambor, 1990).

The microbially mediated oxidation of sulfide minerals within mine-tailings impoundments generates acidic conditions and releases high concentrations of dissolved metals (Table 12). Mill tailings at the Heath Steele Mines in New Brunswick contain up to 85 wt% sulfide minerals (Blowes et al., 1991; Boorman and Watson, 1976). Pore-water pH values as low as 1.0 and concentrations of dissolved SO_4 up to 85 000 $mg\ l^{-1}$ were observed in the shallow pore water of the tailings impoundment (Figure 12; Blowes et al., 1991). This water also contained up to 48 000 $mg\ l^{-1}$ Fe, 3690 $mg\ l^{-1}$ Zn, 70 $mg\ l^{-1}$ Cu, and 10 $mg\ l^{-1}$ Pb. The shallow pore waters at the Waite Amulet tailings impoundment in northwestern Quebec contain 21 000 $mg\ l^{-1}$ SO_4 , 9.5 $mg\ l^{-1}$ Fe, 490 $mg\ l^{-1}$ Zn, 140 $mg\ l^{-1}$ Cu, and 80 $mg\ l^{-1}$ Pb. The pH of this water varies from 2.5 to 3.5 (Blowes and Jambor, 1990). High concentrations of dissolved Zn (48 $mg\ l^{-1}$), Cu (30 $mg\ l^{-1}$), Ni (2.8 $mg\ l^{-1}$), and Co (1.5 $mg\ l^{-1}$) were observed in the shallow groundwater at the inactive Laver Cu mine in northern Sweden (Holmström et al., 1999). These low-pH conditions and high concentrations of dissolved metals occur within the shallowest portions of the tailings impoundment. As this water is displaced downward through the tailings or through adjacent aquifer materials, the pH gradually rises, and many metals are removed from the solution by precipitation, coprecipitation, or adsorption reactions. High concentrations of Fe(II) and SO_4 , however, move down through the tailings and aquifer sediments relatively unattenuated (Dubrovsky et al., 1984; Johnson et al., 2000; Moncur et al., 2005). As this groundwater discharges from the tailings impoundment, Fe(II) oxidizes and precipitates as ferric (oxy)hydroxide and ferric hydroxysulfate minerals. These reactions release H^+ , generating acidic conditions within surface waters. The transport of Fe(II) along the groundwater flow path, therefore, provides the vehicle for transporting acidity long distances from the oxidation zone to the surface-water flow system.

Table 12 Alteration-index criteria for sulfides in oxidized tailings

Index	Alteration
10	Almost complete oxidation of sulfides; traces of chalcopyrite $\pm$ pyrite
9	Only sparse pyrite and chalcopyrite; no pyrrhotite or sphalerite
8	Pyrite and chalcopyrite are common, but chalcopyrite proportion is higher than normal possibly because of pyrite dissolution; no pyrrhotite or sphalerite
7	Pyrite and chalcopyrite proportions are normal; pyrrhotite is absent, but sparse sphalerite is present
6	Pyrrhotite is absent but is sphalerite common
5	Pyrrhotite is represented by marcasite pseudomorphs
4	First appearance of pyrrhotite but only as remnant cores
3	Cores of pyrrhotite are abundant
2	Well-developed cores of pyrrhotite are with narrower alteration rims; replacement by marcasite is decreasing, while pseudomorphs are absent
1	Alteration restricted to narrow rims on pyrrhotite

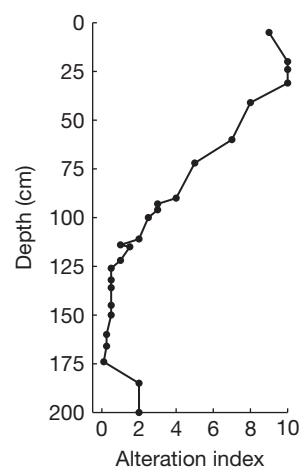


Figure 11 Sulfide alteration index (SAI) for oxidized tailings at the Sherridon mine, Manitoba. Reproduced from Moncur MC, Ptacek CJ, Blowes DW, and Jambor JL (2005) Release, transport and attenuation of metals from an old tailings impoundment. *Applied Geochemistry* 20: 639–659.

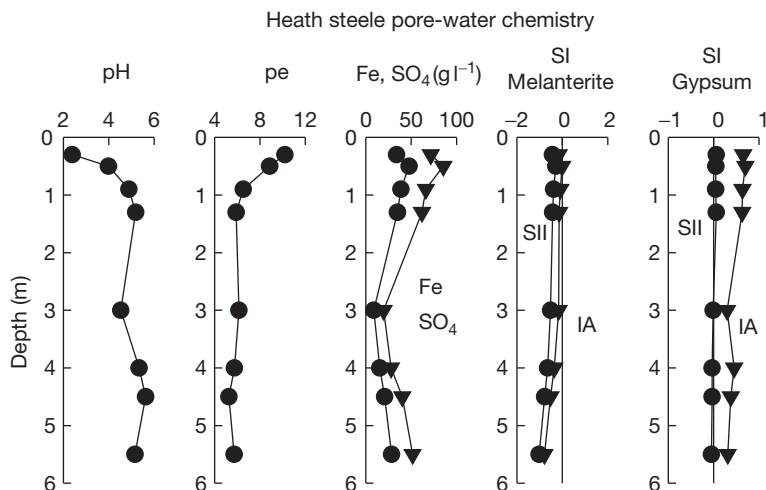


Figure 12 Pore-water chemistry and saturation indices versus depth at the tailings site of the Heath Steele Mines (reproduced from Blowes DW, Reardon EJ, Cherry JA, and Jambor JL (1991) The formation and potential importance of cemented layers in inactive sulfide mine tailings. *Geochimica et Cosmochimica Acta* 55: 965–978). IA represents saturation indices calculated using an ion-association model and SII represents saturation indices calculated using a specific ion-interaction model.

11.5.7 Bioaccumulation and Toxicity of Oxidation Products

Metals released from mines and mine sites, and in some cases from the natural weathering of ore deposits, can harm aquatic biota in adjacent water bodies (Borgmann et al., 2001). Metals are present in aqueous environments in a variety of species, some of which are bioavailable with various toxicities and potential for bioaccumulation. Transformations among these species depend on the physical and chemical characteristics of the water body. Metal toxicity can be acute or chronic. For example, Al can coordinate or precipitate to form species that result in sudden fish kills in waters near mine sites. Other metals, such as Hg, can bioaccumulate, leading to chronic toxicity (Domagalski, 2001). The extent of bioaccumulation and the toxicity of metals within natural environments are controlled by a number of factors, among which are the pH, oxidation–reduction potential, organic carbon content, concentrations and compositions of other dissolved species, and composition of the sediment (Warren and Haak, 2001).

11.5.7.1 Uptake and Bioaccumulation

Bioaccumulation and biomagnification are two terms commonly used for metal toxicity. Bioaccumulation refers to how pollutants (metals) enter a food chain and relates to the accumulation of contaminants, in biological tissues by aquatic organisms, from sources such as water, food, and particles of suspended sediment (Wang and Fisher, 1999). Bioaccumulation involves, relative to the ambient value, an increased concentration of a metal in a biological organism over time. Accumulation in living things can occur whenever metals are taken up and stored faster than they are metabolized or excreted (Markich et al., 2001). Understanding the dynamic processes of bioaccumulation can have important ramifications in protecting human beings and other organisms from

the adverse effects of metal exposure, and hence, bioaccumulation is an important consideration in the regulation and treatment of metals associated with AMD.

Among the terms that are important in conjunction with bioaccumulation are uptake, bioavailability, bioconcentration, and biomagnification. Uptake describes the entrance of a chemical into an organism such as by breathing, swallowing, or absorbing it through the skin without regard to subsequent storage, metabolism, and excretion. Bioavailability refers to the availability of a compound to cross an organism's cellular membrane from the medium the organism inhabits at a given time (Semple et al., 2004). Bioconcentration is the specific bioaccumulation process by which the concentration of a chemical in an organism becomes higher than its concentration in the air or water around the organism. Although the process is the same for natural and anthropogenic chemicals, the term bioconcentration usually refers to chemicals foreign to the organism. For fish and other aquatic animals, bioconcentration after uptake through the gills or, in some circumstances, through the skin is usually the most important bioaccumulation process. Biomagnification refers to the tendency of pollutants to concentrate as they move from one trophic level to the next. The process occurs when a chemical or metal becomes increasingly concentrated as it moves up through a food chain, that is, the dietary linkages between single-celled plants and increasingly larger animal species. The natural bioaccumulation process is essential for the growth and nurturing of organisms (Heikens et al., 2001). Bioaccumulation of substances to harmful levels, however, may also occur.

Acid and alkaline mine waters commonly contain high concentrations of dissolved metals and metal-oxide particulates. The acidification of wetlands can elevate the concentrations of metals and increase the potential bioavailability in aquatic plants and freshwater biota (Albers and Camardese, 1993) and can influence the uptake of metals in both submerged and rooted plants (Sparkling and Lowe, 1998). Arsenic concentrations in freshwater macrophytes affected by effluents

from a Au mine were examined by Dushenko et al. (1995). Macrophytes concentrated As relative to sediment concentrations, with submerged species containing much higher levels of As than those in air-exposed plants. The differences observed were attributed to growth form and the ability of plants to exclude As with increasing sediment concentrations. Plants in the vicinity of high-As values showed clear indications of necrosis of leaf tips and reduced micronutrient levels of Cu, Mn, and Zn in root tissues.

A study of As contamination in wood mice proximal to abandoned mine sites has shown that the extent of accumulation depends on the level of habitat contamination (Erry et al., 2000). Rai et al. (2002) observed that seeds appreciably bioconcentrate toxic metals, in the order $Pb > Cr > Cu > Cd$. The concentrations of these metals correlated positively with metal concentrations in adjacent water and sediments, which had been impacted by domestic and industrial discharges.

11.5.7.2 Toxicity of Oxidation Products

Metals such as Fe, Cu, Cd, Cr, Pb, Hg, Se, and Ni can produce reactive oxygen species, resulting in lipid peroxidation, DNA damage, depletion of sulfhydryls, and Ca homeostasis (Stohs and Bagchi, 1995). The inherent toxicities produced by the oxidation of these metals generally involve symptoms of neurotoxicity and hepatotoxicity. Metal reactions can be influenced by oxidation–reduction reactions, which often occur in aqueous environments impacted by mine-waste effluents. Species that contain more than one oxidation state in natural waters are inherently more mobile, reactive, and will exhibit differences in toxicity (Ahmann et al., 1997; Brown et al., 1999a,b; Ledin and Pedersen, 1996; Lin, 1997; Nordstrom and Alpers, 1999a; Warren and Haak, 2001). Depending on the metal concentrations entering the environment, most oxidation products in excess of natural requirements can produce toxic responses to aquatic biota.

Iron is an essential element for metabolic systems, but in Fe-rich solutions, toxicity can develop in both fish and biota. Iron toxicity has occurred in aquatic plants exposed to Fe-rich groundwater (Lucassen et al., 2000). Iron species can also affect the gill performance in fish, causing acute toxicity and accumulation on the gills (Dalzell and MacFarlane, 1999). In mine-waste discharge, Fe(III) sulfate and oxyhydroxide precipitates can accumulate on the gill epithelium, resulting in clogging and damage, decreasing the available surface area, and increasing the diffusion distance for respiratory exchange (Dalzell and MacFarlane, 1999).

Arsenic in aquatic environments is usually more concentrated in sediments and pore water than in the overlying water column (Ahmann et al., 1997; Smedley and Kinniburgh, 2002; Williams, 2001). The most abundant forms of As are arsenate [As(V)] and arsenite [As(III)], but methylated forms can occur in mine-impacted environments (i.e., methylarsenic acid and dimethylarsenic acid) (Smedley and Kinniburgh, 2002). The principal pathway of As toxicity is through dietary exposure to sediment and suspended particulates by fish, followed by human consumption. Environmental exposure to As is a causal factor in human carcinogenic and other related health issues. Chronic exposure symptoms in humans include hyperkeratosis, hyperpigmentation, skin malignancies, and peripheral

arteriosclerosis. Water provides the dominant pathway for As exposure in humans (Williams, 2001).

Mercury speciation is dependent on the available oxygen, pH, and dominant redox conditions, which often are site specific. Mercury can be present as either elemental Hg or mercuric phases (i.e., Hg^{2+} , HgS, and $HgCl_2$) associated with reduced anoxic environments. Under these conditions, Hg is considered to be relatively insoluble and is less toxic to biota. Under more oxidative conditions, such as those associated with roasting by-products (calcines) from separation procedures, Hg can form soluble sulfates and oxychlorides. Further reduction by SRB can cause the inadvertent methylation of the dissolved Hg (Rytuba, 2000). The methylation of Hg and adsorption of Hg and methylmercury onto Fe oxyhydroxides are important processes that control the fate and transport of Hg species in waters impacted by Hg-containing mine drainage. The primary mechanisms controlling the accumulation of methylmercury and inorganic Hg in aquatic food chains are not sufficiently understood, but it is speculated that bacteria in anoxic sedimentary environments associated with the reduction of SO_4^{2-} and S_2^{2-} are responsible (Domagalski, 2001; Rytuba, 2000). The formation of methylmercury within sediments and suspended particulate matter has the potential to increase bioaccumulation across all trophic levels, resulting in biomagnification up the food web. Mercury concentrations in fish are ultimately determined by methylmercury accumulation at the bottom of the food chain, which is governed by water chemistry, primarily pH and chloride concentration (Mason et al., 1996).

Accumulation of methylmercury in fish is a consequence of the greater trophic transfer efficiency of methylmercury than of inorganic Hg. For example, methylmercury concentrations in phytoplankton accumulate in the cell cytoplasm, and assimilation by zooplankton is four times more efficient than occurs for inorganic Hg, which is bound in cellular membranes (Mason et al., 1996). The toxicity of methylmercury is high because of its increased stability and its affinity for lipid-based compounds and because its ionic properties lead to an increased ability to penetrate the membranes of living organisms. Because methylmercury is lipid-soluble, it can penetrate the blood–brain barrier. This penetration can affect the central nervous systems of most vertebrates by concentrating in the cerebellum and cerebral cortex, binding tightly to sulfhydryl groups. Developing fetuses are subject to risk exposure because methylmercury can cross the placental barrier (Domagalski, 2001; USEPA, 2000).

11.5.7.3 Assessment of Toxicity

11.5.7.3.1 Predictive models

Models are important tools for the prediction of metal toxicity in aquatic systems. The models relate metal toxicity to site-specific differences in the chemical composition of surface waters. For example, Di Toro et al. (2001) and Santore et al. (2001) summarized the biotic ligand model approach to account for compositional effects on the acute toxicity of metals to aquatic organisms. The model is based on the premise that mortality occurs when the metal biotic ligand complex reaches a critical concentration. The biotic ligand in fish is either known or suspected to be the Ca- or Na-channel proteins

within the gill surface that facilitate the ionic composition of the blood. The biotic ligand will therefore interact with metal cations in the solution. In natural systems, the amount of metal that binds to the gill surface is determined by the competition between the toxic metal and other metals such as Ca. The model is a generalization of free ion-activity model that relates toxicity to the concentration of the divalent metal ions in the solution. The difference in this model is the presence of competitive binding at the biotic ligand, which accounts for the protective effects of other cations, and the direct influences from pH. The model is applied using the Windermere humic aqueous model (WHAM) (Tipping, 1994) to describe metal complexation to organic matter, in conjunction with normal chemical speciation models such as MINTEQA2 (Allison et al., 1990).

11.5.7.3.2 Biologic sensors

Metal contaminants can be monitored using biologic sensors. These sensors can be divided into two groups: active and passive. Active monitoring utilizes well-defined species under controlled conditions, whereas passive monitoring refers to direct observation or chemical analysis of indigenous plant and wildlife. Widespread health concerns have led to the adoption and development of a variety of methods for rapid toxicity assessment. These methods include biosensor devices that incorporate biological whole cells on electrode substrates (e.g., cyanobacteria, microalgae, and fish cells) and substrate monitoring (e.g., bivalves, fish parasites, plants, and mosses).

Natural organisms can provide information pertaining to the chemical state within an environment not through their presence or absence but through their ability to concentrate heavy metals within tissues. For example, sentinel organisms, which include bivalves, have been used to monitor the concentrations of bioavailable metals and toxicity in aquatic ecosystems (Byrne and O'Halloran, 2001; Hall et al., 2002; Lau et al., 1998). Bivalves have been used to monitor heavy-metal pollutants from Au-mine operations in Sarawak, Malaysia (Lau et al., 1998).

Plants and algae species have also been used as biosensors to detect high concentrations of metals from contaminated aquatic ecosystems. Long-term evaluation of Zn and Cd concentrations using two species of brown algae was conducted in Sepetiba Bay, Rio de Janeiro, Brazil (Amado Filho et al., 1999).

11.5.7.3.3 Molecular tools

In the last 10 years, there has been rapid advance in the application of molecular based techniques within the geologic science with respect to their use in discerning microbial structure and effect in natural and anthropogenic systems. Although there are abundant applications of molecular techniques to delineate environmental processes, this section concentrates on the emerging use of these techniques to investigate the mechanisms associated with AMD. The biogeochemistry associated with sulfide-rich waste-rock storage and tailings is complex as certain variables including heat generation from sulfide oxidation; pH gradients, oxygen depletion, and nutrient matter can influence microbial colonization. Overall, the anaerobic biogeochemical processes at work in waste-rock piles are poorly understood (Schipper, 2003). Recent studies have

shown that the microbial diversity observed in waste-rock storage sites typically consists of a variety of aerobic and anaerobic species such as autotrophic, heterotrophic, and/or lithotrophic bacteria. The latter have the ability to utilize inorganic compounds as an energy source. An extensive review of this genus has recently been compiled from a variety of mine sites around the globe (Schipper et al., 2010). It is the application of molecular based techniques that makes this possible. Some molecular methods used consist of variations of fluorescence in situ hybridization (FISH) and catalyzed reporter deposition-FISH (CARD-FISH), quantitative polymerase chain reaction (qPCR), denaturing gradient gel electrophoresis, and terminal restriction enzyme fragment length polymorphism. These molecular tools provide an opportunity to profile microbial ecophysiology and quantify individual bacteria groups (Edwards et al., 2003; Fike et al., 2008; Huang et al., 2007; Norland et al., 2009; Schipper, 2007). These tools also can be used to assess microbial community diversity and population. With all techniques there are inherent limitations and caution should be used when interpreting fingerprinting results. Some techniques will only profile more abundant ribotypes within a community but do not detect rare ribotypes.

Metagenomic sequencing of microbial communities in mine-impacted environments promises new perspectives of metabolic activities and pathways in these systems (Hutchens, 2009). Recent applications of these methods have begun to shed light on the complexity of mine-waste dumps. Based on DNA extraction and quantification (qPCR), it has been shown that 'bacteria' typically dominate over 'archaea' and 'Eukarya' with *Acidithiobacillus* spp. being the most abundant Fe(II) and sulfur oxidizers detected (Kock and Schipper, 2006, 2008). Furthermore, the authors report that *Leptospirillum*, an iron oxidizer, and *sulfobacillus* were detected either in low abundance or not at all. These observations can provide valuable insight into how researchers model the biogeochemistry of these systems. Recent work by Norland et al. (2009) has provided further key evidence to describe symbiotic relationships of microbial communities in AMD environments. Based on the application of FISH and high-resolution geochemical imaging using scanning transmission x-ray microscopy, they propose that a microbial cooperative is involved in sulfur cycling in these systems reminiscent of the structure observed for marine consortia. In this study, the authors note the formation of 'sulfur pod cooperative,' which mediate internal redox conditions that can differ from the bulk pore-water solutions. This study provides an interesting interpretation and holds promise for assessing future systems, which are controlling environmental gradients in these AMD sites.

11.5.8 Methods of Prediction

11.5.8.1 Laboratory Static Procedures

Static tests are intended to predict whether a sample, and the rock or soil that it represents, will be acid-producing after exposure to weathering. The distinguishing characteristic of a static test is that it is a one-time determination, whereas kinetic tests involve repeated cycles in which dosages of humidity or aqueous solutions are applied over a period of time. Thus, kinetic tests can provide information on weathering rates and the abundances of ions in the leachates, data not obtainable

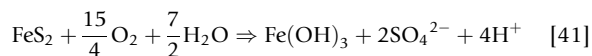
from a static test. Nevertheless, static tests are widely used because they have the advantages of being rapid and simple to perform, and commensurate with those advantages is a relatively low cost per determination. Hence, static tests are commonly performed on large numbers of samples for an individual project and for potentially exploitable mineral and coal deposits, the results are commonly used to provide guidance as to which rocks may merit further study, such as by kinetic tests. At active mines, static tests may be used to monitor the potential of various wastes, such as overburden and barren or sulfide-bearing low-grade rocks that host the ores, to generate acidic drainage. The results may be used to govern how and where those wastes are disposed.

Numerous types of static tests have been developed, and some of the laboratory procedures and the variety of tests or their individual development have been described in various publications, such as those by MEND (1991), Lawrence and Wang (1997), Morin and Hutt (1997), White et al. (1999), and Jambor (2003). By far, the most widely adopted static test, both for metalliferous and coal deposits, is the acid-base account (ABA) of Sobek et al. (1978). As is implied by its name, ABA involves a determination of the acid-producing potential (AP) of a sample and a determination of the base that is potentially releasable; the latter is generally referred to as the neutralization potential (NP). The two chemically determined values therefore provide a net accounting of the expected behavior during weathering. A common form of expressing the result is to obtain the net NP (NNP) by simple subtraction of the two chemically determined values:

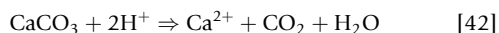
$$\text{NNP} = \text{NP} - \text{AP} \quad [40]$$

If $\text{NP} > \text{AP}$, the resulting value for NNP will be positive, thus indicating that the sample should have some acid-neutralizing capacity; the opposite, with $\text{AP} > \text{NP}$, is taken as an indication that the sample will be acid-generating.

The value for AP is obtained by measuring the sample's weight percentage of sulfur (or wt% S_{sulfide} in some jurisdictions). It is assumed that the S is present as pyrite [FeS_2] and that the pyrite will weather in accordance with the reaction:



Hence, each mole of S produces 2H^+ , or stated alternatively, each mole of pyrite produces two moles of H^+ . Most nonsulfide minerals will react with acid to some extent, and if the effect of the mineral dissolution is to decrease the acidity of the original solution, then the mineral contributes NP, the amount of which is relative to the acidity consumption effected by calcite in the reaction:



The 2H^+ produced per mole of S in eqn [41] can be neutralized by 1 mol of CaCO_3 , as in eqn [42]. Therefore, 1 mol of CaCO_3 is equivalent to 1 mol of S, and their approximate molecular weight values are 100 and 32, respectively; hence, $100 \div 32 = 3.125$, and 1 g of S is equivalent to 3.125 g of CaCO_3 . If S is in kg t^{-1} , which is a unit commonly employed, the AP of a sample is its wt% S multiplied by 31.25 to obtain a value expressed in kg t^{-1} of CaCO_3 equivalence. The NP of

calcite, because it is in g t^{-1} , is 1000. The NP of material is obtained by the addition of acid to a 2 g mass of a sample to determine how much of the acid it consumes (neutralizes). The step-by-step procedures and calculations are given by Sobek et al. (1978), MEND (1991), and Morin and Hutt (1997).

The relationship between the result from a static test and what may occur during weathering is strictly empirical. Many assumptions are involved in the derivation of the NP and AP values, and static tests are a surrogate for weathering, not an emulation of it. Nonetheless, those aspects aside, probably the chief arguments concerning interpretations of static-test results, have been focused on which value constitutes an environmentally 'safe' NNP, on which minerals will provide NP to prevent ARD rather than just mitigate it, and on which minerals contribute to static-test NP even though their effects on ARD are minimal except over longtime periods.

Part of the answer to the arguments has been the measurement of the NP values of specific minerals (Jambor et al., 2002) and the replacement, by some regulatory agencies, of the absolute value of NNP, adopting in its place the ratio of NP/AP for environmental assessments. Like other environmental parameters, the ratio deemed to be appropriate is different in various jurisdictions. Whether NNP values or NP/AP ratios are used, however, they are but one part of a more comprehensive, site-specific environmental evaluation that includes kinetic (dynamic) laboratory tests, commonly.

11.5.8.2 Mineralogical Prediction

Mineralogical factors such as the types, quantity and distribution of sulfide, carbonate and fast-dissolving silicate minerals, and their dissolution and oxidation rates are important in the prediction of AMD. Neutralization capacity and acid generation capacity of the samples can be determined from mineral quantities and the stoichiometry of neutralization and acid-generating reactions (Paktunc, 1999). This mineralogical approach provides an independent assessment of the static ABA test results, which are known to have limitations and uncertainties. Jambor et al. (2007) determined the NP potentials of a range of rocks by the Sobek method and compared the measured NP values with those calculated from the NP of individual minerals (Jambor et al., 2002) and mineral quantities. Although the measured values are always higher than those of the calculated NP values, there is a relationship controlled by the fast-dissolving silicate minerals such as Ca-rich plagioclase and olivine.

Knowledge of the oxidation and dissolution rates of the minerals present in the samples is important for assessing how fast the acidity is produced and whether or not the carbonate and silicate minerals present would dissolve fast enough to effectively neutralize the acidity. Kinetic considerations require that not only the types and quantity of the sulfide and carbonate minerals but also their distribution and availability within the sample need to be known.

The presence of more than one type or generation of pyrite possessing different morphological and chemical features in a deposit such as framboidal and massive types of pyrite and Au and As concentrations in pyrite (Paktunc et al., 2003) would require further considerations on the oxidative dissolution rates of pyrite with orders of magnitude differences.

Differences in the chemical composition of pyrite (Rimstidt and Vaughan, 2003) such as the stoichiometry and minor elements and the particle size would influence the oxidation rates and acid production rates.

The effectiveness of the neutralization capacity can be determined from assessing the particle size and mineral liberation properties. For instance, a study involving long-term column leaching experiments conducted for evaluating the oxidation and leaching characteristics of coarse pyritic uranium tailings with limestone amendments showed that the tailings amended with coarse-grained (6–7 mm) limestone produced acidic drainage after about 2 years of leaching despite the limestone amendment was stoichiometrically adequate for effective neutralization (Paktunc and Dave, 1999). On the contrary, the tailings amended with fine-grained ($-44\ \mu\text{m}$) limestone where there was no acidic drainage during 7.5 years of leaching indicating the importance of availability or effectiveness of the neutralization capacity.

Another important mineralogical feature to consider is the degree of liberation of pyrite or pyrrhotite in the samples. This is especially important for waste rocks but equally applicable to tailings where the sulfide particle size is smaller than average tailings particles. ABA tests inherently assume that sulfide represented by the bulk S concentrations is available. This is acceptable in the case where the sulfide minerals are fully liberated. If sulfide mineral particles are not fully liberated, that means not all of their surfaces are exposed and available for reaction with water and oxygen. The effect of this feature was demonstrated by column leaching studies carried out on three waste-rock samples containing similar quantities of pyrite and calcite but with differing amounts of liberated, locked, and encapsulated pyrite (Paktunc and Davé, 2000). The tailings with essentially liberated pyrite particles produced acidic drainage after about 4 years of leaching, whereas the other two tailings where pyrite particles also occurred as encapsulated and locked particles did not produce acidic drainage during the entire leaching period of 7.5 years.

11.5.8.3 Laboratory Dynamic Procedures

Laboratory dynamic tests are conducted by exposing small volumes of rock to repeated weathering cycles. These tests are commonly referred to as kinetic tests and are intended to assess the potential for acid generation and metal leaching under flow conditions that are designed to emulate field conditions. The objectives of dynamic testing programs vary from confirming the hypothesis developed through static testing to estimating the rates of sulfide oxidation and metal release and to assessing the potential for secondary mineral formation and metal attenuation. Various testing procedures have been developed (DIAND, 1992), among which are well-standardized ones (ASTM, 2001). The test apparatus ranges from humidity cells, which contain approximately 1 kg of sample (ASTM, 2001) to columns containing up to 100 kg of sample (Bennett et al., 1999). The operational steps of the tests vary from procedure to procedure, and the steps may dictate the rate, duration, and volume of sample irrigation, the rate and duration of sample aeration, and the frequency of collection of supernatant samples.

The interpretation of dynamic test data depends on the objective and design of the testing procedure. Although some tests are designed to estimate individual parameters, such as the rate of sulfide-mineral oxidation (Bennett et al., 1999), other tests are intended to provide estimates of the rate of release of dissolved metals and the rate of depletion of sulfide minerals and acid-neutralizing carbonate minerals (ASTM, 2001; DIAND, 1992). It is important that the test design address the objectives. Tests that accelerate the rate of water washing of the sample may not accelerate the rate of sulfide oxidation if that reaction is dependent on the exposed surface area and the extent of bacterial catalysis but independent of the rate of irrigation. Furthermore, many of the products of sulfide oxidation are sparingly soluble at the neutral or moderately acidic pH values that may be observed in the early stages of the test program. Care must be taken to incorporate the masses of these phases into sulfide-oxidation or metal-leaching calculations.

11.5.8.4 Geochemical Models

11.5.8.4.1 Geochemical modeling approaches

A summary of the computer models available to describe inorganic geochemistry in static batch systems was prepared by Alpers and Nordstrom (1998). This summary indicates that a diverse series of models was developed in the 1970s and 1980s, and by the late 1990s, this diversity had decreased to a smaller number of well-supported models. Figure 13 illustrates the current evolution of coupled chemical-hydrological-speciation modeling programs.

Geochemical speciation models and geochemical speciation mass-transfer models are used widely to describe water chemistry in mine-drainage environments. Most of these models are based on the ion-association theory (Garrels and Thompson, 1960) and use the extended Debye-Hückel model or derivative models, such as the Davies or WATEQ equation, for activity corrections (Nordstrom and Munoz, 1994). A variety of geochemical speciation mass-transfer models has been developed over the past decades (Figure 13; Allison et al., 1990; Alpers and Nordstrom, 1998; Ball and Nordstrom, 1987; Parkhurst, 1998; Parkhurst et al., 1985). These models have become highly sophisticated and are able to describe increasingly complex geochemical reactions. Activities derived from the speciation calculations are used to determine mineral saturation indices and predict the extent of mass transfer among aqueous, gaseous, and solid phases, over a range in temperature. Processes described by these models include geochemical speciation, acid-base equilibrium, redox equilibrium, precipitation/dissolution, and adsorption/desorption. The most recent versions of these models are able to incorporate descriptions of kinetically controlled reactions and descriptions of one-dimensional solute transport (e.g., Parkhurst, 1998).

Models based on the ion-pair approach are generally for application to dilute waters, such as those observed in relatively uncontaminated lakes or groundwater systems. Efforts to improve predictions of ion activities and mineral saturation indices for mine-drainage settings (e.g., Ball and Nordstrom, 1987; Nordstrom et al., 1990) have included the addition of extensions to the Debye-Hückel equation to cover a wider concentration range, the inclusion of a large number of

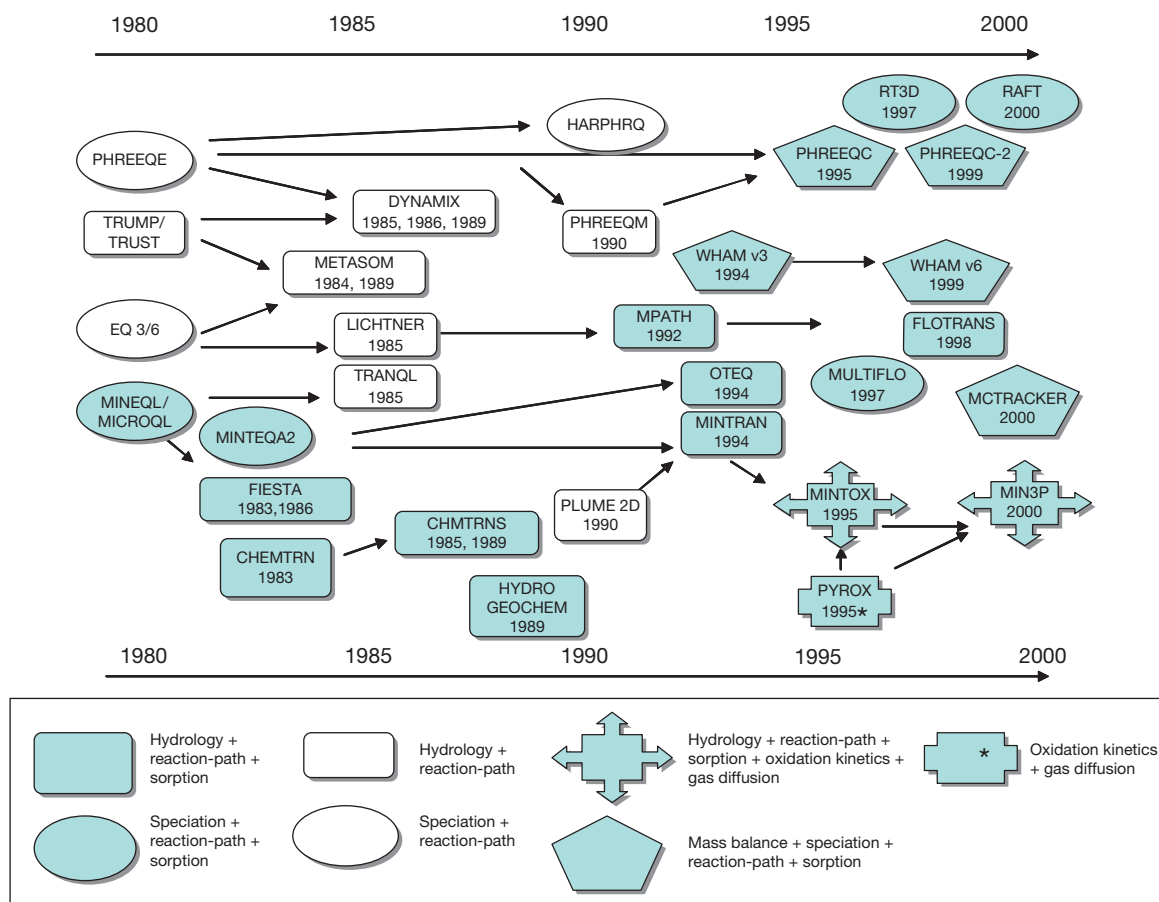


Figure 13 The evolution of coupled chemical-hydrological-speciation modeling programs. Modified from Alpers CN and Nordstrom DK (1998) Geochemical modeling of water-rock interactions in mining environments. In: Plumlee GS and Logsdon MJ (eds.) *The Environmental Geochemistry of Mineral Deposits. Reviews in Economic Geology*, vol. 6A, pp. 289–323. Littleton: Society of Economic Geologists.

ion-association constants and complexation constants specific to mine-drainage settings, and the addition of mineral-solubility constants for phases typically encountered in mine-drainage environments. As a result, the ion-association approach for predicting metal-speciation and mineral-stability fields is generally accurate for mine waters characterized by relatively high concentrations of dissolved solids (ionic strengths up to 1 m) and a relatively broad range in temperature (e.g., Ball and Nordstrom, 1987; Nordstrom et al., 1990).

11.5.8.4.2 Application of geochemical speciation mass-transfer models

Geochemical speciation and geochemical speciation mass-transfer models are used to assist in the interpretation of data collected in mines, deposits of mine wastes, and bodies of receiving water. These models can be used to infer the geochemical reactions that affect the concentrations of major ions and dissolved metals in mine waters and pore waters of mine wastes, to assess the stability of secondary minerals present in mine wastes, and to predict the concentrations of dissolved metals that can be anticipated in mine and mine-waste discharge. The application of these models can be constrained by other observations, particularly characterization of the primary and secondary mineralogy, by measurements of groundwater

and surface-water flow directions and velocity, by water-balance calculations, and by measurements of environmental isotopes (e.g., ^2H , ^3H , ^{18}O , and ^{34}S). Geochemical speciation calculations have been used to assist in the interpretation of water chemistry associated with underground mine workings (Eary et al., 2003), open-pits (Davis and Ashenberg, 1989; Eary, 1999; Ramstedt et al., 2003; Tempel et al., 2000), tailings impoundments (Blowes and Jambor, 1990; Blowes et al., 1992; Gunsinger et al., 2006b; Johnson et al., 2000; Moncur et al., 2005), rivers (Runkel and Kimball, 2002; Runkel et al., 1996), and lakes (Moncur et al., 2006).

Waters associated with mine sites may contain high concentrations of dissolved solids as a result of intense sulfide-oxidation and evaporative processes. Specialized models are required for calculation of ion activities and mineral saturation indices for sites characterized by these high concentrations. The most widely accepted approach for prediction of ion activities and mineral solubilities in complex concentrated electrolyte mixtures is based on the Pitzer ion-interaction formalism (Pitzer, 1973) as summarized in Pitzer (1991). Models based on the Pitzer formalism have been developed for application to various geochemical systems, from acidic to basic, dilute to concentrated, and over a range in temperature, total pressure, and partial pressures of component gases (Clegg and Whitfield,

1991; Pitzer, 1991). The first applications of Pitzer-based models to geochemical systems involved calculations of ion-activity and mineral-solubility relations in waters containing seawater components (e.g., Clegg and Whitfield, 1991; Plummer et al., 1988; Weare, 1987). These efforts have been expanded to include development of ion-interaction-based models for a wide range of chemical compositions, including those derived from sulfide-oxidation processes (Baes et al., 1993; Ptacek and Blowes, 2000; Reardon and Beckie, 1987). Mine drainage containing high concentrations of dissolved metals, sulfate, and acid has been reported for sites in Canada (in excess of 300 000 mg l⁻¹; Blowes et al., 1991; Moncur et al., 2003, 2006) and in the United States (in excess of 900 000 mg l⁻¹; Nordstrom and Alpers, 1999b; Nordstrom et al., 2000).

Geochemical models based on the Pitzer approach have been applied to describe mineral-stability relationships in concentrated mine-drainage waters. At the Heath Steele site in New Brunswick, Canada, high concentrations in pore waters were observed in shallow tailings as a result of sulfide oxidation. Mineral saturation indices calculated using a Pitzer-based model for melanterite, gypsum, and siderite were close to 0.0 where these minerals were observed to occur, whereas saturation index values calculated using the conventional ion-pair model indicated supersaturated conditions (Blowes et al., 1991; Ptacek and Blowes, 1994). Tailings pore waters should have achieved equilibrium with respect to melanterite and gypsum, and possibly siderite, within the observed residence time in the tailings. At the Sherridon mine site in Sherridon, Manitoba, mineral saturation indices for melanterite, gypsum, and siderite also were close to 0.0 where these minerals were observed (Moncur et al., 2005). These results suggest that the Pitzer-based models have the potential to provide predictions of mineral-stability relationships that are consistent with observations at mine-drainage sites with concentrated pore waters.

11.5.8.5 Reactive-Transport Models

Reactive-transport models couple the equations that describe physical transport processes with equations that describe geochemical reactions. These models can be divided into three basic categories: (1) equilibrium models, (2) partial equilibrium models, and (3) kinetic models. The three are differentiated by the approach used to solve the equations for chemical reactions. Equilibrium-based models apply the local equilibrium assumption (LEA) in which all chemical reactions are assumed to proceed to completion or attain equilibrium within each time step. This assumption facilitates coupling between the physical transport step and the geochemical reaction step. In most models, the iteration between the transport step and the chemical step assures convergence between the two modules (Walter et al., 1994a,b). In some cases, however, the two steps proceed sequentially with no iteration between them. A comparison conducted by Walter et al. (1994a) suggests that the error arising from use of the sequential approach is modest. Partial equilibrium models identify a reaction, or a series of reactions, for which the equilibrium assumption is unrealistic. Kinetic expressions are then applied to describe these reactions. The remaining geochemical reactions are described using the LEA (Wunderly et al., 1996). Kinetic models provide the opportunity to consider the rates of all chemical

reactions that occur. In several of these models, the chemical reactions are directly substituted into the transport equations and are solved simultaneously (Lichtner, 1993; Mayer et al., 2002; Steefel and Lasaga, 1994; Steefel and van Capellan, 1998). These models provide a comprehensive description of the geochemical system.

Reactive-transport models have been used to describe sulfide oxidation and transport of its products through mine wastes. Liu and Narasimhan (1989a,b) used a reactive-transport model to describe the transport of dissolved constituents at a uranium mine site. Sulfide oxidation and transport of oxidation products at the inactive Nordic uranium tailings impoundment were described using reactive-transport models (Walter et al., 1994; Wunderly et al., 1996). The transport of metal contaminants at Pinal Creek, Arizona, was modeled by Brown et al. (1998). Sulfide oxidation and transport of oxidation products through the Nickel Rim tailings impoundment, Sudbury, Ontario, were simulated by Bain et al. (2000) and by Mayer et al. (2002). The performance of a PRB at the Nickel Rim mine site was simulated using the reactive-transport model MIN3P (Mayer et al., 2006). These simulations that included the reaction mechanisms proposed by Benner et al. (2000), together with the flow system parameters and temperature conditions measured at the Nickel Rim site, provide good agreement with the field observations. MIN3P also was applied by Brookfield et al. (2006) to describe sulfide oxidation and subsequent reactive transport of oxidation products at the INCO Central Tailings Disposal Area, Copper Cliff, Canada. Acero et al. (2009) applied a coupled model that includes the Pitzer ion-association model and kinetically based geochemical reactions coupled to a dynamic flow and solute transport model to describe sulfide oxidation and reactive transport in laboratory column experiments. The performance of a PRB at the Nickel Rim mine site was simulated using the reactive-transport model MIN3P (Mayer et al., 2006). These simulations that included the reaction mechanisms proposed by Benner et al. (2000), together with the flow system parameters and temperature conditions measured at the Nickel Rim site, provide good agreement with the field observations.

11.5.9 Approaches for Remediation and Prevention

Extensive degradation of surface waters by the effluents from abandoned mines and mine wastes has the treatment of effluents imperative in many mining districts throughout the world. Because the degradation of surface-water resources is obvious, often, the remediation of surface waters is the first step in remediation. However, prevention of oxidation in mine wastes and treatment of contaminated groundwater have become areas of active research in mine rehabilitation efforts over the past two decades. An understanding of the geochemical and hydrological processes that result in the release and transport of the products of sulfide oxidation guides the development and implementation of remedial technologies at many mine sites. Among the various remedial strategies that have been developed for implementation at mine sites are the collection and treatment of contaminated surface water and groundwater, passive treatment of contaminated surface water using constructed wetlands, treatment of contaminated

groundwater using PRBs or other in situ remedial approaches, and emplacement of covers on tailings impoundments and waste-rock piles to prevent oxygen ingress or infiltration of precipitation. Mine-site remediation may focus on application of one of these strategies or, more commonly, the integration of a combination of these approaches. The selection of appropriate technologies depends on the physical conditions of the site, the mineralogy of the wastes, and the extent of existing sulfide oxidation.

11.5.9.1 Collection and Treatment

The most common approach to remediate sulfide-bearing mine wastes is collection of the effluent and treatment by pH neutralization and metal precipitation. At many mine sites, water is collected from ponds and ditches and is conveyed to a central treatment facility. The most common approach to water treatment is through pH neutralization using lime, with the subsequent precipitation of metals as a ferric hydroxide sludge. Over the past decade, substantial improvements have been made in treatment efficiency, particularly in the development of high-density sludge systems that both remove greater amounts of dissolved metals and produce lower volumes of waste sludges. The wastes derived from these treatment plants may be disposed of in dedicated facilities or may be codisposed with mill tailings. Little research has focused on the stability of treatment sludges within tailings impoundments. Zinck (1999) assessed the stability of sludges from wastewater treatment plants from several mine sites in Canada. Although the results suggested that these sludges are stable under the conditions that prevail in many mine-tailings impoundments, the stability of codisposal waste products from metal refineries seems to differ. For example, Al (1996) observed that jarosite residue, derived from the Kidd Creek metallurgical plant at Timmins, Ontario, was unstable in the tailings impoundment at that site. At the Campbell mine at Red Lake, Ontario, As-bearing hematite and maghemite residues derived from the roaster circuit in the Au refinery were mixed with flotation tailings and were codischarged to the tailings impoundment. The biologically mediated reductive dissolution of the roaster residue was determined to release As to the tailings pore water, which has been subsequently displaced into an underlying aquifer (McCreadie et al., 2000; Stichbury et al., 2000).

Alternative water-treatment technologies have recently been developed and applied for the treatment of mine-site effluents. Biologically mediated systems that reduce sulfate and promote the precipitation of insoluble metal sulfides have been developed for treatment of mine-waste streams. Reverse osmosis systems have been applied for treatment of mine-waste effluents or for polishing the effluent from facilities that use lime treatment.

11.5.9.2 Controls on Sulfide Oxidation

11.5.9.2.1 Physical barriers

Figure 14 shows a schematic diagram of a mine-waste impoundment. The generation of acidic drainage results from the exposure of sulfide minerals in the vadose zone of the tailings impoundment to atmospheric O_2 . Oxygen penetrates the sulfide-bearing wastes by downward and inward

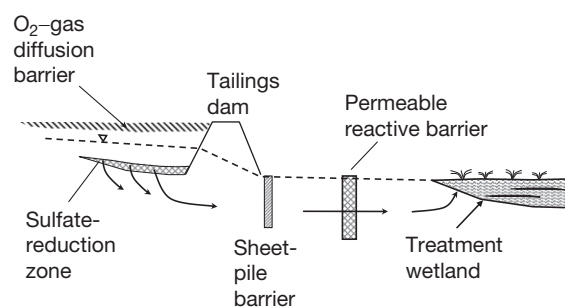


Figure 14 Schematic diagram of a mine-waste impoundment with a combined remediation approach, including a cover to prevent O_2 and water ingress, in situ mixing to geochemically stabilize waste, permeable reactive barrier in aquifer to treat subsurface drainage, and wetland for surface treatment of drainage.

movement. Limiting O_2 transport into mine wastes has the potential to limit the extent of sulfide oxidation and the rate of contaminant release. Various physical barriers have been applied or proposed to prevent oxygen ingress; the most commonly used barriers are covering the wastes with water, which limits oxygen transport because of its low-diffusion coefficient, emplacement of covers composed of soil materials designed to maintain high degrees of saturation and low gas-phase diffusivities, and emplacement of synthetic covers, which can be specified to maintain low rates of oxygen transport. In addition to covers that impede oxygen transport, covers composed of oxygen-consuming materials, such as wood wastes, have been applied to prevent oxidation of underlying sulfide minerals.

To be most effective, physical barriers must be applied either at the time that mine-waste deposition ceases or shortly thereafter. The rate of sulfide oxidation is greatest immediately after mine-waste deposition ceases, whereupon the moisture content declines. Measurements of concentrations of dissolved constituents in inactive tailings impoundments and modeling of the rate of sulfide oxidation in these wastes indicate that extensive oxidation occurs during the first decade after tailings disposal is complete and that the rate of sulfide oxidation rapidly declines thereafter as the zone of active oxidation migrates deeper into the impoundment and the path for oxygen diffusion lengthens (Johnson et al., 2000; Mayer et al., 2002). These observations suggest that, for sites where mine-waste oxidation has progressed for more than a decade, the emphasis of remedial programs should shift from preventing sulfide oxidation to managing the fate of dissolved oxidation products.

11.5.9.2.1.1 Subaqueous disposal

The solubility of O_2 in water is low, approximately $8\text{--}13\text{ mg l}^{-1}$, at normal surficial temperatures. Furthermore, the diffusion coefficient of O_2 in water ($2 \times 10^{-9}\text{ m}^2\text{ s}^{-1}$) is much lower than the diffusion coefficient of O_2 in air ($1.78 \times 10^{-5}\text{ m}^2\text{ s}^{-1}$). These characteristics have been exploited to limit the rate of sulfide-mineral oxidation by covering mine wastes with water. Subaqueous disposal can be achieved by depositing the wastes in natural water bodies, including lakes or marine systems, or in constructed impoundments designed to maintain the water cover. The extent of sulfide oxidation in tailings submerged in lakes in northern Manitoba was observed to be much less extensive than oxidation in adjacent subaerial tailings impoundments

(Pedersen et al., 1993). The development of reduced conditions in the sediments overlying the tailings resulted in the formation of authigenic sulfide minerals.

The environmental implications of dedicating a lake to mine-waste disposal can be significant. Mining companies proposing subaqueous disposal in natural water bodies face regulatory hurdles, and in some jurisdictions, disposal of mine wastes in natural water bodies is not permitted. In addition, mine sites may be remote from water bodies that have the size or depth to accommodate the mass of mine wastes generated through mineral exploitation. In these settings, construction of surface repositories with sufficient retention capacity to maintain water covers over wastes has been constructed (Davé and Vivyurka, 1994). To maintain structural stability of the repository, it is desirable to minimize the depth of the free water cover. Water covers of 1 m or less have been placed on inactive tailings impoundments (Vigneault et al., 2001; Yanful and Verma, 1999). Shallow water covers may be susceptible to resuspension of tailings by wind and wave action (Yanful and Verma, 1999). Furthermore, where sunlight penetrates the water cover, a periphyton layer may develop on the surface of the exposed tailings, producing oxygen at the tailings surface (Vigneault et al., 2001). As a result of these processes, complete prevention of sulfide oxidation may not be attained in repositories with shallow water covers (Vigneault et al., 2001; Yanful and Verma, 1999). For example, Martin et al. (2001) observed more extensive oxidation in shallow lake bed sediments than in similar sediments in deeper portions of a lake in Peru. Although, shallow water covers do not completely prevent sulfide oxidation, these covers do substantially lessen the rate of oxidation relative to rates observed under subaerial conditions. Vigneault et al. (2001) estimate that tailings submerged beneath a 0.3 m water cover oxidized at a rate approximately 2000 times less than the rate observed for in moist tailings exposed to oxygen in humidity cell experiments.

11.5.9.2.1.2 Dry covers

Retaining dams constructed at many mine-waste disposal facilities are composed of coarse-grained mill tailings or mill tailings combined with waste rock. Many of these structures were designed to retain tailings during deposition and to drain and consolidate after deposition is complete. Consolidation enhances the stability of the retaining dam and is a component of the impoundment design. Alternative cover designs may include layers that maintain high moisture contents under negative pressures. These dry covers maintain a near-saturated layer several meters above the water table. Dry covers typically are composed of layers of soil materials with variable grain-sized characteristics (Holmström et al., 2001; Nicholson et al., 1989). In their simplest form, these covers can be composed of a single layer of fine-grained material. In 1988, a fine-grained cover was applied to the surface of waste-rock piles at the Rum Jungle mine site in the Northern Territory, Australia (Harries and Ritchie, 1985, 1987). The performance of the Rum Jungle cover has been monitored regularly since installation (Harries and Ritchie, 1985, 1987; Timms and Bennett, 2000). Analysis and modeling of the field data indicate that the cover decreased the rate of oxidation of the underlying sulfide minerals by one-third to one-half of the initial rate and that recent data

indicates that both the oxidation rate and the rate of infiltration have increased (Timms and Bennett, 2000).

More complex covers may be comprised of several layers, each with different soil characteristics. A cover installed at the Bersbo site in Sweden included layers of fine-grained clay and coarse-grained aggregate material and a protective layer of natural till (Lundgren, 2001).

Dry covers may also be installed with the objective of preventing the infiltration of precipitation into the underlying mine wastes. These covers are designed to retain moisture during wet seasons and promote evapotranspiration of this moisture during dry periods (Bews et al., 1997; Durham et al., 2000).

11.5.9.2.1.3 Synthetic covers

Covers composed of synthetic materials have been emplaced at mine-waste sites to prevent ingress of both oxygen and water. Synthetic materials used include polyethylene, concrete, and asphalt. A complex cover includes a sealing layer consisting of concrete stabilized with fly ash at the Bersbo site in Sweden in 1988. This cover includes a granular stabilization layer, the concrete sealing layer, and a 2-m-thick protective till cover. Instruments were installed through the barrier at the time of implementation, and monitoring since then has shown that, although the cover formed a suitable oxygen barrier, changes in barometric pressure cause migration of O₂ into the waste material (Lundgren, 2001).

To prevent the release of contaminants from an inactive Cu-Zn mine-tailings impoundment near Joutel, Québec, a geomembrane and a protective clay cover overlain by 0.5–1.5 m of till were placed on the impoundment surface (Lewis et al., 2000). The performance of this barrier is the subject of a long-term monitoring program.

11.5.9.2.1.4 Oxygen-consuming materials

Covers containing organic materials have been applied to the surface of mine wastes (Tassé et al., 1994). The principal objective of these covers is to prevent oxygen entry into the mine wastes by intercepting and consuming oxygen in the cover material. Wood wastes and other waste forms of organic carbon have been evaluated to assess their potential as cover materials. Reardon and Poscente (1984) evaluated the potential application of wood wastes as an oxygen-consuming barrier for sulfide-bearing mine tailings. The conclusion is that these materials may provide an effective oxygen barrier, but the life-span of the organic carbon cover would be insufficient for this approach to be viable. Tassé et al. (1994) described the performance of a wood-waste cover installed at a mine site in Québec. At this site, the cover provided an effective barrier to the ingress of O₂ into the underlying tailings.

Organic materials, such as wood waste or waste materials derived from pulp-and-paper operations, have the potential to leach organic acids, which subsequently move into the underlying tailings. For freshly deposited tailings, in which the oxidation of sulfide minerals has been minimal, the presence of these organic acids is not anticipated to have deleterious environmental effects. If an organic cover is emplaced over weathered sulfide-bearing mine tailings, however, accumulations of ferric iron-bearing secondary minerals may be reductively dissolved by the organic acids released from the cover materials. Extraction experiments conducted by Ribet et al. (1995)

indicated that most of the metals present in the shallow portions of the Nickel Rim tailings impoundment, Sudbury, Ontario, were in the oxidized form and would be susceptible to release by reductive dissolution.

11.5.9.2.2 Chemical treatments

A variety of methods has been developed to prevent sulfide-mineral oxidation at the particle scale. Chemical treatments have been proposed to encapsulate sulfide minerals in inert materials, thereby preventing oxidation at the sulfide-mineral surface. Huang and Evangelou (1994) proposed encapsulation with silicate coatings. Silicate binds with ferric iron on the sulfide-mineral surface, rendering the surface inert. Huang and Evangelou (1994) and Ueshima et al. (2004) proposed using phosphate to bind on sulfide surfaces in a similar manner. Laboratory experiments indicated that the rate of oxidation at the sulfide-mineral surface was much less in the presence of phosphate and silicate anions. Silicate coating of sulfide grains was also evaluated by Georgopoulou et al. (1995) and by Fytas and Bousquet (2002). Belzile et al. (1997) also observed that coating pyrite grains with either acetyl acetone or sodium silicate reduced the oxidation rate of pyrite grains. All of these studies indicated that coatings formed on the surfaces of sulfide minerals substantially decreased the rate of sulfide oxidation relative to that of pristine sulfide minerals. In addition to chemical treatments that incorporate the addition of anionic reagents, accelerated oxidation of sulfide particles by using strong oxidants, such as permanganate, has been proposed. The objective is to oxidize the surface of the sulfide grain, creating a protective armor layer.

11.5.9.2.3 Bactericides

The rate of abiotic pyrite oxidation declines as the pH decreases below 3.5. Under these conditions, bacterially mediated pyrite oxidation predominates. Numerous researchers have evaluated the use of bactericides, principally anionic surfactants, to prevent bacterial activity and limit the rate of sulfide oxidation. Bactericides can be applied either directly to mine-waste surfaces or as intimate mixtures with the mine wastes (Ericson and Ladwig, 1985). Preliminary results using this approach were encouraging (Dugan, 1987a,b; Sobek, 1987). Slow release of surfactants from rubber-based pellets is an approach that has been developed to extend the life-span of these surfactants in field applications. The toxicity of low-molecular-weight organic acids to microorganisms is well documented. At low-pH values, they have a neutral charge and are able to diffuse across the membrane and deprotonate in the cytoplasm killing the microorganism (Aston et al., 2009). However, organic acids would be too expensive for use in the field and would also degrade very rapidly. Thiocyanate has also been evaluated as an inhibitor of sulfide-mineral oxidation in waste rock (Olson et al., 2004), but thiocyanate is toxic and it would be difficult to obtain regulatory approval for its use. Stichbury et al. (1995) proposed the use of thiol-blocking agents to inhibit bacteria involved in the early stages of sulfide oxidation. In shake-flask cultures and in column experiments, Stichbury et al. (1995) observed reduced sulfide oxidation that persisted over a 1-year study. In a 2-year field study, the inhibitor mixture completely prevented the growth of

At. ferrooxidans and lessened the growth of *At. thiooxidans* and *Thiobacillus thioparus* (Lortie et al., 1999). Lower concentrations of sulfate, iron, and dissolved metals were observed in the treated field plots than were observed in untreated control plots. The effectiveness of the inhibitors was observed to decline following the first year after application, and thus, they would not be suitable for long-term application. This observation suggests that inhibitors may be best applied in settings where prevention of sulfide oxidation is required for limited periods only. Examples are stockpiling of waste rock before it is placed in underground workings, stockpiles of mill tailings that will be subsequently used for mine backfill, or waste-rock and tailings piles during periods of temporary closure.

11.5.9.3 Passive Remediation Techniques

11.5.9.3.1 Types of passive systems

Passive treatment systems can be based on chemical or biological processes or a combination of both (Neculita et al., 2007). Systems for the treatment of mine drainage will often consist of several treatment cells in series based on different treatment processes (Johnson and Hallberg, 2005). Passive chemical systems include anoxic limestone drains (ALDs), limestone channels, and PRBs, and biological systems include constructed wetlands, bioreactors, and PRBs. PRBs will be discussed in another section of this article.

11.5.9.3.2 Anaerobic bioreactors

Anaerobic bioreactors (ABRs) consist of organic matter to provide substrates for microbial activity, sand and gravel for permeability, and limestone for alkalinity generation (Johnson and Hallberg, 2005; Kaksonen and Puhakka, 2007). They are generally operated in the upflow mode to develop anaerobic conditions more rapidly and to reduce compaction (Mattes et al., 2011). Although a number of processes are responsible for attenuating contaminant concentrations in ABRs, the key process is sulfate reduction. Sulfate reduction generates alkalinity and precipitates metals by the formation of metal sulfides. Because SRB only utilize simple organic compounds such as low-molecular-weight organic acids and alcohols as electron donors, they are dependent on other organisms to provide these substrates (Gould and Kapoor, 2003; Logan et al., 2005; Pruden et al., 2007). The major components of plant-derived organic matter are lignin, cellulose, and more reactive low-molecular-weight organic compounds. Lignin degrades much more slowly than the other materials but will contribute to the longevity of the ABR. The more reactive organic compounds will be consumed rapidly. Most of the electron donors for sulfate reduction will be provided by cellulose degradation. During degradation, cellulose is hydrolyzed to sugars, which are then fermented into alcohols and organic acids which can then be used by the SRB (Gould and Kapoor, 2003). In most ABR systems with the exception of pH extremes, carbon availability is the limiting factor. Microbiological studies of laboratory ABRs indicate that the key microorganisms are the anaerobic cellulose degraders and that cellulose hydrolysis is the rate-determining step (Logan et al., 2005; Pruden et al., 2007).

11.5.9.3.3 Constructed wetlands

Constructed wetlands have been used for water treatment since the early 1950s (Hedin and Naim, 1992). The interest in using constructed wetlands for the remediation of mine drainage stems from their ability to filter particulate matter, thus reducing suspended solids, their ability to reduce biochemical oxygen demand and remove and store nutrients and heavy metals, and their natural buffering capacity (Greenway and Simpson, 1996; Hsu and Maynard, 1999; Machemer and Wildeman, 1992; Mayes et al., 2009; Perry and Kleinmann, 1991). However, wetland systems can lose their effectiveness if the influent pH is too low (Hsu and Maynard, 1999). The influent pH can be increased by treating the influent with an alkalinity-generating system such as anoxic limestone channel or an ALD prior to the wetland (Cravotta, 2010; Zipper and Skousen, 2010). If properly designed and maintained, constructed wetlands can result in the improvement of the quality of drainage water derived from mines (Walton-Day, 1999). The mechanisms that promote acid neutralization and metal retention include dissolution of primary carbonate minerals, formation and precipitation of metal hydroxides, microbially mediated sulfate reduction, formation of metals sulfides, complexation of metals by organic carbon, ion exchange, and direct uptake by plants (Bendell-Young and Pick, 1996). Batty et al. (2008) designed a wetland for the treatment of drainage from an abandoned coal mine with Fe and Mn concentrations of 45 and 2 mg l⁻¹, respectively. The system consisted of two oxidation ponds (total area 800 m²) and three wetlands (total area 3000 m²). The iron was removed in the first part of the system, but manganese was only removed in the latter stages of the system. Wetlands can be designed to function under aerobic or anaerobic conditions. Aerobic wetlands consist of *Typha* and other wetland vegetation planted in shallow relatively impermeable sediments comprised of soil, mine spoil, or clay. The vegetation in anaerobic wetlands is planted in deep (>30 cm) permeable sediments consisting of peat, soil, organically enriched compost, sawdust, etc., which are sequenced between crushed limestone or are mixed (<http://www.wvu.edu/~agexten/landrec/passtr/passtrt.htm>).

Anaerobic constructed wetlands promote bacterially mediated sulfate reduction. These reactions have the potential to remove metals, as well as high concentrations of sulfate, from mine drainage. A growing concern associated with the use of wetlands to attenuate metals is the potential for the wetland soils to become latent sinks of toxic metals (Gopal, 1999). Heal and Salt (1999) monitored a constructed-wetland system impacted by acidic metal-rich drainage from ironstone mine spoils. Mine waters (pH 2.7, 247 mg l⁻¹ total Fe) passing through the wetland were monitored for 12 months. After installation, the acidity declined by 33%, and Fe, Mn, and Al concentrations declined by 20–40%. The performance of the constructed wetland showed favorable response to metal loads during the summer months, but the rates of metal removal decreased significantly in the winter months (Heal and Salt, 1999). Sobolewski (1996) monitored the performance of a constructed peat-based wetland for long-term treatment of mine drainage at the former Bell Copper Mine near Smithers, B.C., Canada. Copper concentrations were used to provide estimates of the long-term potential for removal of Cu from the mine-drainage site. Two Cu species were identified: Cu

sulfides and Cu bound to organic material. Removal from mine effluent containing low concentrations of Cu at near-neutral pH was through sulfide formation. Removal from acidic drainage containing higher Cu concentrations was by complexation to organic matter. The effectiveness of the constructed-wetland system to remove metals under low-pH conditions and with high-metal concentrations was insufficient to be suitable for long-term remediation.

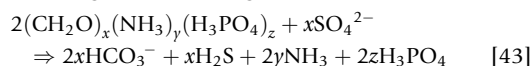
The definition of appropriate operating conditions is required to assure success of wetland treatment systems. Acidic drainage containing high concentrations of dissolved metals discharged from abandoned underground mines in Kentucky, United States, was intercepted by a wetland constructed in 1989 to reduce the concentration of metals (Barton and Karathanasis, 1998). The initial attempt to treat the drainage water using the wetland failed after 6 months because of insufficient utilization of the treatment area, inadequate alkalinity production, and high-metal loadings. A renovation added two ALDs and a series of anaerobic subsurface compartments to promote vertical flow through a successive alkalinity-producing system. The renovated system was monitored over 19 months and showed improved performance in metal removal. Iron concentrations decreased from 787 to 39 mg l⁻¹, the pH increased from 3.38 to 6.46, and the acidity decreased from 2244 to 199 mg l⁻¹ (carbonate equivalent) (Barton and Karathanasis, 1998). The mass loading of metals was reduced by an average of 98% for Al, 95% for Fe, 55% for sulfate, and 49% for Mn.

Improving the efficiency of metal removal using wetlands has proved to be difficult. Successful wetland treatment systems are frequently overdesigned in order to handle variations in influent loading and also to prolong the effective lifetime of the wetland (Hedin, 2008). Detailed assessments of long-term performance of wetland systems with a focus on metal-retention mechanisms, eutrophication, and flushing rates and their effects on downgradient ecosystems are required if wetland treatment is to be applied in temperate regions.

11.5.9.3.4 Permeable reactive barriers

PRBs are used for treatment and prevention of mine-drainage waters within aquifers receiving discharge from mine wastes. These systems are constructed by excavating a portion of the aquifer downgradient from the waste disposal area and filling the excavation with a permeable mixture composed of reactive components (Blowes et al., 2000a,b). Among the mixtures designed for treating mine-drainage waters have been organic carbon in the form of composted municipal wastes, wood wastes, and by-products from pulp-and-paper manufacturing (Benner et al., 1999; Blowes et al., 1994), zerovalent iron (Blowes et al., 2000a; Morrison et al., 2002; Naftz et al., 2002), limestone, and phosphate-based adsorbent materials (Conca et al., 2002).

Reactive mixtures containing organic carbon are designed to support the bacterially mediated reduction of sulfate and the precipitation of metal sulfides (Blowes et al., 1994; Cocos et al., 2002; Waybrant et al., 1998, 2002). SRB oxidize organic carbon by using SO₄ as the electron acceptor, thereby generating H₂S and releasing dissolved inorganic carbon:



where $(\text{CH}_2\text{O})_x(\text{NH}_3)_y(\text{H}_3\text{PO}_4)_z$ represents organic matter undergoing oxidation and x , y , and z are stoichiometric coefficients. The H_2S produced through sulfate reduction combines with metal cations to form metal sulfides:



These metal sulfides are stable below the water table within the reactive barrier. Sulfate reduction also releases dissolved inorganic carbon, which neutralizes the pH and favors the precipitation of metal carbonate minerals, for example, FeCO_3 and MnCO_3 (Waybrant et al., 2002).

The first full-scale reactive barrier for treatment of mine drainage was installed at the Nickel Rim mine site, near Sudbury, Ontario in 1995 (Benner et al., 1997). The barrier contains a reactive mixture composed of 50 vol% organic carbon, 49 vol% pea gravel, and 1 vol% limestone. The organic carbon used was a composted waste of leaf mulch and wood chips derived from a municipal recycling program. Semiannual monitoring of the reactive barrier for 5 years after installation showed consistent removal of dissolved sulfate and dissolved metals and increases in pH and alkalinity (Benner et al., 1999). Enumeration of SRB within the barrier indicated that a large population of SRB had become established within a year following installation (Benner et al., 2000). Measurements of the sulfur isotopic ratio indicated enrichment of ^{34}S - SO_4 in the barrier effluent, which is consistent with the hypothesis that the sulfate removal was through bacterially mediated sulfate reduction. Measurements of the solid-phase sulfide content indicated accumulations of acid-volatile sulfides within the barrier material and lesser accumulations of total reducible sulfide (Herbert et al., 1998, 2000).

The aqueous geochemistry data obtained from the Nickel Rim site indicates that the rate of sulfate reduction is dependent on temperature, with sulfate reduction more rapid during the warm summer months than during the cooler winter months. The relationship between temperature and reaction rate could be explained using an Arrhenius-type relationship (Benner et al., 2002).

A pilot-scale reactive barrier for treating metals and sulfate was installed in 1997 at an industrial site in Vancouver, Canada (Ludwig et al., 2002; McGregor et al., 2002). Monitoring of the barrier between 1997 and 2001 showed that initial concentrations of up to $8570 \mu\text{g l}^{-1}$ Cu were decreased to $17 \mu\text{g l}^{-1}$ after passage through the barrier, Zn decreased from 2780 to $<140 \mu\text{g l}^{-1}$, and Cd and Ni declined from 21 and $210 \mu\text{g l}^{-1}$ to concentrations of <0.2 and $<6 \mu\text{g l}^{-1}$, respectively (McGregor et al., 2002). The performance of the pilot-scale barrier led to the installation, in 2002, of a full-scale reactive barrier that is 400 m long and 17 m deep, with a maximum thickness of 5 m.

PRBs have also been installed at uranium mine-tailings impoundments to remove U and other metals. A reactive barrier containing three reactive components, as part of a side-by-side comparison, was installed at the Fry Canyon Mine site, Utah. The reactor systems included hydrous ferric oxide for the adsorption of U, zerovalent iron to induce U reduction and precipitation, and bone char that is rich in phosphate to induce the precipitation of U-bearing phosphates. Monitoring showed removal of U in all of these systems (Blowes et al., 2000a,b;

Fuller et al., 2002; Naftz et al., 2002). A full-scale permeable barrier using zerovalent iron for reduction and precipitation of U was installed in 1999 in Monticello, Utah (Morrison et al., 2002). This barrier treats U, Mo, Se, and V.

PRBs have significant advantages over the conventional approaches for groundwater remediation. The barriers are passive systems whose effectiveness can persist for several years to decades. Moreover, the reactive material can be adjusted to target specific contaminants from a mine site, and the contaminants precipitated within the barrier are isolated from the surface-water environment and biota (Blowes et al., 2000a,b).

11.5.9.3.5 Other in situ techniques

The addition of organic carbon to mill tailings has the potential to enhance sulfate-reduction reactions and reprecipitation of metal sulfides within tailings impoundments (Blowes and Ptacek, 1992b). The addition of organic carbon below the equilibrium water table position will isolate the organic carbon from oxidation by atmospheric O_2 . The addition of organic carbon to sulfide-rich tailings was evaluated in pilot test experiments conducted at the Kidd Creek Zn-Cu metallurgical site, Timmins, Ontario (Hulshof et al., 2003, 2006), and at the Greens Creek Z-Pb mine, near Anchorage, Alaska (Lindsay et al., 2009, 2011a, b,c). At the Kidd Creek metallurgical site, the addition of organic carbon was added to the sulfide-rich tailings in two test cells, one containing pulp waste from a pulp-and-paper mill and the second containing wood waste, principally tree branches and bark. The organic carbon was mixed thoroughly with the tailings and deposited in test cells, and the changes in water chemistry were monitored over a two-year period. In the test cell containing pulp waste, the organic carbon promoted the growth and activity of SRB, resulting in the precipitation of secondary sulfide minerals and declines in the concentrations of dissolved metals, principally Fe and Zn. The rate and extent of metal removal were greater in the test cell containing pulp waste than in the test cell containing wood waste (Hulshof et al., 2006).

The effectiveness of the addition of organic carbon in the form of peat, spent brewing grain, and municipal sewage sludge was evaluated at the Greens Creek Mine, Alaska, through the construction and monitoring of seven test cells (Lindsay et al., 2009, 2011a,b,c). The test cells were installed by excavating tailings materials, mixing with organic carbon, and placing the mixture within the excavation. A low permeability liner on the sides of the test cells was used to limit horizontal flow in the test cells. The base of the test cells remained open and connected to the underlying tailings impoundment. Monitoring over 4 years indicated that populations of IRB and SRB were established in all of the test cells within the first year of installation, but the largest and most active populations were observed in the test cells containing spent brewing grain (Lindsay et al., 2011b). The precipitation of metal sulfides resulted in substantial decreases in the concentrations of sulfate and dissolved metals, principally Zn. The declines in Zn concentrations were accompanied by an increase in dissolved As, which was attributed to microbially mediated reduction of iron oxyhydroxides and release of coprecipitated As, followed by a subsequent decrease in dissolved As. Mineralogical study of samples collected from the test cells indicated the accumulation of poorly crystalline Zn and Fe sulfides (Lindsay et al., 2011b,c).

11.5.10 Summary and Conclusions

The exposure of sulfide minerals contained in mine wastes to atmospheric oxygen results in the oxidation of these minerals. The oxidation reactions are accelerated by the catalytic effects of Fe hydrolysis and sulfide-oxidizing bacteria. The oxidation of sulfide minerals results in the depletion of minerals in the mine waste and the release of H^+ , SO_4 , $Fe(II)$, and other metals to the water flowing through the wastes. The most abundant solid-phase products of the reactions are typically ferric oxyhydroxide or hydroxysulfate minerals. Other secondary metal sulfate, hydroxide, hydroxy sulfate, carbonate, arsenate, and phosphate precipitates form. These secondary phases limit the concentrations of dissolved metals released from mine wastes.

At many sites, the acid produced by sulfide oxidation is consumed by the pH-buffering reactions of the nonsulfide gangue minerals contained in the wastes. As these acid-neutralization reactions proceed, the pH of the pore water progressively increases, enhancing the potential for attenuation of dissolved metals by adsorption and precipitation reactions. The quality of aqueous effluents from tailings impoundments and waste-rock piles is dependent on the extent of the acid-producing sulfide-oxidation reactions versus the acid-consuming pH-buffering reactions. Regardless of the degree of acid neutralization, the effluent water from the waste material contains increased concentrations of dissolved constituents.

Oxidation products transported from the mine waste can enter streams, lakes, and oceans. A series of further reactions occur upon the discharge of this mine-drainage water, resulting in acidification and metal release. Complex geochemical models and reactive-transport models have been developed and are important tools in the prediction of the environmental impacts of mine wastes and in developing remedial alternatives. The remedial technologies developed and refined over the past two decades are being applied increasingly, thereby decreasing the magnitude of the negative effects of mining operations.

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